

The block below carries no module guard, so docstrip copies it into EVERY generated file. That is what we want for the TeX files and wrong for the latexmkrc, which is Perl and does not understand lines opening with two percent signs – hence the negated guard around it.

The guard lines themselves are hidden behind a false conditional. docstrip reads a guard line either way, but ltxdoc, which typesets this file, would otherwise set the literal text of the guard as a stray paragraph in the body of the manual.

Nothing in this area may name a TeX conditional, not even in prose: outside the meta-comment at the top of the file, every documentation line here IS LaTeX source and is executed, and a conditional named in passing opens or closes a real one. That is not hypothetical – an earlier draft of this very comment swallowed the rest of the manual.

Three conditional primitives are deliberately absent from the lists below: the ones that close or branch a conditional. `\DoNotIndex` scans its argument with TeX's own conditional machinery live, so a bare one of them closes the wrong conditional. The manual ended its run with “Too many }’s” and exited with status 1, which failed the docs step of the harness even though the PDF had been written. Leaving them out of the lists means they get indexed, which is harmless. Note that they cannot be named here either, not even inside . . . shorthand split across a line.

The coppe document class

Version 4.0

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Abstract

In this work, it is described the `coppe` document class as well as other files distributed by the COPPETEX project. This class is suitable for writing academic dissertations, thesis and qualifying exams according to the formatting rules of the Alberto Luiz Coimbra Institute for Graduate Studies and Research in Engineering. The minimalist set of macro commands allows its users to concentrate most of their efforts on text composition rather than on the document layout.

1 Introduction

Writing documents in L^AT_EX may be a laborious task when the authors have to prepare their manuscripts rigorously respecting formatting rules imposed by publishers. Regardless of difficulty, a lot of thesis presented to the Coordination of Graduate Studies and Research in Engineering of the Federal University of Rio de Janeiro (COPPE/UFRJ) is typesetted in L^AT_EX. This demand motivated the creation of the COPPETEX project, which tries to facilitate and encourage the use of L^AT_EX within the COPPE/UFRJ scope.

The `coppe` document class is the main product of COPPETEX. It was designed to be clear and succinct. It enables the creation of dissertations, qualifying exams and thesis in a simple and automatic way. The main goal of the `coppe` class is to maintain authors strictly focused on text composition without worrying about margins sizes, line spacing, paper size, vertical and horizontal alignment, etc. The COPPETEX project comprehends also BibTEX and MakeIndex style files for creating lists of references, symbols and abbreviations. Although there aren't official guidelines to write qualifying exams, we provide this option just for convenience, as this exam is a requisite to obtain the DSc degree and, for some of the programs, the MSc degree.

In which follows, it is described the user interface of the `coppe` class. Some details about using the style files cited above are also given. We use the term *thesis* to generally refer to dissertation, qualifying exam, and thesis itself.

2 License

Each file belonging to this package contains a copyright notice. Its use is protected by the GNU General Public License (GPL) version 3, so that users are free for

copying, distributing or modifying the source code, among other acts covered by this license.

To see the full text of the GNU GPL license, go to the `COPYING` file attached to this package.

3 Support

We maintain a WhatsApp group where users can send questions, comments, and bugs to. More details can be found [here](#).

However, as the project maintenance was transferred to a new repository, bug reports, as well as new feature requests, should be directed to <https://github.com/COPPE-UFRJ/CoppeTeX>.

4 User interface

`\frontmatter` A thesis to be approved by the Academic Registry at COPPE/UFRJ must contain three-parts: *front*, *main* and *back* matters (COPPE/UFRJ Academic Registry guidelines). Each one of these parts is started by calling its corresponding macro `\frontmatter`, `\mainmatter` or `\backmatter`. The front matter of a thesis consists of front cover and face, cataloging page, dedication, acknowledgments, abstracts, table of contents, and lists of tables, algorithms, symbols and abbreviations. The main matter is just composed by chapters, while the back matter usually consists of bibliographic references, appendices and index.

You must invoke the `\frontmatter` macro immediately after the `\maketitle` one. The `\mainmatter` command comes right before the first chapter, and `\backmatter` must be typed before the list of references.

Cover (capa)

The ABNT cover suggested in the manual's Annex A. The `\maketitle` command builds it automatically, ahead of the title page: the university, the full COPPE name, and the graduate program (taken from `\department`), followed by the author, the title, and the city and year, all centred and in uppercase. The UFRJ and COPPE logos head the page. The capa is not counted.

Front face

The front face is unnumbered. There, it is not allowed to use hyphenation (COPPE/UFRJ Academic Registry guidelines). It is constructed by calling `\maketitle`. Next, it is described the commands used to enter the information required to create it.

`\author` The `\author` command was redefined. Here, it takes two arguments: the author's first names and surname, e.g., `\author{First Names}{Surname}`. The words should be typed with only first letters in uppercase.

`\title` The macros `\title` and `\foreigntitle` are used to enter the titles of your monograph in Brazilian Portuguese and English respectively (the two built-in languages of CoppeTeX 4.x). For documents whose main language is neither

brazilian nor english – e.g. spanish, french, italian, or a user-supplied language –, also call `\titlein{<lang>}{<text>}` to register the title in that language; the cover and folha de rosto print whichever title matches the main language declared in the class option. See the section “Multilingual support” below for the complete picture. The `babel` package is loaded automatically by `coppe.cls`, so you do not need to load it yourself.

`\advisor` Every COPPE student is coordinated by at least one advisor. M.Sc. and D.Sc. students can have at most 2 and 3 advisors, respectively. Their names must be provided by issuing the command `\advisor` as below:

```
\advisor{Name}{Surname}{Degree}{Institution}
\advisor{Second Advisor's Name}{Surname}{Degree}{Institution}
\advisor{Third Advisor's Name}{Surname}{Degree}{Institution}
```

`\coadvisor` takes exactly the same four arguments.

The advisors are not necessarily members of the thesis examination board. Thus, it is required to enter the names of all examiners using the `\examiner` macro. The examiners' names are entered differently – given name and surname go in one argument:

```
\examiner{First Examiner's Name Surname}{Degree}{Institution}
\examiner{Second Examiner's Name Surname}{Degree}{Institution}
...
\examiner{N-th Examiner's Name Surname}{Degree}{Institution}
```

The argument order changed in v4.1. The treatment (“Prof.”, “Prof.^a”) used to be the first mandatory argument and the institution an optional one; now the treatment is the optional argument and is EMPTY by default, and the institution is the last mandatory argument. 3.1.2.1.3(e) asks for name, titulacao and institution, and asks for no treatment at all, so that is what the folha de aprovacao now prints. The institution may be left empty – `\advisor{Ana}{Lima}{D.Sc.}{}{}` compiles and simply prints nothing in its place.

To print a treatment anyway, pass it in the optional argument:

```
\advisor[Prof.]{Ana}{Lima}{D.Sc.}{UFRJ}
\examiner[Prof.\textsuperscript{a}]{Bia Sousa}{Ph.D.}{UFF}
```

A document written for v4.0 or earlier has to be rewritten. See `MIGRATION_v3_to_v4.md` for the mechanical rule.

Who appears on the folha de aprovacao. Since v4.1, only the EXAMINERS. The advisors stay on the capa and the folha de rosto, where they belong, but the approval sheet records who examined the work, and the advisor conducted it. Where the Programa does seat the advisor on the board, the `orientadorexamina` class option puts the advisors and coadvisors back, ahead of the examiners.

`\dataaprovacao` The date the board approved the work, printed on the folha de aprovacao after “Aprovada em”. Give it in words, the way the sheet should read: `\dataaprovacao{15 de setembro de 2026}`. **With no date given the sheet says “a ser determinada”**, in the main language of the work – there is no blank rule to fill in by hand, because the deposit is digital and nobody writes on a PDF.

Remember that all names must be given before calling `\maketitle`.

`\department` The Alberto Luiz Coimbra institute is divided into 12 academic units: Biomedical Engineering (PEB), Civil Engineering (PEC), Electrical Engineering (PEE), Mechanical Engineering (PEM), Metallurgical and Materials Science Engineering (PEMM), Nuclear Engineering (PEN), Ocean Engineering (PENo), Energy Planning (PPE), Production Engineering (PEP), Chemical Engineering (PEQ), Systems Engineering and Computer Science (PESC), and Transportation Engineering (PET). You must specify your department using one of the above abbreviations, e.g., `\department{PEC}`.

`\date` This macro is used to set the month and year of defense. This information is required to create the front face, cataloging details page and abstracts. For example, October 2007 should be entered as `\date{10}{2007}`.

`\keyword` The keywords should describe the concentration areas of your work. You must provide them as follows:

```
\keyword{First Keyword}
\keyword{Second Keyword}
...
\keyword{N-th Keyword}
```

Usually, six words are enough.

Folha adicional: Coleta CAPES

Since August 2026 every thesis and dissertation must carry a *folha adicional* immediately after the folha de rosto, holding the fields the CAPES/MEC collection requires plus the ficha catalográfica (SiBI manual, 9th rev. ed. 2026, 3.1.2.1.2 and Annex H). The sheet is filled in by the graduate programme, and is neither counted nor numbered.

The macros below feed that sheet. They are institutional data, so the sheet is always typeset in Portuguese regardless of the main language of the work, as the COPPE norm requires of institutional identity.

Since v4.1 the sheet is drawn as Annex H shows it: a CLOSED FRAME the width of the text block, one cell per field, cells separated by a horizontal rule, and **no rules to write on** – the space to fill in is the blank space of the cell. A field left empty simply leaves its cell blank. The ficha catalográfica sits centred below the frame, on the same sheet.

`\areaconcentracao` The area of concentration, e.g. `\areaconcentracao{Engenharia de Sistemas e Computacao}`. It appears on the folha de rosto, on the folha de aprovacao and on the folha adicional.

`\linhapesquisa` The research line, required by the CAPES model of the folha de rosto, where it is printed at the left margin just above the advisors. Its label follows the MAIN LANGUAGE of the work (“Linha de pesquisa”, “Research line”, “Línea de investigación”, “Axe de recherche”, “Linea di ricerca”). Leave it out, or give it an empty argument, and the folha de rosto comes out without the line and without any error.

`\tipoproducao` The kind of intellectual output, one of `bibliografica`, `artistica`, `tecnologica` or `tecnica`. Anything else raises a warning.

`\projetovinculado` Whether the work belongs to a research project (`sim` or `nao`) and, if so, its name.

`\agenciafomento` A funding agency, given as full name and acronym; repeat once per agency:

```

\agenciafomento{Conselho Nacional de Desenvolvimento
                Cientifico e Tecnologico}{CNPq}
\agenciafomento{Fundacao Carlos Chagas Filho de Amparo a
                Pesquisa do Estado do Rio de Janeiro}{FAPERJ}

```

Cataloging details

This page contains cataloging information useful for librarians. Fortunately, it is automatically generated from the data you entered at the time you call `\maketitle`. It is not needed in qualifying exams, though.

Dedication (optional)

`\dedication` This macro was added for convenience. The input text is placed at the right bottom of a blank page. It is emphasized and in normal size.

Abstracts

`abstract (env.)` As stated by the COPPE/UFRJ Academic Registry, abstracts must be in one page each, with at most 250 words. We recommended that they should be only one paragraph long. They must be defined inside the environments `abstract` and `foreignabstract`.

Lists of symbols and abbreviations (optional)

`\abbrev` The lists of symbols and abbreviations are optional, although highly recommended.

`\syml` It is a good practice to define a symbol/abbreviation in its first occurrence in the text. To define a symbol use `\syml[alphabetic symbol]{Symbol}{Symbol Definition}`, and for abbreviations `\abbrev[alphabetic symbol]{Abbreviation}{Abbreviation Definition}`. These commands are called *dummy*, since they don't output anything at the place they are executed, just an entry in the correspondent list.

`\makeloabbreviations` These lists are lexicographically sorted by using the MakeIndex program, which is part of any L^AT_EX implementation. For `\syml`, if the optional parameter is provided, it will be used as sort key. This was later, in 2024, implemented also for `\printloabbreviations` and `\printlosymbols`. `\abbrev`, otherwise `Symbol`, or `Abreivation` will be used as sort key, what can result in an undesirable order if it contains L^AT_EX commands, mathematical symbols, or mix of uppercase and lowercase. MakeIndex needs two commands to create a final sorted list: one which generates a list of entries and the other that indicates the position where the list will be printed out. To generate the lists of symbols and abbreviations, the `coppe` class provides the commands `\makeloabbreviations` and `\makelosymbols`, respectively. They must be called in the document preamble. The commands `\printlosymbols` and `\printloabbreviations` have to be invoked at the point where you want these lists appear, e.g., following the list of tables as showed in the example. Once you call `latex`, it will be created two files with extensions `abx` and `syx`, which contain MakeIndex input data. They must be processed with `makeindex` in order to get the lists correctly produced, redirecting the output to files with extension `lab` and `los` respectively:

```

makeindex -s coppe.ist -o example.lab example.abx
makeindex -s coppe.ist -o example.los example.syx

```

Note the `-s` option for specifying the style `coppe.ist`. Now, rerun `latex` twice to get the references solved and you are done.

Illustrations: caption on top, source below

`\source` 2.10 of the Manual puts the CAPTION ABOVE every illustration – figure, table, quadro, code listing, algorithm – with the designating word, the number in arabic figures, an em dash and the title; and it makes the SOURCE mandatory, BELOW the illustration, in the smaller size. The class does that for you: the caption goes on top wherever you write `\caption`, and `\source{...}` (or `\fonte{...}`, the same command) sets the source line. The source advances no counter and appears in no list.

```
\begin{figure}[ht]
\centering
\caption{0 que a figura mostra}    % em cima
\label{fig:algo}
\includegraphics[width=0.5\textwidth]{arquivo}
\fonte{Elaboracao propria.}        % embaixo
\end{figure}
```

If another package has already defined `\source` or `\fonte`, use `\cpsource` and `\cpfonte`, which are always available.

A **quadro** is an illustration whose data is bordered on all sides, unlike a *tabela*, which is ruled only at the top and the bottom. The `quadro` environment behaves like `figure` and `table` and has its own list, `\listofquadros`.

For an illustration the Manual does not name – a map, a photograph, a score – declare your own float in the preamble:

```
\newcoppefloat{mapa}{Mapa}{Lista de Mapas}
```

You get the `mapa` environment, numbered within the chapter like every other illustration, the caption on top, `\source` below, `\autoref` support, and a `\listofmapas` command to put with the other pre-textual lists. The example does exactly this.

Long quotations, footnotes, captions and the folio all come out in the smaller uniform size 2.2(b) prescribes; you do not set that yourself.

References

COPPE_{TEX} produces citations and the reference list with `biblatex` and the `biber` backend, through its own bibliography style implementing the post-2020 ABNT NBR 6023 rules. The class loads this style automatically: there is no `\bibliographystyle` to set and no Bib_{TEX} `.bst` file to choose (the former `coppe-plain` and `coppe-unsrt` Bib_{TEX} styles have been retired).

By default, citations follow the author–date style, e.g., “Chartier (2002)”. Passing the `numbers` class option switches to the numeric style, in which citations are bracketed numbers and the reference list is numbered in order of citation (unsorted).

Declare your bibliography database in the preamble with `\addbibresource` and print the list where it should appear with `\printbibliography`:

```
\addbibresource{example.bib}    % in the preamble
...
\printbibliography              % where the list goes
```

Entry types and fields may be written in English (`@book`, `title`, `author`) or with their unaccented Portuguese synonyms (`@livro`, `titulo`, `autor`).

Run in sequence `LATEX`, `biber`, and twice again `LATEX` to resolve the citations and references. The style is `natbib` compatible (via the `biblatex natbib=true` option), so you may freely issue `\citet` and `\citep`, as well as the other `natbib` commands.

Reference entry types

Every ABNT document category maps to a `biblatex` entry type, each with an unaccented Portuguese synonym (a worked example of every type is in the sample manuscript’s reference list):

Academic `@book` (`@livro`); `@article` (`@artigo`); `@artigodejornal` (newspaper); `@incollection` (`@partedelivro`); `@inbook` (`@emlivro`); `@inreference` (`@verbete`); `@proceedings` (`@anais`); `@inproceedings` (`@trabalhodeevento`); `@thesis` (`@tese/@dissertacao`, with the presets `@dissertacaomestrado`, `@tesedoutorado`, `@tesephD`, `@tcc`); `@report` (`@relatorio`); `@periodical` (`@periodico`); `@manual`; `@misc` (`@diverso`); `@online`.

Special (NBR 6023 §4.2.5–4.2.14) `@patent` (`@patente`); `@legislation` (`@legislacao/@atonormat`); `@jurisdiction` (`@jurisprudencia`, with a `relator` field); `@legal` (`@documentojuridico`); `@standard` (`@norma`); `@music` (`@partitura`); `@map` (`@mapa`); `@image` (`@iconografico`); `@artwork` (`@obradearte`); `@video` (`@audiovisual`); `@movie` (`@filme`); `@letter` (`@correspondencia`); `@dataset` (`@conjuntodedados`).

Games and TV `@game` (`@jogo`); `@boardgame` (`@jogodetabuleiro`); `@videogame` (`@jogoeletronico`); `@tvshow` (`@programadetv`).

When an entry has no `author` or `editor`, it is entered by title, with the first significant word (and a leading article) in capitals. A periodical is the exception: 4.2.3 sets its title in capitals throughout.

Fields the ABNT categories need

Beyond the usual `biblatex` fields, each with an unaccented Portuguese synonym:

Events `eventtitle` (`evento`) — the name of the event, set in capitals ahead of the title of the proceedings, as 4.2.4 requires; `number` (`numero`) the edition; `eventdate` (`dataevento`) the year it was held; `venue` (`localevento`) the city. The title (`@anais`) or `booktitle` (`@trabalhodeevento`) then carries only “Anais [...]”, which is the element in bold.

Patents `filingdate` (`deposito`) and `grantdate` (`concessao`), printed as “Depósito: Concessão:” (4.2.5).

Theses `course` (`curso`), `institution` (`instituicao`), `pagetotal` — rendered as “1997. 203 f. Dissertação (Mestrado em ...) — Instituição, Local, 1997.” (4.2.1.1).

Legal documents `relator`, plus `volume`, `number` and `pages` of the vehicle in which the act was published (4.2.6).

Anything electronic `url` and `urldate` (`dataacesso`), closing the entry as “Disponível em: Acesso em:”

Every one of these is exercised, with the manual’s own example data, by `adversativa/referencias-manual.bib` — one entry per category of 4.2 — and checked against the manual’s printed form by `tools/conferir-referencias.py`. Neither is distributed: they prove the styles, they are not part of them.

Appendix and Annex (Optional)

`\appendix` Appendices and annexes are optional chapters that are part of the back matter.

`\annex` The `\appendix` command is a standard L^AT_EX command used before all appendices. COPPETEX introduces the `\annex` command, which should be used only in the back matter (i.e., after the `\backmatter` command) and only after all existing `\appendix` chapters. This restriction is due to implementation constraints.

Therefore, the order for backmatter is:

```
\backmatter
\printbibliography
\appendix
\chapter{An Appendix}
\chapter{Another Appendix}
\annex
\chapter{First Annex}
\chapter{Second Annex}
```

Annex was introduced in v3.5

Final colophon page (`\coppetexfinalpage`)

`\coppetexfinalpage` Optional closing page modelled on the printer’s *colophon*: it records, in Portuguese, the tooling and edition used to typeset the thesis. Call it once, after every other piece of content, immediately before `\end{document}`. The macro

- forces a verso (even) page, so in a two-sided print the colophon sits opposite the rear cover;
- types the body at the foot of the page (`\vfill`);
- renders the text in Portuguese unconditionally (Brazilian babel), even when the thesis main language is English, Spanish, French or Italian — the colophon is institutional metadata, not body content.

Three slot macros are user-overridable. Redefine them with `\renewcommand` before `\coppetexfinalpage` if the document deviates from the defaults:

`\coppefinalengine` the T_EX engine identifier. Default: auto-detected via `iftex` (`\hologo{pdfLaTeX}`, `\hologo{LuaLaTeX}` or `\hologo{XeLaTeX}`).

`\coppefinalfont` the text body font. Default: *Latin Modern* (`lmodern`), which is what the class loads. Redefine if you loaded `mathptmx`, an OpenType math font, etc.

`\coppefinalmanual` the reference manual title (typeset in italics). Default: the current COPPE/UFRJ norm. Redefine if your work follows an earlier edition.

The macro also prints the COPPET_EX class version (captured into `\coppe@version` at class-load time, so it survives any later package that may redefine the global `\fileversion`) and the GitHub repository URL, <https://github.com/COPPE-UFRJ/CoppeTeX>.

5 Class options

There are some options users can specify in order to customize the appearance of the output produced by the `coppe` class. These options can be passed to `coppe` as follows: `\documentclass[option1, option2]{coppe}`. In which follows, we give a brief description of all supported options.

dsc, msc, dsceexam, msceexam The `coppe` class is able to produce thesis, dissertations, and qualifying exams, which are enabled by the `dsc`, `msc`, `msceexam`, and `dsceexam` options, respectively.

doublespacing The default line spacing is one-and-a-half. For enabling double spacing between lines, use the `doublespacing` option.

numbers The default citation style is the author–date scheme. For numbered citations, specify the `numbers` option to the `coppe` class; the reference list is then numbered in order of citation (unsorted). The matching `biblatex` style is selected automatically in either case.

brazilian, english, spanish, french, italian Pick the document’s *main* language. The class loads Babel with that language as the main one (the last in the Babel option list); inside the document `\selectlanguage{...}` and `\begin{otherlanguage}{...}` keep working as usual. The default, when no language option is given, is **brazilian**.

- **brazilian** and **english** are *built-in*: their strings ship inside `coppe.cls` and need no extra file.
- **spanish, french, italian** are shipped as *language packs*: two files per language (`coppe-lang-<lang>.def` and `<lang>-coppe.lbx`) that must sit next to `coppe.cls` (or in your project directory). They are auto-loaded by the class. Picking one of them as the main language without the matching pack files installed raises a clear class error.
- Any other Babel-known language can be plugged in by writing the same two files and either passing the language name as a class option or calling `\usecoppelanguage{<lang>}` in the preamble.

See the section “Multilingual support” below for the three-slot architecture, the `\titlein / \brazilianabstract` API, and how to create a new language pack.

pdfa Produce the deposited file as **PDF/A-2b**, which 2.2(d) of the manual requires of the digital version. The class loads `pdfx` at level **a-2b** and writes the XMP metadata from the document’s own `\title`, `\author`, `\keyword` and `\department`. Level **a-2b** rather than **a-1b**: **a-1b** forbids transparency, which `tcolorbox` and several graphics packages produce. The metadata reaches the PDF on the **SECOND** run, because `pdfx` reads it from a file the class writes at the end of the first one – which is no burden, since three runs are needed for cross-references anyway.

orientadorexamina Put the advisors and coadvisors back on the folha de aprovacao, ahead of the examiners, as president of the board (3.1.2.1.3(e)). **Off by default since v4.1**: the folha de aprovacao records who **EXAMINED** the work, and the advisor conducted it rather than examined it, so only the examiners are listed unless the Programa says otherwise. Before v4.1 there was no choice – the advisors always appeared, and the only way out was not to declare them, which also removed them from the capa and the folha de rosto, where they belong.

assinaturas **Gone in v4.1, and now a no-op that only warns.** It used to draw a signature rule above each board member. The deposit has been digital only since Resolucao CEPG n. 246/2023, so there is nothing to sign by hand; the folha de aprovacao now always uses the compact list. The option is still accepted so a document written before v4.1 still compiles.

coorientador List the coadvisors on the three abstract pages, under the advisors, in the same shape the folha de rosto uses. **Off by default**: the abstract page listing only the advisors is a COPPE tradition, and 3.1.2.1.4 does not require either form.

rascunhoficha While writing, fill the folha adicional with a ficha catalografica typeset by the class itself, so the page is not an empty frame. **Never valid for deposit**: the real ficha comes from the SiBI generator or the Programa’s library, and goes in with `\fichacatalografica`.

twoside, oneside The 2026 edition dropped the verso margins and the deposit is digital, so the class sets one-sided pages with a 3 cm left margin on every folha. **twoside** restores the mirrored layout for anyone who is going to **PRINT** and bind the work; **oneside** is the default and is there to be explicit.

semmorewrites, morewrites $\TeX$ has sixteen output streams and a work that asks for every list runs out of them; the failure, “No room for a new `\write`”, lands on whichever list came last and explains nothing. Under pdf $\TeX$ the class therefore loads the `morewrites` package by itself when it is installed, which removes the ceiling. **semmorewrites** turns that off, for the rare case where the package disagrees with something else in the document – it buffers what is written until shipout, so a package that writes a file and reads it back in the same run can misbehave. **morewrites** means “I insist”: a missing package becomes an error instead of a warning. Under Lua $\LaTeX$, which has 128 streams, none of this is loaded and none of it is needed.

comserifa, semserifa Since v4.1 the class sets the whole document in the sans-serif family by default; **comserifa** goes back to the serif one. Neither the manual nor the COPPE norm prescribes a typeface: 2.2(b) fixes the colour, the body size 12 and the smaller uniform size of the four items it lists, and says nothing about the family. The choice is the author's, and the SiBI manual that carries the rule is itself set in Arial – which is why sans is now the default. The class loads **lmodern**, which brings both families as vector fonts, so either way the document is PDF/A material. **semserifa** is still accepted and now does nothing, so a document written before v4.1 still compiles. For a DIFFERENT sans face – Helvetica, say – load its package in the preamble and redefine **\familydefault** there; the default is for the common case of an author who does not want to choose a font at all. Redefine **\coppefinalfont** if you do load another face, so the colophon does not announce Latin Modern.

listasnosumario Put the pre-textual lists – figures, tables, quadros, programas, abreviaturas, simbolos – back into the sumario. **Contrary to 3.1.2.1.6**, which tells the author to follow the manual's own sumario, and that one opens at “1 INTRODUCAO”: the lists are pre-textual elements and come BEFORE the sumario, so listing them inside it is listing what the reader has already passed. The option exists because every CoppeTeX document written before this release had them there, and a work in its final weeks should not have its sumario change under it.

numeraisromanos Number the pre-textual folhas in lowercase roman numerals. **Contrary to 2.7**, which has the count running from the folha de rosto with nothing printed until the textual part. Kept only so that documents written under the older norm still compile; do not use it for a new work.

5.1 Changing document identification

\freeconfig The user could *optionally* use the command **freeconfig** to modify the parameters that print the document identification. The command **freeconfig** needs all those parameters, which are degree initials, degree name, title, foreign title, local doctype, and foreign doctype as in the following example:

```
\freeconfig{Dr.}{Philosophiae Doctor}{PhD}{Doutor}{Dissertation}{Tese}
```

5.2 Multilingual support

CoppeTeX 4.x reserves three fixed language slots:

main the language of the body, captions, TOC labels and the **abstract** environment. Picked by the class option (**brazilian** by default; one of **english**, **spanish**, **french**, **italian** also accepted out of the box; any other Babel language possible with a user-supplied pack).

foreign the language of the **foreignabstract** environment. **english** by default for every main language; switches automatically to **brazilian** when **main = english**.

third optional. Loaded on demand with `\usecoppelanguage{<lang>}` in the preamble, useful when the body quotes or cites in a third language (e.g. a Spanish-main thesis citing a French source). Does not have a class option of its own.

<code>\titlein</code>	Two pieces of user-facing API come with the multilingual model:
<code>brazilianabstract (env.)</code>	
<code>\usecoppelanguage</code>	• <code>\titlein{<lang>}{<text>}</code> records the title in any language. <code>\title{...}</code> still writes the Brazilian-Portuguese title and <code>\foreigntitle{...}</code> still writes the English one (back-compat); for a main language outside the <code>brazilian/english</code> pair, also call <code>\titlein{<lang>}{...}</code>. The cover and folha de rosto print whichever title matches <code>\coppe@mainlang</code>. • The <code>brazilianabstract</code> environment is the optional third abstract used when the main language is neither <code>brazilian</code> nor <code>english</code> – typically a Spanish/French/Italian thesis that also wants a Portuguese resumo for UFRJ examiners. It wraps its body in <code>\begin{otherlanguage}{brazilian}</code> and uses the Portuguese template strings. A document with <code>main = brazilian</code> can simply keep using <code>abstract</code>; <code>brazilianabstract</code> is redundant then. • <code>\usecoppelanguage{<lang>}</code> loads an extra language pack in the preamble without making it the main or foreign language.

Institutional names stay Portuguese

The cover, folha de rosto and ficha catalográfica are a Brazilian institutional template (NBR 14724 + COPPE manual). The strings `universityname`, `cityname`, `statename`, `countryname`, the departments declared with `\department` and the surrounding “apresentada/submetida ao...” wording are deliberately kept in Portuguese for every main language. Only the title (printed via `\coppe@selecttitle`) follows the main language.

Writing a new language pack

A language pack is just two files dropped next to `coppe.cls` (or anywhere on the TEXINPUTS path):

`coppe-lang-<lang>.def` class-level strings. Populate the dispatcher table with `\copperdefstring{<lang>}{<key>}{<value>}` for every key already populated for `brazilian/english` in `coppe.cls` (appendix, annex, frame, listframe, source, program, listprogram, references, in:, art, platform, directedby, producedby, listabbreviation, listsymbol, glossary, advisor, advisors, dept, approvedmale, approvedfemale, universityname, cityname, statename, countryname, monthname1...monthname12, algorithm2eopt, babelname). Finish with

```
\AtBeginDocument{\DeclareLanguageMapping{<lang>}{<lang>-coppe}}
```

so the `biblatex` mapping is registered after the bibliography engine is loaded by `coppe.cls` (which happens *after* the language pack is read).

<lang>-coppe.lbx biblatex localization strings. \InheritBibliographyExtras{<lang>} then \DeclareBibliographyStrings for the keys availablefrom, urlseen, in, editor, editors, depositor, coppescale, msdiss, dscthis, phdthesis, tcc, coppemscdeg, coppedscdeg, coppephddeg, coppetccdeg, coppefiling, coppegrant. The shipped spanish-coppe.lbx, french-coppe.lbx and italian-coppe.lbx are the reference templates to copy from.

Tip: the algorithm2eopt key holds whatever algorithm2e calls the language. Mind the spelling: algorithm2e ships portuguese, english, spanish, french and italiano – not italian.

Example: a Spanish-main thesis

```
\documentclass[spanish,dsc]{coppe}
\addbibresource{example.bib}
\begin{document}
  \title{T'\i tulo em portugu\~es}      % Brazilian abstract slot
  \foreigntitle{English title}          % English abstract slot
  \titlein{spanish}{T'\i tulo en espa\~nol} % cover/folha
  \author{Juan}{P\erez}
  \advisor{Mar'\i a}{Garc'\i a}{D.Sc.}{UFRJ}
  \department{PESC}\date{01}{2026}
  \keyword{multilenguaje}
  \maketitle
  \frontmatter
  \begin{abstract}      Resumen principal en espa\~nol. \end{abstract}
  \begin{foreignabstract} English foreign abstract. \end{foreignabstract}
  \begin{brazilianabstract} Resumo em portugu\~es para a banca. \end{brazilianabstract}
  \tableofcontents
  \mainmatter
  \chapter{Introducci\on} Cuerpo en espa\~nol\dots
\end{document}
```

6 Files in your project folder

In a normal install where CoppeTeX is part of your TEXINPUTS path (e.g. inside the local texmf tree of MiKTeX or TeX Live), you do not need to think about this list at all — the files travel with the class. This section is for two situations where you do:

- shipping the class together with your manuscript (a self-contained submission package); or
- installing CoppeTeX by hand, dropping the files next to coppe.cls in the project directory.

Always required

These files must be findable at every compilation, regardless of the main language or any class option:

coppe.cls the document class itself.

`coppe.dbx` custom `biblatex` datamodel (declares the ABNT-specific entry types `@game`, `@videogame`, `@boardgame`, `@tvshow` and `@map`, plus a few custom fields).

`coppe.bbx` and `coppe.cbx` `biblatex` bibliography and citation styles for the author–date scheme (the default).

`brazilian-coppe.lbx` and `english-coppe.lbx` `biblatex` localisation tables for Portuguese and English. Both are required *even when neither language is the main one*, because `coppe.bbx` registers

```
\DeclareLanguageMapping{brazilian}{brazilian-coppe}
\DeclareLanguageMapping{english}{english-coppe}
```

unconditionally, so `biblatex` tries to read both LBX files at every compilation.

Conditional on class options

`coppe-numeric.bbx` and `coppe-numeric.cbx` only when the `numbers` class option is given (`\documentclass[numbers]{coppe}`). Without it, `biblatex` never tries to read them.

`coppe.ist` only when the glossary, list-of-symbols, or list-of-abbreviations features are used. It is the `makeindex` style file invoked by the `makeindex -s coppe.ist ...` step described in subsection *Lists of symbols and abbreviations* above.

Conditional on the main language

The main language is chosen by the matching class option (`brazilian`, `english`, `spanish`, `french`, `italian`). The two built-in languages ship their strings inside `coppe.cls` and so need no extra file. The three language packs add two files each — one class-level (`coppe-lang-<lang>.def`) and one `biblatex`-level (`<lang>-coppe.lbx`):

Main language	Extra files needed
<code>brazilian</code> (default)	— (built into <code>coppe.cls</code>)
<code>english</code>	— (built into <code>coppe.cls</code>)
<code>spanish</code>	<code>coppe-lang-spanish.def</code> , <code>spanish-coppe.lbx</code>
<code>french</code>	<code>coppe-lang-french.def</code> , <code>french-coppe.lbx</code>
<code>italian</code>	<code>coppe-lang-italian.def</code> , <code>italian-coppe.lbx</code>

Loading a third language at preamble time via `\usecoppelanguage{<lang>}` brings in the same pair of files for that language. A user-supplied language pack follows the same scheme: drop `coppe-lang-<lang>.def` and `<lang>-coppe.lbx` next to `coppe.cls`.

Cover assets

The institutional title page produced by `\maketitle` needs two PDF logo files at compile time:

`ufrj-logo.pdf` and `coppe-logo.pdf` the UFRJ shield and the COPPE/UFRJ wordmark. Both ship with the CoppeTeX distribution; they must be findable from the project directory (or on `TEXINPUTS`).

7 Required L^AT_EX packages

The class is built on the standard `book` class (`12pt`, `a4paper`, `oneside`) and brings in the stock CTAN packages listed below with `\RequirePackage`. On a full T_EX Live or MiK_TE_X install they are all present already, so this list matters only for a minimal install — or when you simply want to know which commands are already in scope. You do *not* need to `\usepackage` any of these yourself; every command and environment they provide is available in your manuscript. Options in brackets are the ones the class passes.

Engine and encoding

`iftex` detects the running engine, so the class can branch its setup between `pdfTEX` and the Unicode engines (`LuaTEX`/`XeTEX`).

`inputenc (utf8)` UTF-8 input; loaded *only under pdfT_EX* (the Unicode engines read UTF-8 natively).

`fontenc (T1)` T1 font encoding, for correctly composed accented glyphs and hyphenation.

Mathematics and symbols

`amsmath` the AMS mathematics environments.

`amssymb` the AMS symbol fonts; loaded *only under pdfT_EX*. Under a Unicode engine load `unicode-math` (or `amssymb`) yourself — the class warns at end of document if neither is present.

Page layout and spacing

`geometry` A4 one-sided margins (`3/3/2/2` cm) and header height.

`fancyhdr` running headers and footers.

`titlesec` chapter and section title formatting.

`setspace` one-and-a-half line spacing (double with the `doublespacing` option).

Lists, tables and rules

`enumitem` customisable list spacing and labels.

`tabularx` tables with automatic-width columns.

`booktabs` professional horizontal rules (`\toprule`, `\midrule`, `\bottomrule`).

`longtable` tables that break across pages.

Floats, graphics and captions

`graphicx` image inclusion.

`float` float placement control (e.g. the `[H]` specifier).

`caption` caption typography (the ABNT caption-on-top layout).

`subcaption` subfigures and subtables.

Code, verbatim and algorithms

`listings` source-code listings.

`fancyvrb` extended verbatim.

`algorithm2e` (`linesnumbered`, `lined`, `algochapter`, `ruled`) typeset algorithms;
the natural-language option tracks the main language.

Bibliography, quotations and language

`babel` multilingual typesetting; the language options are computed from the main
and foreign language slots.

`csquotes` context-sensitive quotation marks (the recommended companion of
`biblatex`).

`biblatex` (`backend=biber`) the bibliography engine and the ABNT `coppe` styles
(`coppe` or, with the `numbers` option, `coppe-numeric`). The backend is
`biber`, not `bibTeX`.

Hyperlinks

`hyperref` clickable cross-references, PDF bookmarks and document metadata.

Low-level utilities

`etoolbox` e-TeX programming toolkit.

`ltxcmds` low-level command helpers (`\ltx@ifpackageloaded`, ...).

`ifthen` the `\ifthenelse` conditional.

`hyphenat` hyphenation control (`\nohyphens`, used on the cover).

`lastpage` `\pageref{LastPage}` for the cataloging page count.

`hologo` TeX-family logos (`\hologo{LaTeX}`, ...).

`xcolor` colour support (used by `listings` and `hyperref`).

8 Quick, useful tips

Pictures. The default picture format of L^AT_EX is the Encapsulated PostScript (EPS). If you use pdfL^AT_EX, the default format becomes the PDF, but you can equally load PNG files. For such, you must enter the name of your image file without extension, e.g., `\includegraphics{filename}`, and pdf_latex will firstly look for a file called `filename.pdf` and after for file `filename.png`. For producing high quality pictures with embedded fonts we recommend the Ipe drawing software available [here](#).

Fonts. The default font in L^AT_EX is the Computer Modern. If you would like to try its enhanced version, consider using the `lmodern` package. To use Times, it is recommended to load the package `mathptmx`, rather than the deprecated `times`. There is also an enhanced Times version available with the `tgtermes` package. You can still use the Arial font face with the package `uarial`.

Hyperref. When working with PDF's, there is the possibility to add extra information to the file as the author's name, document title, subject, keywords, etc. This is easily done with the `hyperref` package, which is loaded by the `coppe` class itself (since v4.0): the user must *not* load it again in the manuscript preamble. Author, title, subject and keyword metadata are filled in automatically by `\maketitle`.

Math (amssymb vs. unicode-math). The class always loads `amsmath`. The companion AMS symbol package `amssymb`, however, is loaded *only* under pdfL^AT_EX: it is a legacy 8-bit package built around the `msam/msbm` Type 1 fonts and clashes with `unicode-math`, the modern OpenType-math package used on Unicode engines (LuaL^AT_EX, XeL^AT_EX). On those engines the class skips `amssymb` and schedules an *end-of-document* check: if by then the author has loaded neither `unicode-math` nor `amssymb` (which they could have loaded by hand, though it is discouraged), a `ClassWarning` reminds them to add

```
\usepackage{unicode-math}
\setmathfont{Latin Modern Math} % or STIX Two Math, ...
```

to the manuscript preamble. The class deliberately does *not* auto-load `unicode-math`, because picking an OpenType math font is a typographic choice that belongs to the author. **Never load amssymb together with unicode-math:** it would re-establish the 8-bit AMS fonts on top of the OpenType math font and split symbols across two incompatible encodings.

Printing. To get your work correctly printed, you must ensure that any page scaling option (e.g., fit or shrink to printable area) isn't enabled. This kind of option often comes in print dialogs of document visualization softwares.

Quotation A quotation longer than three lines is INDENTED BY 4 cm from the left margin – not set with a 4cm left margin, which is a different and much narrower thing. 2.2(b) of the manual says “recuo de 4 cm da margem esquerda”, so on an A4 page with a 3 cm margin the quoted text starts 7 cm from the edge of the

paper. It carries no quotation marks and is set in the smaller font, single spaced. The `coppe` class provides the `longquote` environment, which does all of that; the indentation is `\recuolongquote`, should a Programa ever need to change it.

9 A simple example

```

1 \example
2 \documentclass[dsc]{coppe}
3
4 \usepackage{rotating}          % rodando coisas, como tabelas
5 \usepackage[most]{tcolorbox} % caixas de texto
6 \usepackage{tikz}             % gráficos vetoriais (o tcolorbox já o traz)
7 % Sob LuaLaTeX/XeLaTeX, descomente as duas linhas a seguir para obter os
8 % símbolos AMS (\mathbb, \hbar, \varnothing, ...) via unicode-math. Sob
9 % pdfLaTeX, a classe já carrega amssymb automaticamente; deixe comentado.
10 %\usepackage{unicode-math}
11 %\setmathfont{Latin Modern Math} % ou STIX Two Math, TeX Gyre Termes Math...
12 \DeclareMathOperator{\RMSE}{RMSE} % operadores próprios (cap. de matemática)
13 \DeclareMathOperator{\diag}{diag}
14 % A classe coppe já carrega: amsmath, booktabs, longtable, subcaption,
15 % hyperref, biblatex, listings, fancyvrb, algorithm2e -- não recarregue esses
16 % pacotes aqui (em particular, hyperref deve ficar a cargo da classe para que
17 % a ordem com setspace e biblatex permaneça correta).
18
19 \addbibresource{example.bib}% base de referências (biblatex/biber)
20
21 \makelosymbols
22 \makeloabbreviations
23 % Flutuante criado pelo autor: mesma regra dos demais (legenda acima, fonte
24 % abaixo) e uma lista própria, \listofmaps.
25 \newcoppefloat{mapa}{Mapa}{Lista de Mapas}
26 % Fonte: a norma fixa o corpo 12, e não a família. A classe compõe sem serifa
27 % por padrão; para compor com serifa, passe a opção de classe 'comserifa' lá
28 % em cima -- NÃO redefine \familydefault aqui, porque isso venceria a opção.
29 % Para uma sem serifa diferente da Latin Modern, carregue o pacote da fonte e
30 % troque a família:
31 %\usepackage{helvet}\renewcommand{\familydefault}{\sfdefault}
32
33
34 \begin{document}
35   \title{Título da Tese}
36   \foreigntitle{Thesis Title}
37   \author{Nome do Autor}{Sobrenome}
38   % Orientador: nome, sobrenome, titulacao e instituicao. O tratamento
39   % ("Prof.") e o argumento OPCIONAL entre colchetes e vem vazio por padrao --
40   % a norma nao pede tratamento nenhum. A instituicao e obrigatoria, mas pode
41   % ficar vazia: \advisor{Ana}{Lima}{D.Sc.}{} compila sem erro.
42   \advisor{Nome do Primeiro Orientador}{Sobrenome}{D.Sc.}{UFRJ}
43   % O segundo orientador vai com tratamento, so para mostrar o argumento
44   % opcional funcionando; a norma nao pede tratamento nenhum.
45   \advisor[Prof.]{Nome do Segundo Orientador}{Sobrenome}{Ph.D.}{UFRJ}
46   % Coorientador: mesmos quatro argumentos de \advisor. Aparece na folha de
47   % rosto e na folha de aprovacao; para ve-lo tambem nas paginas de resumo,

```

```

48 % passe a opcao de classe 'coorientador'.
49 \coadvisor{Nome do Coorientador}{Sobrenome}{D.Sc.}{UFF}
50 % A classe aceita de um a tres \advisor; se houver apenas um, o rotulo na
51 % capa e na folha de rosto sai como "Orientador" (singular), com dois ou
52 % mais sai como "Orientadores" (plural). Para o caso 1/3 orientadores,
53 % veja tests/test_brazilian_one_advisor.tex e tests/test_brazilian_three_advisors.tex.
54
55 \examiner{Nome do Primeiro Examinador Sobrenome}{D.Sc.}{UFRJ}
56 \examiner{Nome do Segundo Examinador Sobrenome}{Ph.D.}{UFRJ}
57 \examiner{Nome do Terceiro Examinador Sobrenome}{D.Sc.}{UFF}
58 \examiner{Nome do Quarto Examinador Sobrenome}{Ph.D.}{UNICAMP}
59 % Instituicao vazia: o argumento e obrigatorio, mas aceita ficar em branco e
60 % entao nada e impresso no lugar dela.
61 \examiner{Nome do Quinto Examinador Sobrenome}{Ph.D.}{}
62 \department{PESC}
63 \date{06}{2026}
64 % Data da defesa, na folha de aprovacao. Sem esta linha a folha sai com
65 % "a ser determinada", que e o que se quer enquanto a data nao existe.
66 % \dataaprovacao{15 de setembro de 2026}
67 %
68 % A folha de aprovacao lista so os EXAMINADORES. Se no seu Programa o
69 % orientador integra a banca, passe a opcao de classe 'orientadorexamina'
70 % la em cima e ele volta para a lista, na frente dos examinadores.
71
72 \keyword{Primeira palavra-chave}
73 \keyword{Segunda palavra-chave}
74 \keyword{Terceira palavra-chave}
75
76 % Dados da folha adicional (Coleta CAPES), obrigatoria desde agosto de 2026.
77 % A folha e preenchida pelo Programa; estes campos apenas a compoem.
78 \areaconcentracao{Engenharia de Sistemas e Computa\c c\~ao}
79 \linhapesquisa{Engenharia de Dados e Conhecimento}
80 \tipoproducao{bibliografica} % bibliografica | artistica | tecnologica | tecnica
81 \projetovinculado{sim} % sim | nao
82 \nomeprojeto{Nome do Projeto de Pesquisa ao qual o trabalho se vincula}
83 \agenciafomento{Conselho Nacional de Desenvolvimento Cientifico e
84 % Tecnologico}{CNPq}
85 \agenciafomento{Fundacao Carlos Chagas Filho de Amparo a Pesquisa do
86 % Estado do Rio de Janeiro}{FAPERJ}
87 % A ficha oficial vem de fichacatalografica.sibi.ufrj.br; tendo o arquivo,
88 % descomente a linha abaixo e a moldura vazia da folha adicional some.
89 % \fichacatalografica{ficha.pdf}
90
91 \maketitle
92
93 \frontmatter
94 \dedication{A alguém cujo valor é digno desta dedicatória.}
95
96 \chapter*{Agradecimentos}
97
98 Gostaria de agradecer a todos os que contribu\iram para este trabalho:
99 orientadores, colegas, familiares e \as ag\~encias de fomento. Lorem ipsum
100 dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt
101 ut labore et dolore magna aliqua.

```

```

102
103 \begin{abstract}
104
105 Apresenta-se, nesta tese, um exemplo de resumo. Lorem ipsum dolor sit amet,
106 consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et
107 dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation
108 ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor
109 in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla
110 pariat. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui
111 officia deserunt mollit anim id est laborum. Sed ut perspiciatis unde omnis
112 iste natus error sit voluptatem accusantium doloremque laudantium, totam rem
113 aperiam, eaque ipsa quae ab illo inventore veritatis et quasi architecto beatae
114 vitae dicta sunt explicabo, nemo enim ipsam voluptatem quia voluptas sit
115 aspernatur aut odit aut fugit.
116
117 \end{abstract}
118
119 \begin{foreignabstract}
120
121 In this work, we present an example abstract. Lorem ipsum dolor sit amet,
122 consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et
123 dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation
124 ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor
125 in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla
126 pariat. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui
127 officia deserunt mollit anim id est laborum. Sed ut perspiciatis unde omnis
128 iste natus error sit voluptatem accusantium doloremque laudantium, totam rem
129 aperiam, eaque ipsa quae ab illo inventore veritatis et quasi architecto beatae
130 vitae dicta sunt explicabo, nemo enim ipsam voluptatem quia voluptas sit
131 aspernatur aut odit aut fugit.
132
133 \end{foreignabstract}
134
135 % Ordem pre-textual (NBR 14724/6027, R55): listas (ilustracoes, tabelas,
136 % abreviaturas e siglas, simbolos) ANTES do sumario; sumario por ultimo.
137 \listoffigures
138 \listoftables
139 \listofquadros
140 \listofmapas
141 \listofprogramas
142 \listofalgorithms
143 \printloabbreviations
144 \printlosymbols
145 \tableofcontents
146
147 \mainmatter
148 \chapter{Introdução}
149
150 Este é um documento exemplo para o uso da classe CoppeTeX, destinado a ajudar os alunos do
151
152 A classe \verb|coppe| foi criada originalmente por Vicente Helano e George Ainsworth, poré
153 Em 2026, Geraldo Xexéo, com forte ajuda das ferramentas Claude 4.7 Opus e Max e 4.8 Opus M
154
155 Provavelmente Geraldo Xexéo, professor do Programa de Engenharia de Sistemas e Computação

```

156

157 A versão mais atual dessa classe é mantida no GitHub, no repositório `\url{https://github.com}`

158

159 Esse documento segue a norma de formatação de teses e dissertações da COPPE. Ele também po

160

161 Há um grupo de WhatsApp para perguntas e discussões do estilo: `\href{https://github.com/COPPE-UFRJ/CoppeTeX/}`

162

163 Bugs devem ser reportados na `\href{https://github.com/COPPE-UFRJ/CoppeTeX/issues}{página d`

164

165 Esse documento é usado como exemplo de coisas que podem ser feitas. Ele está configurado p

166

167 É importante de notar que essa classe não foi construída sobre a classe `\LaTeX \` para a A

168

169 Essa classe implementa citações ABNT modernas e vários tipos de entrada descritos na norma

170

171 Este documento não substitui, mas complementa, o documento que descreve a classe.

172

173

174

175 `\chapter{Configurações Iniciais}`

176

177 A primeira coisa a fazer é escolher o tipo de documento. Isso é feito como uma opção no c

178

179 Como pode ser visto nesse documento, muita coisa pode ser configurada, o que gerará o tra

180

181 Recomendo ler o documento “‘The `\verb*|coppe|` document class’” para entender melhor todas

182

183 `\section{Linguagem principal do texto}`

184

185 Essa classe considera que o texto principal está em português e algumas partes específicas

186 `\begin{itemize}`

187 `\item` A opção `\verb|english|` deve ser usada no comando `\verb|\documentclass|`.

188 `\item` A bibliografia segue automaticamente o idioma do Babel (português ou inglês); não é p

189 `\end{itemize}`

190

191 A variação de linguagem, em inglês ou português apenas, já é suportada pela classe `\CoppeT`

192

193

194 `\section{Por que usar o \LaTeX}`

195

196 Há uma grande discussão entre usuário de Word e `\LaTeX`, principalmente, quanto ao uso dess

197

198 Nós escolhemos o `\LaTeX` por alguns motivos: grande facilidade de seguir um estilo sem se p

199

200 As principais desvantagens são: idiossincrasias que podem gastar tempo, pouco controle sob

201

202 `\section{Como e onde usar o \LaTeX}`

203

204 Existem muitos tutoriais de `\LaTeX`, mas basicamente, em 2024, ele é usado em dois ambiente

205 `\begin{enumerate}`

206 `\item` Na sua máquina, instalando uma versão completa como o `\hologo{MiKTeX}`, típico do Win

207 `\item` Usar um ambiente na rede, como o Overleaf.

208 `\end{enumerate}`

209

210 Em todo caso, recomendo fortemente que, ao mesmo tempo, mantenha versões no Git e faça o b
211
212
213 `\chapter{Algumas Regras da COPPE}`
214
215 Todas abreviaturas e símbolos devem ser definida antes de utilizada. Isso é facilmente fei
216
217 É imprescindível definir os símbolos, tal como o
218 conjunto dos números reais $\mathbb{R}$ e o conjunto vazio $\emptyset$.
219 `\syml{\mathbb{R}}{Conjunto dos números reais}`
220 `\syml{\emptyset}{Conjunto vazio}`. Usamos esse exemplo aqui justamente para mostrar como
221
222 Para as listas de abreviaturas e símbolos funcionarem no Overleaf é necessário rodar o `\ve`
223
224 Como as listas de símbolos e de abreviaturas usamo o mesmo comando usado para criar índice
225
226 `\section{Citações}`
227 `\label{sec:citacoes}`
228
229 Citações curtas podem ser feitas `\enquote{o comando quote}` ou direto com ‘‘duas crases e do
230
231 `\begin{longquote}`
232 Um exemplo de citação longa nas regras da ABNT (4cm de recuo e fonte menor)
233 feita com o ambiente `\verb*|longquote|` The primary objective of this
234 investigation was to determine the feasibility of detecting corrosion in
235 aluminum Naval aircraft components with neutron radiographic interrogation
236 and the use of standard corrosion penetrameters. Secondary objectives
237 included the determination of the effect of object thickness on image quality,
238 the defining of minimum levels of detectability and a preliminary investigation
239 of a means whereby the degree of corrosion could be quantified with neutron
240 radiographic data. `\citep{article-example}`
241 `\end{longquote}`
242
243 Citações devem apontar as referências. Para isso, está disponível o ótimo pacote `\verb*|na`
244
245 Em todo caso, `\textbf{deve se tomar enorme atenção com as citações, para evitar ocorrer em`
246
247 Quando se cita uma obra por meio de outra (`\textit{apud}`), pela ABNT entra na lista de ref
248
249 `\chapter{Floats}`
250
251 Grande parte dos problemas de iniciantes, e veteranos, em `\LaTeX` é da localização dos `\texti`
252
253 A regra geral de posicionamento é que uma figura ou quadro só pode aparecer a partir da mesm
254
255 `\textbf{Pela norma atual (manual2024), a legenda}` `\verb|\caption|` `\textbf{de toda ilustração}`
256
257 Quadros são ilustrações cujos dados são delimitados por linhas em `\emph{todas}` as bordas (ao
258
259
260 `\section{Tabelas e Figuras Padrão}`
261
262 Vamos ver uma tabela padrão, como a `\autoref{tab:exemplo_numeros}`.
263

```

264 \begin{table}[ht]
265 \centering % Centraliza a tabela
266 \caption{Exemplo de Tabela de Números}
267 \label{tab:exemplo_numeros}
268 \begin{tabular}{ccc} % Define a quantidade de colunas
269 \hline % Linha superior
270 \textbf{Col. 1} & \textbf{Col. 2} & \textbf{Col. 3} \\ % Cabeçalhos
271 \hline % Linha média
272 1 & 2 & 3 \\ % Primeira linha de dados
273 \hline
274 4 & 5 & 6 \\ % Segunda linha de dados
275 \hline
276 7 & 8 & 9 \\ % Terceira linha de dados
277 \hline
278 10 & 11 & 12 \\ % Quarta linha de dados
279 \hline % Linha inferior
280 \end{tabular}
281 \fonte{Elabora\c{c}\~ao pr'opria.} % fonte obrigatoria, abaixo do float
282 \end{table}
283
284
285
286 Já a \autoref{fig:exemplo_figura} é uma figura padrão, com controle da largura.
287
288 \begin{figure}[ht]
289 \centering % Centraliza a figura
290 \caption{Exemplo de Figura com Legenda Acima} % Legenda em cima (norma atual)
291 \label{fig:exemplo_figura} % Etiqueta para referência cruzada
292 \includegraphics[width=0.5\textwidth]{coppe-logo.pdf} % Inclui a imagem com metade da largura
293 \source{COPPE/UFRJ.} % fonte obrigatoria, abaixo da figura
294 \end{figure}
295
296 Quadros têm a mesma regra (legenda acima, fonte abaixo), mas com linhas em todas as bordas.
297
298 \begin{quadro}[ht]
299 \centering
300 \caption{Exemplo de Quadro}
301 \label{qua:exemplo}
302 \begin{tabular}{|l|l|}
303 \hline
304 \textbf{Ilustração} & \textbf{Delimitação} \\
305 \hline
306 Tabela & Linhas apenas no topo e na base. \\
307 \hline
308 Quadro & Linhas em todas as bordas. \\
309 \hline
310 \end{tabular}
311 \fonte{Elabora\c{c}\~ao pr'opria.}
312 \end{quadro}
313
314 Um segundo quadro, no \autoref{qua:segundo}, prova que o contador anda e que a lista de quadros
315
316 \begin{quadro}[ht]
317 \centering

```



```

318 \caption{Segundo Quadro, para provar a numeração}
319 \label{qua:segundo}
320 \begin{tabular}{|l|l|}
321 \hline
322 \textbf{Elemento} & \textbf{Onde fica} \\
323 \hline
324 Legenda & Acima da ilustração. \\
325 \hline
326 Fonte & Abaixo da ilustração. \\
327 \hline
328 \end{tabular}
329 \fonte{Adaptado do manual do SiBI, 9.\textsuperscript{a} ed. rev., 2026.}
330 \end{quadro}
331
332 \section{Criando um float novo}
333
334 \verb|\newcoppefloat{nome}{Nome}{Titulo da Lista}| cria um flutuante que segue a mesma regra
335
336 \begin{mapa}[ht]
337 \centering
338 \caption{Primeiro mapa, num float criado pelo autor}
339 \label{mapa:primeiro}
340 \begin{tikzpicture}[scale=0.8]
341 \draw[thick] (0,0) rectangle (4,2.4);
342 \draw[fill=gray!20] (0.4,0.4) rectangle (1.8,2);
343 \draw[fill=gray!40] (2.2,0.4) rectangle (3.6,1.2);
344 \node at (2,-0.4) {\footnotesize esquema de ocupação};
345 \end{tikzpicture}
346 \fonte{Elabora\c{c}\~ao pr\'opria.}
347 \end{mapa}
348
349 \begin{mapa}[ht]
350 \centering
351 \caption{Segundo mapa, para provar o contador e a lista}
352 \label{mapa:segundo}
353 \begin{tikzpicture}[scale=0.8]
354 \draw[thick] (0,0) circle (1.2);
355 \draw[->] (-1.6,0) -- (1.6,0);
356 \draw[->] (0,-1.6) -- (0,1.6);
357 \node at (0,-2) {\footnotesize esquema de alcance};
358 \end{tikzpicture}
359 \fonte{Elabora\c{c}\~ao pr\'opria.}
360 \end{mapa}
361
362
363 \section{Tabelas mais elegantes}
364
365 Atualmente a tendência é usar tabelas mais leves, como \autoref{tab:exemplo_numerosbom}. Iss
366
367 \begin{table}[ht]
368 \centering % Centraliza a tabela
369 \caption{Exemplo de Tabela de Números mais elegantes}
370 \label{tab:exemplo_numerosbom}
371 \begin{tabular}{ccc} % Define a quantidade de colunas

```

```

372 \toprule % Linha superior
373 \textbf{Col. 1} & \textbf{Col. 2} & \textbf{Col. 3} \\ % Cabeçalhos
374 \midrule % Linha média
375 1 & 2 & 3 \\ % Primeira linha de dados
376 4 & 5 & 6 \\ % Segunda linha de dados
377 7 & 8 & 9 \\ % Terceira linha de dados
378 10 & 11 & 12 \\ % Quarta linha de dados
379 \bottomrule % Linha inferior
380 \end{tabular}
381 \fonte{Elabora\c{c}\~ao pr\'opria.}
382 \end{table}
383
384 \section{Tabelas Longas ou Largas}
385
386 Se sua tabela é muito longa ou larga, existem várias opções.
387 \begin{itemize}
388     \item alterar o tamanho da letra
389     \item Usar o longtable
390     \item rodar a tabela, fazendo ela em \textit{landscape}
391     \item fazer a tabela dentro de um minibox
392 \end{itemize}
393
394
395 \subsection{Tabelas largas demais}
396
397 É comum em teses que as tabelas sejam largas demais. Há várias formas de resolver isso.
398
399 A \autoref{tab:tabela_largafns} é larga demais, e nela isso é resolvido diminuindo a fonte p
400
401 \begin{table}[ht]
402 \centering % Centraliza a tabela
403 \caption{Exemplo de Tabela Larga com Fonte Menor}
404 \label{tab:tabela_largafns}
405 \footnotesize % Aplica uma fonte menor para a tabela
406 \begin{tabular}{ccccccc} % Aumente o número de colunas conforme necessário
407 \toprule
408 \textbf{Col. 1} & \textbf{Col. 2} & \textbf{Col. 3} & \textbf{Col. 4} & \textbf{Col. 5} & \textbf{Col. 6} & \textbf{Col. 7} \\
409 \midrule
410 Dado 1.1 & Dado 1.2 & Dado 1.3 & Dado 1.4 & Dado 1.5 & Dado 1.6 & Dado 1.7 & Dado 1.8 \\
411 Dado 2.1 & Dado 2.2 & Dado 2.3 & Dado 2.4 & Dado 2.5 & Dado 2.6 & Dado 2.7 & Dado 2.8 \\
412 Dado 3.1 & Dado 3.2 & Dado 3.3 & Dado 3.4 & Dado 3.5 & Dado 3.6 & Dado 3.7 & Dado 3.8 \\
413 \bottomrule
414 \end{tabular}
415 \fonte{Elabora\c{c}\~ao pr\'opria.}
416 \end{table}
417
418 O comando \verb|\resizebox{width}{height}{content}| permite ajustar o tamanho de qualquer co
419
420 \begin{table}[ht]
421 \centering
422 \caption{Exemplo de Tabela Redimensionada}
423 \label{tab:examplerb}
424 \resizebox{\textwidth}{!}{%
425 \begin{tabular}{llll}

```

```

426 \toprule
427 Coluna 1 & Coluna 2 & Coluna 3 & Coluna 4 \\
428 \midrule
429 Dados 1 & Dados 2 & Dados 3 & Dados 4 \\
430 Dados 5 & Dados 6 & Dados 7 & Dados 8 \\
431 \bottomrule
432 \end{tabular}%
433 }
434 \fonte{Elabora\c{c}\~ao pr\'opria.}
435 \end{table}
436
437
438 Para rodar uma tabela muito larga em 90 graus no LaTeX, voc\ea pode usar o pacote \verb*|rotat
439
440 Aqui est\aa um exemplo de como usar o ambiente \verb*|sidewaystable| para girar uma tabela. Pr
441
442 \begin{sidewaystable}
443 \centering
444 \caption{Sua Legenda Aqui}
445 \label{tab:sua_tabela}
446 \begin{tabular}{lll}
447 \toprule
448 Coluna 1 & Coluna 2 & Coluna 3 \\
449 \midrule
450 Item 1 & Item 2 & Item 3 \\
451 Item 4 & Item 5 & Item 6 \\
452 \bottomrule
453 \end{tabular}
454 \fonte{Elabora\c{c}\~ao pr\'opria.}
455 \end{sidewaystable}
456
457 Se a tabela for muito longa, o ambiente \verb|longtable| \e o ideal. Ele fornece comandos par
458
459 % Exemplo de tabela longa que se estende por v\arias p\aginas
460 \begin{longtable}{|c|c|c|}
461 % primeiro cabe\c{a}lho (\e o caption)
462 \caption{Exemplo de Tabela Longa}\label{tab:longa} \\
463 \hline \textbf{Col. 1} & \textbf{Col. 2} & \textbf{Col. 3} \\ \hline
464 \endfirsthead
465 % cabe\c{a}lho normal
466 \multicolumn{3}{c}%
467 {\table\ \thetable} -- continua\c{a}o da p\agina anterior} \\
468 \hline \textbf{Col. 1} & \textbf{Col. 2} & \textbf{Col. 3} \\ \hline
469 \endhead
470 % p\ea normal
471 \hline \multicolumn{3}{r|}{Continua na pr\oxima p\agina} \\ \hline
472 \endfoot
473 \hline
474 % \c{u}ltimo p\ea
475 \multicolumn{3}{r|}{Continua na pr\oxima p\agina}} \\
476 \hline \hline
477 \endlastfoot
478
479 % Conte\c{u}do da tabela

```

480 1 & 2 & 3 \\
 481 4 & 5 & 6 \\
 482 1 & 2 & 3 \\
 483 4 & 5 & 6 \\
 484 1 & 2 & 3 \\
 485 4 & 5 & 6 \\
 486 1 & 2 & 3 \\
 487 4 & 5 & 6 \\
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 494 1 & 2 & 3 \\
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 520 4 & 5 & 6 \1 & 2 & 3 \\
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542 4 & 5 & 6 \\
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580 4 & 5 & 6 \\
581 1 & 2 & 3 \\
582 4 & 5 & 6 \\
583 1 & 2 & 3 \\
584 4 & 5 & 6 \\
585 1 & 2 & 3 \\
586 4 & 5 & 6 \\
587

```

588 % Repetir linhas semelhantes conforme necessário para estender a tabela por 3 páginas
589 \end{longtable}
590 \fonte{Elabora\c{c}\~ao pr\'opria.}% a longtable não é float: a fonte vem após o ambiente
591
592 \section{Subfiguras (subcaption)}
593
594 Para colocar várias imagens (ou tabelas) lado a lado, cada uma com sua própria legenda, use
595
596 \begin{figure}[ht]
597   \centering
598   \begin{subfigure}[b]{0.42\textwidth}
599     \centering\includegraphics[width=\linewidth]{example-image-a}
600     \caption{Primeira vista}\label{fig:sub-a}
601   \end{subfigure}
602   \hfill
603   \begin{subfigure}[b]{0.42\textwidth}
604     \centering\includegraphics[width=\linewidth]{example-image-b}
605     \caption{Segunda vista}\label{fig:sub-b}
606   \end{subfigure}
607   \caption{Figura com duas subfiguras}
608   \label{fig:subs}
609   \fonte{Elabora\c{c}\~ao pr\'opria.}
610 \end{figure}
611
612 Referencie a figura inteira com \verb|\autoref{fig:subs}|. Cada parte tem
613 rótulo próprio: \verb|\autoref{fig:sub-a}| leva à \autoref{fig:sub-a}.
614
615 \section{Imagens (graphicx)}
616
617 A classe já carrega o \verb|graphicx|. Insira imagens com \verb|\includegraphics|; as opções
618 \begin{itemize}
619   \item \verb|[width=0.8\textwidth]| ou \verb|[scale=0.5]| --- tamanho;
620   \item \verb|[angle=90]| --- rotação;
621   \item \verb|[trim=l b r t, clip]| --- recorte (corta as bordas \emph{esquerda, baixo, direi}
622 \end{itemize}
623 Formatos aceitos com \hologo{pdfLaTeX}/\hologo{LuaLaTeX}: \texttt{pdf}, \texttt{png}, \texttt{t}
624
625 \section{Introdução ao tikz}
626
627 O \verb|tikz| (disponível também via \verb|tcolorbox|) desenha gráficos vetoriais no próprio
628
629 \begin{figure}[ht]
630   \centering
631   \begin{tikzpicture}
632     \node[draw, rounded corners] (a) at (0,0) {Início};
633     \node[draw, rounded corners] (b) at (4,0) {Fim};
634     \draw[->, thick] (a) -- (b) node[midway, above] {processo};
635   \end{tikzpicture}
636   \caption{Diagrama simples em tikz}
637   \label{fig:tikz}
638   \fonte{Elabora\c{c}\~ao pr\'opria.}
639 \end{figure}
640
641 Para gráficos de dados (curvas, barras), veja o \verb|pgfplots|, construído sobre o \verb|ti

```

```

642
643 \section{Float ou direto no texto?}
644
645 Colocar a tabela ou imagem dentro de \verb|figure|/\verb|table| (um \emph{float}) deixa o \L
646
647 \chapter{Using BibLaTeX}
648 \label{cap:biblatex}
649
650 Esta classe gera as referências com BibLaTeX e o \emph{backend} \texttt{biber}, no padrão
651
652 \textbf{É} obrigatório compilar com \texttt{biber}} (e não com \hologo{BibTeX}). A sequênci
653
654 Os comandos do \texttt{natbib} (\verb|\citet|, \verb|\citep|) continuam disponíveis porque
655
656 \begin{table}[h]
657 \caption{Exemplos de cita{\c c}\~oes utilizando o comando padr\~ao
658 \texttt{\textbackslash cite} do \LaTeX\ e
659 o comando \texttt{\textbackslash citep},
660 disponível pela opção \texttt{natbib} do BibLaTeX.}
661 \label{tab:citation}
662 \centering
663 {\footnotesize
664 \begin{tabular}{|c|c|c|}
665 \hline
666 Tipo da Publicação & \verb|\cite| & \verb|\citep|\\
667 \hline
668 Livro & \cite{book-example} & \citep{book-example}\\
669 Artigo & \cite{article-example} & \citep{article-example}\\
670 Relatório & \cite{techreport-example} & \citep{techreport-example}\\
671 Relatório & \cite{techreport-exampleIn} & \citep{techreport-exampleIn}\\
672 Anais de Congresso & \cite{inproceedings-example} &
673 \citep{inproceedings-example}\\
674 Séries & \cite{incollection-example} & \citep{incollection-example}\\
675 Em Livro & \cite{inbook-example} & \citep{inbook-example}\\
676 Dissertação de mestrado & \cite{mastersthesis-example} &
677 \citep{mastersthesis-example}\\
678 Tese de doutorado & \cite{phdthesis-example} & \citep{phdthesis-example}\\
679 \hline
680 \end{tabular}}
681 \fonte{Elabora\c{c}\~ao pr\'opria.}
682 \end{table}
683
684 \begin{table}[h]
685 \caption{Exemplos de cita{\c c}\~oes utilizando o comando padr\~ao
686 \texttt{\textbackslash cite} do \LaTeX\ e
687 o comando \texttt{\textbackslash citet},
688 disponível pela opção \texttt{natbib} do BibLaTeX. Além disso, usando o booktabs.}
689 \label{tab:citation1}
690 \centering
691 {\footnotesize
692 \begin{tabular}{ccc}
693 \toprule
694 Tipo da Publicação & \verb|\cite| & \verb|\citet|\\
695 \midrule

```

```

696 Livro & \cite{book-example} & \citet{book-example}\\
697 Artigo & \cite{article-example} & \citet{article-example}\\
698 Relatório & \cite{techreport-example} & \citet{techreport-example}\\
699 Relatório & \cite{techreport-exampleIn} & \citet{techreport-exampleIn}\\
700 Anais de Congresso & \cite{inproceedings-example} &
701 \citet{inproceedings-example}\\
702 Séries & \cite{incollection-example} & \citet{incollection-example}\\
703 Em Livro & \cite{inbook-example} & \citet{inbook-example}\\
704 Dissertação de mestrado & \cite{mastersthesis-example} &
705 \citet{mastersthesis-example}\\
706 Tese de doutorado & \cite{phdthesis-example} & \citet{phdthesis-example}\\
707 \bottomrule
708 \end{tabular}}
709 \fonte{Elabora\c{c}\~ao pr'opria.}
710 \end{table}
711
712 \section{Comandos de citação}
713
714 Os principais comandos são: \verb|\cite| (referência simples), \verb|\citet| (autor por ex
715
716 \subsection{Parâmetro opcional: página e “tradução nossa”}
717
718 Todos os comandos aceitam um parâmetro opcional para a localização (página, seção) e obser
719
720 \subsection{Citação de citação (\emph{apud})}
721
722 Para citar uma obra através de outra, use \verb|\citetapud| e \verb|\citeapud| (demonstra
723
724 \subsection{Múltiplas bibliografias}
725
726 Para separar as referências (por tipo, capítulo ou tema), use o ambiente \verb|refsection|
727
728 \section{Tipos de entrada bibliográfica}
729
730 Cada tipo de documento da ABNT corresponde a um tipo de entrada do BibLaTeX, com um sinôni
731
732 \subsection{Tipos acadêmicos principais}
733 \begin{itemize}
734 \item Livro / monografia --- \verb|@book| (\verb|@livro|)
735 \item Artigo de periódico --- \verb|@article| (\verb|@artigo|); matéria de jornal --- \v
736 \item Capítulo de livro --- \verb|@incollection| (\verb|@partedelivro|), \verb|@inbook|
737 \item Evento no todo (anais) --- \verb|@proceedings| (\verb|@anais|); trabalho em evento
738 \item Tese/dissertação --- \verb|@thesis| (\verb|@tese|, \verb|@dissertacao|); com tipo
739 \item Relatório --- \verb|@report| (\verb|@relatorio|, \verb|@relatoriotecnico|); períod
740 \item Manual --- \verb|@manual|; diversos --- \verb|@misc| (\verb|@diverso|); página/sit
741 \end{itemize}
742
743 \subsection{Tipos especiais (ABNT §4.2.5--4.2.14)}
744 \begin{itemize}
745 \item Patente --- \verb|@patent| (\verb|@patente|)
746 \item Legislação / ato normativo --- \verb|@legislation| (\verb|@legislacao|, \verb|@ato
747 \item Norma técnica --- \verb|@standard| (\verb|@norma|); partitura --- \verb|@music| (\v
748 \item Documento cartográfico (mapa) --- \verb|@map| (\verb|@mapa|, \verb|@cartografico|)
749 \item Audiovisual --- \verb|@video| (\verb|@audiovisual|); filme --- \verb|@movie| (\verb

```



```

750 \end{itemize}
751
752 \subsection{Jogos e TV (tipos próprios da CoppeTeX)}
753 \begin{itemize}
754 \item Jogo --- \verb|@game| (\verb|@jogo|); de tabuleiro --- \verb|@boardgame| (\verb|@j|
755 \item Programa de TV / série --- \verb|@tvshow| (\verb|@programatv|, \verb|@serietv|)
756 \end{itemize}
757
758 Os campos também aceitam sinônimos em português (sem acentos): \verb|autor|, \verb|titulo|
759
760 \chapter{Alguns outros exemplo úteis}
761
762 \begin{tcolorbox}[title=Meu Textbox]
763 Este é o conteúdo do meu textbox. Você pode adicionar qualquer texto aqui, bem como incluir
764 \end{tcolorbox}
765
766 \begin{tcolorbox}
767 Este é o conteúdo do meu textbox sem título. Você pode adicionar qualquer texto aqui, bem co
768 \end{tcolorbox}
769
770 \begin{figure}[ht]
771 \centering
772 \begin{tikzpicture}
773 \node[anchor=south west,inner sep=0] (image) at (0,0) {\includegraphics[width=0.5\textwidth]{
774 \begin{scope}[x={(image.south east)},y={(image.north west)}]
775 % Definindo o textbox dentro da figura
776 \node[anchor=north west, text width=0.3\textwidth, fill=white, opacity=0.7, text=
777 \begin{tcolorbox}[colback=red!5!white,colframe=red!75!black,title=Textbox de
778 Este textbox fala sobre como inserir um textbox dentro de uma figura usa
779 \end{tcolorbox}
780 };
781 \end{scope}
782 \end{tikzpicture}
783 \caption{Figura com Textbox}
784 \label{fig:figura_com_textbox1}
785 \fonte{Elabora\c{c}\~ao pr\'opria.}
786 \end{figure}
787
788
789 \begin{figure}[ht]
790 \centering
791 \begin{tcolorbox}
792 Este é o conteúdo do meu textbox sem título. Você pode adicionar qualquer texto aqui, bem co
793 \end{tcolorbox}
794 \caption{Figura com Textbox simples}
795 \label{fig:figura_com_textbox}
796 \fonte{Elabora\c{c}\~ao pr\'opria.}
797 \end{figure}
798
799 \section{Listagens de código e algoritmos}
800
801 O CoppeTeX traz embutido o suporte a listagens de programas (pacote \verb|listings|) e a alg
802
803 \begin{python}[caption={Exemplo de programa em Python},label=prog:exemplo]

```

```

804 def saudacao(nome):
805     # imprime uma saudação acentuada
806     print(f"Olá, {nome}!")
807
808 saudacao("COPPE")
809 \end{python}
810 \source{Elaboração própria.}
811
812 Para inserir código no meio do texto use \verb|\pythoninline|, como em \pythoninline{saudacao}
813
814 \begin{Verbatim}[frame=lines,numbers=left,numbersep=0.5em]
815 Olá, COPPE!
816 \end{Verbatim}
817
818 Há estilos prontos também para \verb|java|, \verb|xml|, \verb|html| e \verb|prolog|. O \auto
819
820 \begin{java}[caption={Exemplo de programa em Java},label=prog:java]
821 public class Saudacao {
822     public static void main(String[] args) {
823         System.out.println("Olá, COPPE!");
824     }
825 }
826 \end{java}
827 \fonte{Elaboração própria.}
828
829 Há também \verb|\pythoninline| e \verb|\javainline| para código no meio do
830 texto, como em \javainline{new Saudacao()}, e o ambiente \verb|Verbatim| (com as opções acima)
831
832 \begin{algorithm}[H]
833 \caption{Soma dos elementos de um vetor}
834 \label{alg:soma}
835 \Dados{vetor  $v$  com  $n$  elementos}
836 \Resultado{soma  $s$  dos elementos de  $v$ }
837  $s \leftarrow 0$ ;
838 \Para{ $i \leftarrow 1$  \Ate{ }  $n$ }{
839      $s \leftarrow s + v[i]$ ;
840 }
841 \Retorna  $s$ ;
842 \end{algorithm}
843
844 O \autoref{alg:busca} é o segundo algoritmo, e usa \verb|\Enquanto|. Como no caso das listas
845
846 \begin{algorithm}[H]
847 \caption{Busca sequencial em um vetor}
848 \label{alg:busca}
849 \Dados{vetor  $v$  com  $n$  elementos e uma chave  $k$ }
850 \Resultado{posição de  $k$  em  $v$ , ou  $0$  se  $k$  não estiver em  $v$ }
851  $i \leftarrow 1$ ;
852 \Enquanto{ $i \leq n$  \textbf{e}  $v[i] \neq k$ }{
853      $i \leftarrow i + 1$ ;
854 }
855 \Se{ $i > n$ }{\Retorna  $0$ ; }
856 \Retorna  $i$ ;
857 \end{algorithm}

```

```

858
859 \chapter{Escrevendo matemática}
860
861 A classe carrega \texttt{amsmath} em qualquer motor. Sob \hologo{pdfLaTeX} ela também carrega
862
863 \section{Alinhamento (align)}
864 O ambiente \verb|align| alinha equações pelo \verb|&| e numera cada linha (\verb|\\| quebrar a linha)
865 \begin{align}
866   f(x) &= (x+1)^2 \\
867         &= x^2 + 2x + 1 .
868 \end{align}
869
870 \section{Matrizes (matrix)}
871 O \texttt{amsmath} traz \verb|pmatrix|, \verb|bmatrix|, \verb|vmatrix|, entre outros:
872 \begin{equation}
873   A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \\
874   \quad \quad \quad \\
875   \det A = \begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix} \\
876 \end{equation}
877
878 \section{Subequações (subequations)}
879 Para numerar um grupo de equações sob o mesmo número, com letras (a), (b):
880 \begin{subequations}
881 \begin{align}
882   a &= b + c , \\
883   d &= e - f .
884 \end{align}
885 \end{subequations}
886
887 \section{Estruturas alinhadas (array)}
888 O \verb|array| monta estruturas dentro do modo matemático, como definições por partes:
889 \begin{equation}
890   |x| = \left\{ \begin{array}{l} x, \quad \text{se } x \geq 0, \\ -x, \quad \text{caso contrário.} \end{array} \right. \\
891   \quad \quad \quad \\
892   \end{equation}
893
894 \end{equation}
895
896 \section{Operadores (\textbackslash DeclareMathOperator)}
897 Para operadores com a tipografia correta (romano, espaçamento adequado), declare-os no preâmbulo:
898 \begin{verbatim}
899 \usepackage{amsmath}
900 \DeclareMathOperator{\RMSE}{RMSE}
901 \DeclareMathOperator{\diag}{diag}
902 \end{verbatim}
903 Assim:
904 \begin{itemize}
905   \item \verb|\RMSE(y,\hat{y})| produz  $\text{RMSE}(y,\hat{y})$ ;
906   \item \verb|\diag(\lambda_1,\lambda_2)| produz  $\text{diag}(\lambda_1,\lambda_2)$ .
907 \end{itemize}
908
909 \section{Unicode e LuaLaTeX (unicode-math)}
910 \label{sec:unicode-math}
911

```

```

912 Ao migrar para \hologo{LuaLaTeX} (ou \hologo{XeLaTeX}), \textbf{não carregue} \texttt{amss
913 \begin{verbatim}
914 \usepackage{amsmath}      % a classe já carrega
915 \usepackage{unicode-math} % NÃO use \usepackage{amssymb} junto
916 \setmathfont{Latin Modern Math} % ou STIX Two Math
917 \end{verbatim}
918 O código clássico continua valendo ( $\mathbb{R}$ ,  $\int_a^b f(x)dx$ ,  $\alpha$ ) e você ai
919
920 \chapter{Referências cruzadas, notas e índices}
921 \label{cap:refs}
922
923 Este capítulo reúne os recursos para \emph{apontar} para coisas: rótulos, notas de rodapé,
924
925 \section{Rótulos e referências cruzadas}
926
927 Marque um elemento com \verb|\label{tipo:nome}| e aponte para ele com \verb|\ref| (só o nú
928
929 \section{Notas de rodapé}
930
931 Use \verb|\footnote{texto}|\footnote{Como esta nota, que a classe imprime em espaçamento s
932
933 \section{hyperref e marcadores}
934
935 A classe carrega o \verb|hyperref|: o sumário, as citações e os \verb|\ref|/\verb|\autoref
936
937 \section{URLs}
938
939 Para endereços, use \verb|\url{https://exemplo.org}|: o \verb|hyperref| cria o link e queb
940 use \verb|\href{https://exemplo.org}{texto}|. Evite digitar URLs como texto comum --- elas n
941
942 \section{Índice remissivo}
943
944 Marque os termos com \verb|\index{termo}| e imprima o índice com \verb|\printindex| (carre
945
946 \section{Traduções (NBR 10520)}
947
948 Ao citar um trecho que você mesmo traduziu, coloque a tradução no corpo do texto e escreva
949
950 \chapter{Truques de \LaTeX}
951
952 \section{Comentários mágicos (\texttt{\% !TeX})}
953 Linhas \verb|\% !TeX| no topo do arquivo dizem ao editor (\hologo{TeX}studio, Overleaf, \ho
954 \begin{verbatim}
955 % !TeX program = lualatex
956 % !TeX encoding = UTF-8
957 % !TeX spellcheck = pt_BR
958 % !BIB program = biber
959 \end{verbatim}
960 Com isso o documento compila igual em qualquer ambiente. O \texttt{latexmkrc} cumpre papel
961
962 \section{Espaço engolido}
963 Um comando sem argumento engole o espaço seguinte: \verb|\LaTeX é bom| sai como “\LaTeX é
964
965 \section{Escopo (grupos)}

```

```

966 Chaves \verb|{...}| (ou \verb|\begingroup|\dots\verb|\endgroup|) criam um escopo: mudanças
967
968 \section{Descarregar figuras: \texttt{\textbackslash clearpage} × \texttt{\textbackslash newpage}
969 \verb|\newpage| apenas quebra a página e leva os floats pendentes adiante, acumulando-os.
970
971 \section{\texttt{latexmkrc}, perl e Overleaf}
972 0 \verb|latexmk| automatiza a compilação (roda \LaTeX, \verb|biber| e \verb|makeindex| qua
973
974 \chapter{Método Proposto}
975 \chapter{Resultados e Discussões}
976
977 \section{Algumas Demonstrações}
978
979 A Lista de Símbolos precisa usar comandos específicos. Aqui vamos usar os símbolos $\alpha$
980 \syml[beta]{Beta}{A palavra Beta more e corrigida}
981 \syml[zzbeta]{\beta}{A letra $\beta$ corrigida}
982 \syml[beta]{A palavra beta}
983 \syml[alpha]{A palavra alpha}
984 \syml[alpha]{Alpha}{A palavra Alpha}
985 \syml[zzalpha]{\alpha}{A letra $\alpha$ corrigida}
986 \syml[marco]{Marco}{A palavra Marco corrigida}
987
988 A Lista de Abreviações segue, a partir de 2024, a mesma regra, e aqui seguem alguns exemplos
989 \abbrev{GoT}{Game of Thrones}
990 \abbrev[GOT]{GoT}{Game of Thrones ordenado como GOT}
991 \abbrev[iot]{IoT}{IoT ordenado como iot}
992 \abbrev[IOT]{IoT}{IoT ordenado como IOT}
993 \abbrev[IOT]{IoT}{IoT ordenado como IOT}
994 \abbrev{IoT}{IoT com ordenação default}
995 \abbrev[ITU]{ITU}{ITU mesmo}
996
997 \newsigla{ufrj}{UFRJ}{Universidade Federal do Rio de Janeiro}
998 \newsigla{cpgp}{CPGP}{Comissão de Pós-Graduação e Pesquisa de Engenharia}
999
1000 As siglas (manual2024, seção 2.8) são expandidas na primeira ocorrência e registradas automaticamente
1001
1002
1003
1004
1005 \chapter{Conclusões}
1006
1007 \backmatter
1008
1009 % Tipos exóticos da ABNT (NBR 6023): patente, legislação, norma, mapa,
1010 % partitura --- declarados via sinônimos em português em example.bib.
1011 \nocite{patente-example,legislacao-example,norma-example,mapa-example,partitura-example}
1012 \nocite{jurisprudencia-example}
1013 \nocite{videogame-example,filme-example,semautor-example,semautorart-example}
1014 \nocite{jornalcaderno-example,jornal-example}
1015 \printbibliography
1016
1017
1018
1019 \appendix

```

```

1020
1021 \chapter{Um apêndice}
1022
1023 Segundo a norma da ABNT (Associação Brasileira de Normas Técnicas), a definição e utilização
1024
1025 Apêndice: O apêndice é um texto ou documento elaborado pelo autor do trabalho com o objetivo
1026
1027
1028 \chapter{Outro apêndice}
1029
1030 \annex
1031
1032
1033 \chapter{Um Anexo}
1034 Segundo a norma da ABNT (Associação Brasileira de Normas Técnicas), a definição e utilização
1035
1036
1037
1038 Anexo: O anexo, por sua vez, consiste em um texto ou documento não elaborado pelo autor, que
1039
1040 No modelo \CoppeTeX os anexos devem obrigatoriamente vir depois dos apêndices e usam o comand
1041
1042
1043 \chapter{Outro Anexo}
1044
1045
1046 \coppetexfinalpage% colofão institucional (página par, face do contracapa)
1047
1048 \end{document}
1049
1050 </example>

```

10 Implementation

10.1 The ‘coppe.cls’ file

```

1051 < *class>
1052 \def\filename{coppe.dtx}
1053 \def\fileversion{v4.1}
1054 \def\filedate{2026/09/10}
1055 \NeedsTeXFormat{LaTeX2e}[1995/12/01]
1056 \ProvidesClass{coppe}[\filedate\ \fileversion\ COPPE Dissertations and Thesis]
1057 % Save the class version and date into private macros immediately, before any
1058 % required package can redefine the very common \fileversion / \filedate
1059 % global names (LaTeX has no convention reserving them for the host class).
1060 % \coppe@version / \coppe@date are what user-facing macros (e.g.
1061 % \coppetexfinalpage) should reference.
1062 \let\coppe@version\fileversion
1063 \let\coppe@date\filedate
1064 % Document date stored in private slots so \date can set them without ever
1065 % touching TeX's \month/\year primitives (which would break \today). They
1066 % default to the current system month/year, so a document without \date
1067 % still renders sensibly on the cover and folha de rosto.
1068 \edef\coppe@docmonth{\the\month}

```

```

1069 \edef\coppe@docyear{\the\year}
1070 \LoadClass[12pt,a4paper,oneside]{book}
1071 % setspace, hyperref and the bibliography engine (biblatex+biber) are loaded at
1072 % the END of the class (after every class hook is defined), in that order:
1073 % setspace -> hyperref -> biblatex. hyperref must come late so all the macros
1074 % it patches (TOC, refs, lists) are already defined, and biblatex must come
1075 % after hyperref so its citation links and back-references work.
1076 \RequirePackage{hyphenat}
1077 \RequirePackage{lastpage}
1078 \RequirePackage{ifthen}
1079 \RequirePackage{graphicx}
1080 \RequirePackage{tabularx}
1081 \RequirePackage{booktabs} % regras de tabela profissionais (toprule/midrule/bottomrule)
1082 \RequirePackage{longtable} % tabelas que quebram entre páginas
1083 \RequirePackage{etoolbox}
1084 \RequirePackage{ltxcmds}
1085 \RequirePackage{hologo} % TeX-family logos: \hologo{TeX}, \hologo{LaTeX}, ...
1086 % Render \TeX, \LaTeX and friends via hologo (\CoppeTeX is a custom logo, kept).
1087 \def\TeX{\hologo{TeX}}
1088 \def\LaTeX{\hologo{LaTeX}}
1089 \def\LaTeXe{\hologo{LaTeXe}}
1090 \def\BibTeX{\hologo{BibTeX}}
1091 \def\pdfLaTeX{\hologo{pdfLaTeX}}
1092 \def\pdfTeX{\hologo{pdfTeX}}
1093 \def\LuaLaTeX{\hologo{LuaLaTeX}}
1094 \def\XeLaTeX{\hologo{XeLaTeX}}
1095 \def\MiKTeX{\hologo{MiKTeX}}
1096 % Engine-aware UTF-8 input: pdfLaTeX needs inputenc(utf8); LuaLaTeX (and XeTeX)
1097 % are UTF-8 native, so inputenc must NOT be loaded there (it would clash). Source
1098 % files may use accented Latin characters (áéíóú) directly. Under pdfLaTeX we also
1099 % suggest LuaLaTeX, which handles Unicode/fonts better (at the cost of speed).
1100 \RequirePackage{iftex}
1101 \ifPDFTeX
1102 \RequirePackage[utf8]{inputenc}
1103 \AtEndOfClass{\ClassWarningNoLine{coppe}{%
1104 Although slower, you should try compiling with LuaLaTeX}}
1105 \fi
1106 % lmodern MUST come with the T1 encoding. Without it, T1 falls back to the EC
1107 % fonts, which exist only as METAFONT bitmaps in most installations, and the
1108 % whole document is typeset in Type3 bitmap fonts. Measured before this change:
1109 % example.pdf carried 87 Type3 fonts against 6 Type1, and coppe.pdf 133 against
1110 % 2. The consequences are not cosmetic:
1111 %
1112 % * the text renders blurred at any zoom and prints poorly;
1113 % * copying and searching text in the PDF is unreliable;
1114 % * PDF/A -- required for deposit by 2.2(d) of the 2026 manual -- is not
1115 % attainable with bitmap text.
1116 %
1117 % Latin Modern is the vector successor of Computer Modern, metrically and
1118 % visually near-identical, and ships with every current distribution, so the
1119 % page layout does not move.
1120 \RequirePackage{lmodern}
1121 \RequirePackage[T1]{fontenc}
1122 % Math support. amsmath is universal and works on every engine, so the class

```

```

1123 % brings it in unconditionally (it must precede hyperref, which is loaded at
1124 % the end of the class). amssymb is the legacy AMS symbol package, built
1125 % around the 8-bit Type 1 fonts msam/msbm; on a Unicode engine (LuaLaTeX or
1126 % XeLaTeX) it clashes with unicode-math, which already provides all of its
1127 % symbols through Unicode math codes. So amssymb is loaded ONLY under
1128 % pdfLaTeX; Unicode-engine users get a ClassWarning pointing them at
1129 % unicode-math (the class does not auto-load unicode-math because choosing
1130 % an OpenType math font is a typographic decision that belongs to the author).
1131 \RequirePackage{amsmath}
1132 \ifPDFTeX
1133   \RequirePackage{amssymb}
1134 \else
1135   % Unicode engine: defer the check to end of document. Only warn if the
1136   % user did NOT add either unicode-math (the modern OpenType-math route)
1137   % or amssymb (the legacy route, which they would have loaded manually).
1138   % This avoids nagging authors who already followed the documented
1139   % \usepackage{unicode-math} workflow.
1140   \AtEndDocument{%
1141     \@ifpackageloaded{unicode-math}{%}{%
1142       \@ifpackageloaded{amssymb}{%}{%
1143         \ClassWarningNoLine{coppe}{%
1144           You are compiling with a Unicode engine (LuaLaTeX or XeLaTeX) and\MessageBreak
1145           neither unicode-math nor amssymb is loaded. AMS-style symbols\MessageBreak
1146           (\string\mathbb, \string\hbar, \string\varnothing, ...) will be undefined.\MessageBreak
1147           Add to your preamble:\MessageBreak
1148           \space\space\string\usepackage{unicode-math}\MessageBreak
1149           \space\space\string\setmathfont{Latin Modern Math}\MessageBreak
1150           Do NOT load amssymb together with unicode-math -- it would clash\MessageBreak
1151           with the OpenType math font setup}}}%
1152 \fi
1153 % Margins per the SiBI manual, 9th rev. ed. (2026), 2.3: left 3cm, top 3cm,
1154 % right 2cm, bottom 2cm -- ONE-SIDED. The 2025 edition also specified verso
1155 % margins (right 3cm / left 2cm) and recommended printing the body on both
1156 % sides; the 2026 revision DELETED the verso margins entirely, following CEPG
1157 % Resolution 246/2023, which ended the printed deposit. A digital-only document
1158 % has no gutter, so the mirrored layout (previously inner=outer=2cm plus a 1cm
1159 % bindingoffset) no longer has any normative basis and is now opt-in via the
1160 % \texttt{twoside} class option, handled after \ProcessOptions below.
1161 %
1162 % headheight=15pt is given here (not patched later) so geometry's own page
1163 % arithmetic stays self-consistent: fancyhdr needs >= 14.49998pt of header
1164 % height to host the running head, and the standard 12pt default triggers
1165 % its "headheight is too small" warning. Letting geometry know up front
1166 % keeps \topmargin a derived value -- we never touch it by hand.
1167 % (geometry has no \texttt{oneside} key -- only \texttt{twoside}; with the key
1168 % absent it follows the class, which \LoadClass set to one-sided above.)
1169 % headsep is given explicitly to place the folio. 2.7 measures it the way it
1170 % measures the right margin -- to the digits themselves: "a 2cm da borda
1171 % superior, ficando o ultimo algarismo a 2cm da borda direita". The right edge
1172 % already landed at 2.02cm; the top of the digits sat at 1.87cm, too high.
1173 % geometry derives \topmargin from top - 1in - headheight - headsep, so
1174 % shortening headsep lowers the header and leaves the body top untouched:
1175 % topmargin + headheight + headsep is 13.088pt either way, and 3cm is where the
1176 % body still starts. The value is CALIBRATED, not computed -- the folio sits a

```



```

1177 % little lower than the arithmetic alone predicts, because fancyhdr's box does
1178 % not put the digits flush with the top of \headheight. Two measured points,
1179 % taken from the rendered page rather than from the PDF's font metrics:
1180 %   headsep 18.07pt -> digits at 1.8669cm   headsep 14.29pt -> 2.0616cm
1181 % which put 2.0000cm at 15.49pt while the folio was still set at the body size.
1182 % It is \footnotesize now (2.2(b) puts pagination in the smaller font), the
1183 % digits are shorter, and their top therefore sits lower for the same baseline:
1184 % re-measured, 15.49pt gave 2.0659cm and the value moved to 16.87pt. That is
1185 % precisely why this is a measured constant and not a computed one -- change
1186 % the folio's size or headheight and it has to be measured again, not scaled.
1187 % Verified on the rendered page: the top of the digits and their right edge come
1188 % out at the same 2.019cm, and the right margin is 2cm by construction, so that
1189 % shared 0.019cm is the ruler's own bias and not a residual error.
1190 \RequirePackage[a4paper,headheight=15pt,headsep=16.87pt,%
1191 top=3cm,bottom=2cm,left=3cm,right=2cm]{geometry}
1192 % The first paragraph of every section must be indented like any other. LaTeX's
1193 % default is the Anglo-Saxon convention -- no indent after a heading -- which is
1194 % not what Brazilian typography (nor the SiBI manual's own examples) uses.
1195 % indentfirst was mentioned only as a commented suggestion in the example
1196 % preamble, so every document built with this class came out with the first
1197 % paragraph of each section flush left. Loading it in the class fixes it for
1198 % everybody instead of asking each author to remember.
1199 %
1200 % Loaded BEFORE titlesec so titlesec's \@afterindent handling sees the patched
1201 % \@afterindentfalse.
1202 % Long unbreakable tokens -- \verb|...| in running prose, command names, URLs --
1203 % cannot be hyphenated, so TeX overfills the line rather than stretching the
1204 % spaces. example.tex alone produced 17 overfull boxes this way, the worst 86pt,
1205 % which is the "text does not always respect the margins" a recent review
1206 % reported. \emergencystretch gives TeX a final pass in which it may stretch
1207 % interword spaces instead of running into the margin. It cannot save a single
1208 % token longer than the measure, but it clears the ordinary cases.
1209 \setlength\emergencystretch{3em}
1210 \RequirePackage{indentfirst}
1211 \RequirePackage{titlesec}
1212 % --- Section heading formats (manual2024 2.6 / proposta P22) -----
1213 % chapter (numbered): "N TITLE", ALL CAPS, bold, flush left (no "Capitulo N").
1214 % APENDICE/ANEXO chapters print "APENDICE A -- Title" (R75/R76); normal chapters
1215 % keep "N Title". \appendix and \annex raise the flag (see below).
1216 \newif\if@coppeappendix \@coppeappendixfalse
1217 \titleformat{\chapter}[hang]
1218   {\normalfont\Large\bfseries}
1219   {\if@coppeappendix\MakeUppercase{\appendixname}\ \thechapter\ \textendash\else\thechapter\
1220    \lex}{\MakeUppercase}
1221 \titlespacing*{\chapter}{0pt}{10pt}{1.5\baselineskip}
1222 % 2.6 says the numeric indicative of a section precedes its title flush left,
1223 % and that titles WITHOUT a numeric indicative are centred -- and then lists
1224 % them, "apendice(s)" and "anexo(s)" among them. A letter is not a numeric
1225 % indicative, and the manual sets its own Annex A centred. The chapter format is
1226 % therefore re-issued when \appendix or \annex is called: same label, same
1227 % travessao, but a centred block instead of a flush-left hang. Numbered chapters
1228 % are untouched -- they DO carry a numeric indicative and stay flush left.
1229 \newcommand\coppe@appendixheadings{%
1230   \titleformat{\chapter}[block]

```

```

1231     {\normalfont\Large\bfseries\filcenter}
1232     {\MakeUppercase{\appendixname}\ \thechapter\ \textendash}
1233     {lex}{\MakeUppercase}%
1234     \titlespacing*{\chapter}{0pt}{10pt}{1.5\baselineskip}}
1235 \let\coppe@orig@appendix\appendix
1236 \renewcommand\appendix{%
1237   \coppe@orig@appendix\@coppeappendixtrue\coppe@appendixheadings}
1238 % Sumario (TOC) entries for APENDICE/ANEXO chapters read "APENDICE A -- Title"
1239 % / "ANEXO A -- Title" (R75/R76), not the bare "A". The label word is chosen by
1240 % which macro is written (so it is correct at .toc-render time even though the
1241 % appendixname has changed back); the chapter letter is baked in at write time.
1242 \newif\if@coppeannex \@coppeannexfalse
1243 \newcommand\coppe@tocapp[1]{%
1244   \MakeUppercase{\coppestring{appendix}}%
1245   \nobreakspace#1\nobreakspace\textendash\nobreakspace}
1246 \newcommand\coppe@tocanx[1]{%
1247   \MakeUppercase{\coppestring{annex}}%
1248   \nobreakspace#1\nobreakspace\textendash\nobreakspace}
1249 \def\@chapter[#1]#2{%
1250   \ifnum \c@secnumdepth >\m@ne
1251     \if@mainmatter
1252       \refstepcounter{chapter}%
1253       \typeout{\@chapapp\space\thechapter.}%
1254       \if@coppeappendix
1255         \if@coppeannex
1256           \addcontentsline{toc}{chapter}{\protect\coppe@tocanx{\thechapter}#1}%
1257         \else
1258           \addcontentsline{toc}{chapter}{\protect\coppe@tocapp{\thechapter}#1}%
1259         \fi
1260       \else
1261         \addcontentsline{toc}{chapter}{\protect\numberline{\thechapter}#1}%
1262       \fi
1263     \else
1264       \addcontentsline{toc}{chapter}{#1}%
1265     \fi
1266   \else
1267     \addcontentsline{toc}{chapter}{#1}%
1268   \fi
1269   \chaptermark{#1}%
1270   \addtocontents{lof}{\protect\addvspace{10\p@}}%
1271   \addtocontents{lot}{\protect\addvspace{10\p@}}%
1272   \if@twocolumn
1273     \@topnewpage[\@makechapterhead{#2}]%
1274   \else
1275     \@makechapterhead{#2}%
1276     \@afterheading
1277   \fi}
1278 % chapter (unnumbered): centred ALL CAPS bold (errata, lists, resumo, sumario,
1279 % referencias, glossario, ... -- unnumbered headings are centred, P29-P30).
1280 \titleformat{name=\chapter,numberless}[hang]
1281   {\normalfont\Large\bfseries\filcenter}{\0pt}{\MakeUppercase}
1282   \titlespacing*{name=\chapter,numberless}{0pt}{10pt}{1.5\baselineskip}
1283 % section: ALL CAPS, normal weight -- the \bfseries here contradicted this very
1284 % comment until v4.1, and made the secondary level as heavy as the primary one,

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1285 % defeating the "destaque gradativo" of 2.6. The gradation is now
1286 % chapter = ALL CAPS bold, section = ALL CAPS regular, subsection = Title Case
1287 % bold, subsubsection = Title Case bold italic, as read from the manual.
1288 \titleformat{\section}
1289 {\normalfont\large}{\thesection}{1ex}{\MakeUppercase}
1290 % subsection: bold (author writes Title Case).
1291 \titleformat{\subsection}
1292 {\normalfont\bfseries}{\thesubsection}{1ex}{\bfseries}
1293 % subsubsection: bold italic.
1294 \titleformat{\subsubsection}
1295 {\normalfont\bfseries\itshape}{\thesubsubsection}{1ex}{\bfseries}
1296 % paragraph = quinary section: italic, the author capitalising only the first
1297 % word. 2.6 recommends going as deep as the quinary, and the annotation on the
1298 % 2025 manual (specs/ANOTACOES.md) describes exactly this shape.
1299 %
1300 % book makes \paragraph a RUN-IN heading, so the text continued on the same
1301 % line -- 2.6 requires "o texto deve ser iniciado em outra linha". The positive
1302 % "after" spacing in \titlespacing turns it into a displayed heading.
1303 \titleformat{\paragraph}[hang]
1304 {\normalfont\itshape}{\theparagraph}{1ex}{\bfseries}
1305 \titlespacing*{\paragraph}{0pt}{2.5ex plus 1ex minus .2ex}{1.5ex plus .2ex}
1306 % book defaults secnumdepth to 2, so \subsubsection and \paragraph came out
1307 % UNNUMBERED however carefully \titleformat named \thesubsubsection. 2.6 has
1308 % the numbering following the whole chain down to the quinary, and the manual's
1309 % own sumario is four levels deep, so tocdepth follows.
1310 \setcounter{secnumdepth}{4}
1311 \setcounter{tocdepth}{4}
1312 % --- 0 sumario (2.6 e 3.1.2.1.6) -----
1313 %
1314 % Duas exigencias o governam, e a segunda tem um exemplo obrigatorio.
1315 %
1316 % A 2.6: "no sumario, as secoes devem ser grafadas conforme apresentadas no
1317 % corpo do trabalho". A gradacao do corpo -- primaria em CAIXA ALTA negrito,
1318 % secundaria em CAIXA ALTA, terciaria em negrito, quaternaria em negrito
1319 % italico, quinary em italico -- tem de reaparecer aqui. So as duas primeiras
1320 % reapareciam: da terciaria em diante o sumario saia em redondo, contradizendo
1321 % o corpo.
1322 %
1323 % A 3.1.2.1.6: os titulos "devem ser alinhados a esquerda... pela margem do
1324 % titulo do indicativo mais extenso", e "utilize como exemplo o sumario deste
1325 % trabalho". Medido no PDF do manual: o indicativo comeca em 3,00 cm nos cinco
1326 % niveis -- a margem -- e o titulo em 4,75 cm nos cinco. Uma coluna so.
1327 %
1328 % 0 \@dottedtocline do LaTeX faz o contrario. Recua progressivamente (3,00 /
1329 % 3,62 / 4,57 / 5,89 / 7,13 cm nesta classe) e da a cada nivel uma caixa de
1330 % indicativo de largura fixa. Isso empurrava o titulo quinario para 9,20 cm --
1331 % 6,2 dos 16 cm de mancha -- e, pior, a caixa fixa ESTOURA: com o indicativo
1332 % "10.10.10.10.10", que uma tese de dez capitulos alcanca, o numero entrava
1333 % 4,6 mm POR CIMA do titulo. A 2.6 quer o titulo "separado por um espaco" do
1334 % indicativo; ali nao havia espaco nenhum.
1335 %
1336 % A coluna abaixo e medida, nao arbitrada: \coppe@tocnumberline mede cada
1337 % indicativo enquanto compoe o sumario e guarda o maior; o valor vai para o
1338 % .aux e governa a passada seguinte. E literalmente "a margem do titulo do

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1339 % indicativo mais extenso". Custa uma passada -- a mesma que as referencias
1340 % cruzadas ja custam.
1341 \newlength\coppe@tocnumwidth
1342 \newlength\coppe@tocnummax
1343 \newlength\coppe@tocnumsep
1344 % 0 espaco entre o indicativo e o titulo (2.6, "separado por um espaco").
1345 \setlength\coppe@tocnumsep{0.5em}
1346 % Piso: a coluna do proprio manual. Pela regra literal, um trabalho so com
1347 % capitulos teria uma coluna de meio centimetro e o titulo quase colado na
1348 % margem; o piso evita isso. A coluna nunca fica MENOR que o indicativo mais
1349 % extenso mais o espaco -- o piso so a impede de ficar estreita demais.
1350 \setlength\coppe@tocnumwidth{1.75cm}
1351 \setlength\coppe@tocnummax{\z@}
1352 \newcommand\coppe@tocnumberline[1]{%
1353   \settowidth\@tempdima{#1}%
1354   \ifdim\@tempdima>\coppe@tocnummax \global\coppe@tocnummax\@tempdima \fi
1355   \hb@xt@\coppe@tocnumwidth{#1\hfil}}
1356 % A largura medida na passada anterior chega pelo .aux, que o \document le
1357 % antes dos ganchos de \AtBeginDocument.
1358 \AtBeginDocument{%
1359   \@ifundefined{coppe@tocnumsaved}{}{%
1360     \setlength\@tempdima{\coppe@tocnumsaved}%
1361     \addtolength\@tempdima\coppe@tocnumsep
1362     \ifdim\@tempdima>\coppe@tocnumwidth
1363       \setlength\coppe@tocnumwidth\@tempdima
1364     \fi}}
1365 \AtEndDocument{%
1366   \ifdim\coppe@tocnummax>\z@
1367     \if@filesw
1368       \immediate\write\@auxout{\string\gdef\string\coppe@tocnumsaved
1369         {\the\coppe@tocnummax}}%
1370     \fi
1371     \setlength\@tempdima\coppe@tocnummax
1372     \addtolength\@tempdima\coppe@tocnumsep
1373     \ifdim\@tempdima>\coppe@tocnumwidth
1374       \ClassWarning{coppe}{A coluna do sumario cresceu; rode LaTeX de novo}%
1375     \fi
1376   \fi}
1377 % Uma linha do sumario. #1 nivel, #2 fonte, #3 entrada, #4 folha. A entrada
1378 % traz \numberline{N} seguido do titulo; o \hskip-\leftskip puxa a primeira
1379 % linha de volta a margem, e o \leftskip deixa as linhas seguintes de um
1380 % titulo longo alinhadas pela coluna do titulo -- nunca debaixo do numero.
1381 % A folha herda a fonte do nivel, como no sumario do manual.
1382 \newcommand\coppe@tocline[4]{%
1383   \ifnum #1>\c@tocdepth\else
1384     \vskip\z@\@plus.2\p@
1385     {\leftskip\z@ \rightskip\@tocrmarg \parfillskip-\rightskip
1386     \parindent\z@ \interlinepenalty\@M
1387     \leavevmode
1388     #2%
1389     \advance\leftskip\coppe@tocnumwidth
1390     \null\nobreak\hskip-\leftskip
1391     #3\nobreak
1392     \leaders\hbox{$\m@th\mkern\@dotsep mu\hbox{.}\mkern\@dotsep mu$}%

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1393      \hfill\nobreak
1394      \hb@xt@{\pnumwidth{\hss #4}\par}%
1395  \fi}
1396 % 0 pontilhado do manual e mais fechado que o do book; \@dotsep e a distancia
1397 % entre os pontos, em unidades matematicas.
1398 \renewcommand\@dotsep{2}
1399 % Os cinco niveis. 0 \MakeUppercase vai so nos dois primeiros porque so eles
1400 % saem em caixa alta no corpo; ele atravessa os algarismos sem toca-los e
1401 % preserva comandos robustos, inclusive os do hyperref.
1402 %
1403 % 0 nivel 1 nao ganha mais o \addvspace{1.0em} do book: o sumario do manual
1404 % tem espacamento uniforme, sem folga extra antes da secao primaria. As
1405 % penalidades ficam, para que uma entrada de capitulo nao seja abandonada
1406 % sozinha no pe da folha.
1407 \renewcommand*\l@chapter[2]{%
1408   \addpenalty{-\@highpenalty}%
1409   \coppe@tocline{0}{\bfseries}{\MakeUppercase{#1}}{#2}%
1410   \penalty\@highpenalty}
1411 \renewcommand*\l@section[2]{%
1412   \coppe@tocline{1}{\normalfont}{\MakeUppercase{#1}}{#2}}
1413 \renewcommand*\l@subsection[2]{%
1414   \coppe@tocline{2}{\normalfont\bfseries}{#1}{#2}}
1415 \renewcommand*\l@subsubsection[2]{%
1416   \coppe@tocline{3}{\normalfont\bfseries\itshape}{#1}{#2}}
1417 \renewcommand*\l@paragraph[2]{%
1418   \coppe@tocline{4}{\normalfont\itshape}{#1}{#2}}
1419 % --- Captions (manual2024 2.10-2.11 / proposta P6): smaller font, ABNT
1420 % "Figura 1 -- titulo" travessao separator (sources handled by \source). ---
1421 \RequirePackage{caption}
1422 % justification=centering makes MULTI-LINE captions centre like the one-line
1423 % ones. Without it, caption's singlelinecheck centres a caption that fits on one
1424 % line and justifies one that wraps, so in a document with mixed caption lengths
1425 % a couple of tables looked left-aligned while the rest looked centred -- which
1426 % is what a recent review noticed about tables 5.1 and 5.2. 2.10 does not
1427 % prescribe the alignment; it does not tolerate two alignments either.
1428 \captionsetup{font=footnotesize,labelsep=endash,justification=centering}
1429 % subcaption (NÃO subfig/subfigure) for subfigures / subtables; load right after
1430 % caption so it picks up the ABNT captionsetup above.
1431 \RequirePackage{subcaption}
1432 % --- Floats: caption on TOP and a mandatory "Fonte:" line at the bottom.
1433 % manual2024 2.10-2.11 (R46/R51: identification on top; R48/R52: the source
1434 % at the bottom is mandatory). The float package's plaintop style forces the
1435 % caption above the body for figure, table and the new quadro float. -----
1436 \RequirePackage{float}
1437 \floatstyle{plaintop}
1438 \restylefloat{figure}
1439 \restylefloat{table}
1440 % New "quadro" float -- "Quadro" (pt) / "Frame" (en) (R53) -- with its own list
1441 % file (.loq). A quadro's data is bordered on all sides (drawn by the author),
1442 % unlike a tabela (rules on top and bottom only).
1443 \providecommand{\quadroname}{\coppestring{frame}}
1444 \providecommand{\listquadroname}{\coppestring{listframe}}
1445 \newfloat{quadro}{htbp}{loq}[chapter]
1446 \floatname{quadro}{\quadroname}

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1447 \providecommand{\quadroautorefname}{\quadroname}
1448 % Format the Lista de Quadros entries exactly like the Lista de Figuras ones
1449 % (float's own \l@quadro lays the number/title out incorrectly).
1450 \let\l@quadro\l@figure
1451 \newcommand\listofquadros{%
1452   \chapter*{\listquadroname}%
1453   \coppe@listtoc{\listquadroname}%
1454   \@mkboth{\MakeUppercase\listquadroname}{\MakeUppercase\listquadroname}%
1455   \@starttoc{loq}}
1456 % \source / \fonte: the illustration source (ABNT-mandatory). Typeset below the
1457 % float as a smaller, single-spaced, centred line; it does NOT advance any
1458 % caption counter. \cpsource/\cpfonte are clash-safe aliases (the rename rule);
1459 % the label is "Fonte" (pt) / "Source" (en).
1460 \newcommand{\cpsourcenamename}{\coppestring{source}}
1461 \newcommand{\cpsource}[1]{%
1462   \par\addvspace{\smallskipamount}%
1463   {\footnotesize\singlespacing\centering\cpsourcenamename: #1\par}}
1464 \let\cpfonte\cpsource
1465 \@ifundefined{source}{\let\source\cpsource}{}%
1466 \@ifundefined{fonte}{\let\fonte\cpsource}{}%
1467 % \newcoppefloat{env}{caption name}{list title}: declare an extra caption-on-top
1468 % float (e.g. "picture") that supports \source and gains a \listof<env>s command.
1469 % \floatstyle is NOT wrapped in a group here: \newfloat defines the environment
1470 % locally, so a group around it threw the new environment away as soon as it
1471 % closed and the document then failed with "Environment undefined". plaintop is
1472 % the class-wide style anyway, set once above and never changed, so leaving it
1473 % selected costs nothing.
1474 % The new float is numbered WITHIN THE CHAPTER, like figure, table and quadro,
1475 % and its list entries are laid out by \l@figure -- float's own \l@<env> runs
1476 % the number into the title and drops the leaders, which is what the Lista de
1477 % Mapas of the example showed the first time.
1478 \newcommand{\newcoppefloat}[3]{%
1479   \floatstyle{plaintop}\newfloat{#1}{htbp}{lo#1}[chapter]%
1480   \floatname{#1}{#2}%
1481   \expandafter\let\csname l@#1\endcsname\l@figure
1482   \expandafter\providecommand\csname #1autorefname\endcsname{#2}%
1483   \expandafter\newcommand\csname listof#1s\endcsname{%
1484     \chapter*{#3}\coppe@listtoc{#3}%
1485     \@mkboth{\MakeUppercase{#3}}{\MakeUppercase{#3}}%
1486     \@starttoc{lo#1}}}
1487 % --- Footnotes (manual2024 / R16, R8, R21): single spacing and a 5 cm rule
1488 % (filete) from the left margin; footnotesize is the book default. -----
1489 \let\coppe@orig@footnotetext\@footnotetext
1490 \long\def\@footnotetext#1{\coppe@orig@footnotetext{\singlespacing#1}}
1491 \renewcommand\footnoterule{\kern-3\p@\hrule\@width 5cm\kern 2.6\p@}
1492 % --- alineas: list keyed by lowercase letters a) b) c); subitems with "--"
1493 % (manual2024 2.6 / proposta P24). -----
1494 \RequirePackage{enumitem}
1495 \newlist{alineas}{enumerate}{2}
1496 \setlist{alineas,1}{label=\alph*},leftmargin=1.8em,itemsep=0pt}
1497 \setlist{alineas,2}{label=--,leftmargin=1.8em,itemsep=0pt}
1498 % --- Page numbering: arabic in the top OUTER corner on textual pages
1499 % (manual2024 2.7 / proposta P25-P26). Pre-textual numbering is set by
1500 % \frontmatter (unnumbered by default; roman with the numeraisromanos

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1501 %      class option). -----
1502 % Every list goes through \@starttoc, which allocates a stream the first time
1503 % it sees an extension. That is where the ceiling is hit in practice, so that
1504 % is where the class checks -- one \ifnum before the allocation, and an error
1505 % that names the three ways out instead of the kernel's bare "No room".
1506 %
1507 % \count17 is the write allocation counter, and 15 is the last usable stream.
1508 % Under LuaTeX there are 128 and the counter is managed differently, so the
1509 % check is skipped there; the same is true when morewrites is in charge.
1510 % The wrap waits for \begin{document} on purpose: hyperref redefines
1511 % \@starttoc as well, and it is loaded at the very end of this class, so a
1512 % \let taken here would be thrown away.
1513 \newcommand\coppe@checkwrites[1]{%
1514   \ifdefined\directlua\else
1515     \if@coppemorewrites\else
1516       \@ifundefined{tf@#1}{%
1517         \ifnum\count17>14
1518           \ClassError{coppe}{Acabaram os 16 fluxos de escrita do TeX}{%
1519             Este trabalho pede mais listas do que o pdfTeX consegue manter
1520             abertas ao mesmo tempo.\MessageBreak
1521             Tres saidas, nesta ordem de preferencia:\MessageBreak
1522             (1) compile com lualatex, que tem 128 fluxos;\MessageBreak
1523             (2) passe a opcao 'morewrites' a classe coppe;\MessageBreak
1524             (3) remova uma lista que o trabalho nao usa.\MessageBreak
1525             A lista que faltou agora foi a de extensao '#1', mas o problema
1526             nao e dela: e a soma de todas.}%
1527           \fi}{}%
1528     \fi
1529   \fi}
1530 % The wrapping itself lives in a macro of its own rather than inside the hook:
1531 % \AtBeginDocument stores its argument by expansion, and a parameter written
1532 % there does not survive it -- the document died with "Parameters must be
1533 % numbered consecutively" at \begin{document}. One level of nesting, one
1534 % doubling, and the hook only has to call it.
1535 \newcommand\coppe@wrapstarttoc{%
1536   \let\coppe@orig@starttoc\@starttoc
1537   % \numberline mede o indicativo (ver \coppe@tocnumberline) apenas no sumario:
1538   % a lista de figuras e a de tabelas numeram ilustracoes, nao secos, e nao tem
1539   % por que compartilhar a coluna.
1540   \def\@starttoc##1{%
1541     \coppe@checkwrites{##1}%
1542     \begingroup
1543       \def\coppe@@ext{##1}\def\coppe@@toc{toc}%
1544       \ifx\coppe@@ext\coppe@@toc \let\numberline\coppe@tocnumberline \fi
1545       \coppe@orig@starttoc{##1}%
1546     \endgroup}}
1547 \AtBeginDocument{\coppe@wrapstarttoc}
1548 \RequirePackage{fancyhdr}
1549 % \headheight is set to 15pt by geometry up above (see the geometry call),
1550 % which is what fancyhdr needs (>= 14.49998pt) for the running head.
1551 % 2.2(b) lists what is set in "fonte menor e uniforme": long quotations,
1552 % footnotes, PAGINATION, and the captions of illustrations and tables. The
1553 % folio was the one item on that list still coming out at the body size of 12pt.
1554 % It is \footnotesize now, which is the same smaller size the other three

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1555 % already use -- "uniforme" is part of the requirement, not just "menor".
1556 % Pre-textual lists reach the sumario only under the listasnosumario option
1557 % (see the option's declaration for the reason). Post-textual material calls
1558 % \addcontentsline directly and is unaffected.
1559 \newcommand\coppe@listtoc[1]{%
1560   \if@coppe@listastoc\addcontentsline{toc}{chapter}{#1}\fi}
1561 % \coppe@glosspre marks the two PRE-textual uses of the glossary environment --
1562 % the list of abbreviations and the list of symbols -- apart from the genuine
1563 % post-textual Glossario, which shares the same environment but must appear in
1564 % the sumario.
1565 \newif\if@coppeglosspre \@coppeglossprefalse
1566 \newcommand\coppe@folio{{\footnotesize\thepage}}
1567 \fancypagestyle{coppe}{{fancyhf}{}\fancyhead[R0,LE]{\coppe@folio}}%
1568 \renewcommand{\headrulewidth}{0pt}}
1569 % Blank pages inserted by \cleardoublepage (two-sided) must be empty: no page
1570 % number may leak onto them (e.g. the verso right after the cover).
1571 \renewcommand{\cleardoublepage}{%
1572   \clearpage
1573   \if@twoside
1574     \ifodd\c@page\else
1575       \hbox{}\thispagestyle{empty}\newpage
1576       \if@twocolumn\hbox{}\newpage\fi
1577     \fi
1578   \fi}
1579 \def\CoppeTeX{{\rm C\kern-.05em{\sc o\kern-.025em p\kern-.025em
1580 p\kern-.025em e}}\kern-.08em
1581 T\kern-.1667em\lower.5ex\hbox{E}\kern-.125emX\spacefactor1000}

1582 \newboolean{maledoc}
1583 \setboolean{maledoc}{false}
1584 % maledoc diz apenas que o substantivo do tipo de documento e masculino
1585 % ("Exame de Qualificacao ... apresentado", "SUBMETIDO"): serve a concordancia,
1586 % e nada mais. Quem NAO vai a deposito -- e portanto nao tem ficha
1587 % catalografica nem folha da Coleta CAPES -- e o exame de qualificacao, que e
1588 % outra coisa. As duas condicoes coincidem hoje por acaso, e por isso ganham
1589 % cada uma o seu booleano.
1590 \newif\if@coppeexame \@coppeexamefalse
1591 %
1592 % Class options.
1593 % Multilingual model (CoppeTeX 4.x, branch nlinguas). The class holds three
1594 % fixed language slots:
1595 % * main      -- the language of the body, captions, and the main abstract.
1596 %               Set by a class option: brazilian (default), english,
1597 %               spanish, french, italian, or any user-supplied language
1598 %               that ships a coppe-lang-<name>.def file.
1599 % * foreign   -- the language used by the foreignabstract environment and
1600 %               by \foreign@... registers. Defaults to english (or to
1601 %               brazilian when main=english) for back-compat.
1602 % * third     -- optional, loaded on demand with \usecoppelanguage{<name>}
1603 %               in the preamble (for citations or quotations in a third
1604 %               language). Has no class option of its own.
1605 % Note: the legacy \if@english boolean (defined and toggled here in
1606 % v3.x for backward compatibility) was removed in v4.0 -- nothing
1607 % inside the class or in the biblatex style files (.bbx/.cbx/.dbx/.lbr)
1608 % tests it. Code that needs to branch on the main language should test

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1609 % \ifx\coppe@mainlang\coppe@englishstr instead.
1610 \def\coppe@mainlang{brazilian}
1611 \def\coppe@foreignlang{english}
1612 \DeclareOption{brazilian}{%
1613   \def\coppe@mainlang{brazilian}\def\coppe@foreignlang{english}}
1614 \DeclareOption{english}{%
1615   \def\coppe@mainlang{english}\def\coppe@foreignlang{brazilian}}
1616 \DeclareOption{spanish}{%
1617   \def\coppe@mainlang{spanish}\def\coppe@foreignlang{english}}
1618 \DeclareOption{french}{%
1619   \def\coppe@mainlang{french}\def\coppe@foreignlang{english}}
1620 \DeclareOption{italian}{%
1621   \def\coppe@mainlang{italian}\def\coppe@foreignlang{english}}
1622 \DeclareOption{msc}{%
1623   \newcommand{\@degree}{M.Sc.}
1624   \newcommand{\@degreeName}{Mestrado}
1625   \newcommand{\local@degName}{Mestre}
1626   \newcommand{\foreign@degName}{Master}
1627   \newcommand\local@doctype{Disserta{\c c}{\~ a}o}
1628   \newcommand\foreign@doctype{Dissertation}
1629 }
1630 \DeclareOption{dsceexam}{%
1631   \newcommand{\@degree}{D.Sc.}
1632   \newcommand{\@degreeName}{Doutorado}
1633   \newcommand{\local@degName}{Doutor}
1634   \newcommand{\foreign@degName}{Doctor}
1635   \setboolean{maledoc}{true}\@coppeexamtrue
1636   \newcommand\local@doctype{Exame de Qualifica{\c c}{\~ a}o}
1637   \newcommand\foreign@doctype{Qualifying Exam}
1638 }
1639 \DeclareOption{mscexam}{%
1640   \newcommand{\@degree}{M.Sc.}
1641   \newcommand{\@degreeName}{Mestrado}
1642   \newcommand{\local@degName}{Mestre}
1643   \newcommand{\foreign@degName}{Master}
1644   \setboolean{maledoc}{true}\@coppeexamtrue
1645   \newcommand\local@doctype{Exame de Qualifica{\c c}{\~ a}o}
1646   \newcommand\foreign@doctype{Qualifying Exam}
1647 }
1648 \DeclareOption{dsc}{%
1649   \newcommand{\@degree}{D.Sc.}
1650   \newcommand{\@degreeName}{Doutorado}
1651   \newcommand{\local@degName}{Doutor}
1652   \newcommand{\foreign@degName}{Doctor}
1653   \newcommand\local@doctype{Tese}
1654   \newcommand\foreign@doctype{Thesis}
1655 }
1656 \newif\if@coppenumeric\@coppenumericfalse
1657 \DeclareOption{numbers}{\@coppenumerictrue}
1658 % numeraisromanos: show roman numerals on the pre-textual pages. Default: the
1659 % pre-textual pages are counted but not numbered (manual2024 2.7).
1660 \newif\if@copperoman\@copperomanfalse
1661 \DeclareOption{numeraisromanos}{\@copperomantrue}

```

Default line spacing is one-and-a-half; the doublespacing option switches to dou-

ble spacing. Since `setspace` is loaded at the end of the class (after `hyperref`), the actual `\onehalfspacing/\doublespacing` call is deferred: the option only flips a flag, which the late-load block reads.

```

1662 \newif\if@coppedoublespace\@coppedoublespacefalse
1663 \DeclareOption{doublespacing}{\@coppedoublespacetrue}
1664 % Two-sided layout is opt-in since v4.1: the 2026 manual dropped the verso
1665 % margins and the deposit is digital (CEPG 246/2023). Given \texttt{twoside},
1666 % the mirrored 3cm inner / 2cm outer layout is restored after \ProcessOptions.
1667 % With no \fichacatalografica file given, the folha adicional leaves a framed
1668 % blank where the ficha goes. The rascunhoficha option fills that blank with the
1669 % class's own LaTeX-composed ficha instead -- useful while writing, never valid
1670 % for deposit: the official ficha comes from the SiBI generator.
1671 % PDF/A output. Opt-in for now: 2.2(d) of the 2026 manual requires the deposited
1672 % file to be PDF/A, but the constraints (embedded fonts, colour profile, no
1673 % encryption, restricted transparency) can trip up a document that pulls in
1674 % arbitrary images or packages, and breaking existing theses to enforce it would
1675 % be worse than letting the author switch it on when depositing.
1676 \newif\if@coppepdfa \@coppepdfafalse
1677 \DeclareOption{pdfa}{\@coppepdfatrue}
1678 % A opcao 'assinaturas' punha uma regua para assinatura a mao acima do nome de
1679 % cada membro da banca. Com a entrega so digital (Resolucao CEPG n. 246/2023)
1680 % nao ha o que assinar a mao, e a v4.1 tirou as reguas. A opcao continua aceita,
1681 % sem efeito, para nao quebrar documento escrito antes disso.
1682 % Quem examina o trabalho e a BANCA EXAMINADORA, e a folha de aprovacao lista
1683 % quem examinou. O orientador nao e, por definicao, examinador: ele conduziu o
1684 % trabalho. Por isso a folha lista, por padrao, so os examinadores.
1685 %
1686 % Ha Programas em que o orientador integra a banca e assina a aprovacao. Para
1687 % esses existe 'orientadorexamina': com ela, orientadores e coorientadores
1688 % voltam para a lista, na frente dos examinadores, na ordem que a 3.1.2.1.3(e)
1689 % prescreve para o presidente da banca.
1690 %
1691 % Ate a v4.1 nao havia escolha: os orientadores saiam sempre, e um Programa que
1692 % nao os quisesse ali tinha de nao os declarar -- o que os tirava tambem da capa
1693 % e da folha de rosto, onde eles tem de estar.
1694 \newif\if@coppeorientexam \@coppeorientexamfalse
1695 \DeclareOption{orientadorexamina}{\@coppeorientexamtrue}
1696 \DeclareOption{assinaturas}{\ClassWarningNoLine{coppe}{A opcao 'assinaturas'
1697   nao faz mais nada: a folha de aprovacao nao tem mais linhas de assinatura,
1698   porque a entrega e digital desde a Resolucao CEPG n. 246/2023}}
1699 % As paginas de resumo listam os ORIENTADORES abaixo do titulo. O coorientador
1700 % nao aparece ali, e isso e tradicao da COPPE, nao exigencia do manual: a
1701 % 3.1.2.1.4 descreve o resumo e nao lista quem assina a orientacao. Como ha
1702 % Programas que preferem ve-lo, a opcao acrescenta os coorientadores nas tres
1703 % paginas de resumo, no mesmo formato da folha de rosto. Sem a opcao nada muda.
1704 \newif\if@coppecoadvresumo \@coppecoadvresumofalse
1705 \DeclareOption{coorientador}{\@coppecoadvresumotrue}
1706 % 3.1.2.1.6 says the sumario enumerates the divisions "na mesma ordem e grafia
1707 % em que a materia nele se sucede" and adds "inclusive os elementos
1708 % pos-textuais" -- and then settles the question by pointing at its own sumario
1709 % as the model. That sumario opens at "1 INTRODUCAO": the lists of figures, of
1710 % tables, of abbreviations and of symbols come BEFORE the sumario and are not
1711 % listed in it, which is also the rule of NBR 6027 for pre-textual elements.
1712 % The class was adding all of them, so every thesis carried six entries the

```

```

1713 % manual's own example does not have. They come out. Post-textual elements --
1714 % Referencias, Glossario, Apendices, Anexos, Indice -- stay, as required.
1715 %
1716 % Programas that want the old behaviour ask for it by name.
1717 \newif\if@coppelistastoc \@coppelistastocfalse
1718 \DeclareOption{listasnosumario}{\@coppelistastoctrue}
1719 \newif\if@copperascunho \@copperascunhofalse
1720 \DeclareOption{rascunhoficha}{\@copperascunhotrue}
1721 \newif\if@coppetwoside \@coppetwosidefalse
1722 \DeclareOption{twoside}{\@coppetwosidetrue}
1723 \DeclareOption{oneside}{\@coppetwosidefalse}
1724 % --- Fluxos de escrita -----
1725 % TeX has SIXTEEN \write streams and no more. Every list costs one for the
1726 % whole run: the .toc, the .lof, the .lot, and one for each list this class
1727 % adds (.loq for quadros, .lol for programas, .loa for algoritmos), plus the
1728 % .abx and .syx behind the lists of abbreviations and of symbols, the .idx of
1729 % the index, hyperref's .out, biblalex's .bcf and the kernel's .aux and friends.
1730 % A work that asks for all of them sits exactly at the ceiling.
1731 %
1732 % The kernel's failure in that situation -- "No room for a new \write" -- names
1733 % neither the cause nor the way out, and lands on whichever list happens to be
1734 % requested last, which is usually \tableofcontents and has nothing to do with
1735 % it. The class therefore checks before the allocation and says what is
1736 % happening (see \coppe@starttoc below).
1737 %
1738 % There are three ways out and the class chooses none of them for the author:
1739 %
1740 % * compile with LuaLaTeX. It has 128 streams; nothing else has to change,
1741 %   and the class supports it.
1742 % * let the class load 'morewrites', which multiplexes the streams so the
1743 %   ceiling disappears under pdfLaTeX.
1744 % * drop a list the work does not use. Each one returns a stream.
1745 %
1746 % The DEFAULT is the second one, and it is on. An error telling the author to
1747 % pass an option is advice for someone who already knows: a student two days
1748 % from the deposit reads "No room for a new \write" and concludes the class is
1749 % broken. So under pdfTeX the class loads morewrites by itself.
1750 %
1751 % Three things keep that from becoming a burden:
1752 %
1753 % * it is loaded ONLY IF INSTALLED. Missing, the class carries on without it
1754 %   and warns; morewrites does not become a hard dependency of a work that
1755 %   would never have reached the ceiling anyway.
1756 % * 'semmorewrites' turns it off, for whoever hits a conflict. morewrites
1757 %   patches \newwrite, \openout and \write and buffers what is written until
1758 %   shipout, so a package that writes a file and reads it back in the SAME run
1759 %   can misbehave. That is the real cost of this default, and it is why the
1760 %   escape exists.
1761 % * 'morewrites' still exists and now means "I insist": with it, a missing
1762 %   package is an error rather than a warning.
1763 %
1764 % Under LuaTeX none of this applies and nothing is loaded.
1765 \newif\if@coppemorewrites \@coppemorewritetrue
1766 \newif\if@coppemwforced \@coppemwforcedfalse

```

```

1767 \DeclareOption{morewrites}{\@coppemorewritestrue\@coppemwforcedtrue}
1768 \DeclareOption{semmorewrites}{\@coppemorewritestruefalse}
1769 % A 2.2(b) fixa a cor, o corpo 12 e o corpo menor uniforme dos quatro itens
1770 % listados nela, e nao diz uma palavra sobre familia de fonte -- nem a Norma da
1771 % COPPE. A escolha e do autor. Desde a v4.1 o padrao da classe e a SEM SERIFA,
1772 % que e como o proprio manual do SiBI, que fixa a regra, esta composto;
1773 % 'comserifa' volta para a serifada, que e a do LaTeX. 'semserifa' continua
1774 % aceita, agora sem efeito, para nao quebrar documento escrito antes da v4.1.
1775 \newif\if@coppesansserif \@coppesansseriftrue
1776 \DeclareOption{semserifa}{\@coppesansseriftrue}
1777 \DeclareOption{comserifa}{\@coppesansseriffalse}

1778 \ProcessOptions\relax
1779 % A troca da familia padrao vai aqui porque so agora se sabe se a opcao foi
1780 % pedida; o lmodern ja esta carregado la em cima e traz as duas familias, a
1781 % serifada e a sem serifa, ambas vetoriais -- que e o que o PDF/A exige.
1782 % Quem quiser uma sem serifa DIFERENTE (Helvetica, por exemplo) carrega o
1783 % pacote da fonte no preambulo e refaz \familydefault ali; o padrao existe
1784 % para o caso comum, em que ninguem quer escolher fonte.
1785 \if@coppesansserif
1786   \renewcommand{\familydefault}{\sfdefault}%
1787 \fi
1788 % morewrites has to come before anything else allocates, so it goes first
1789 % thing after the options are read.
1790 \if@coppemorewrites
1791   \ifdefined\directlua
1792     % Under LuaTeX there is nothing to solve: 128 streams already. The flag is
1793     % cleared too, so the \@starttoc guard further down knows the ceiling is not
1794     % in play here either.
1795     \ClassInfo{coppe}{morewrites dispensado: o LuaTeX ja tem 128 fluxos}%
1796     \@coppemorewritestruefalse
1797   \else
1798     \IfFileExists{morewrites.sty}
1799       {\RequirePackage{morewrites}}
1800       {\if@coppemwforced
1801         \ClassError{coppe}{Opcao 'morewrites' pedida, mas morewrites.sty nao
1802           esta instalado}{Instale o pacote morewrites, ou compile com lualatex,
1803           que nao precisa dele.}%
1804         \else
1805           \ClassWarningNoLine{coppe}{O pacote morewrites nao esta instalado.\MessageBreak
1806             Sem ele o pdfTeX so aguenta 16 fluxos de escrita, e um trabalho\MessageBreak
1807             com todas as listas passa disso. Se a compilacao parar por falta\MessageBreak
1808             de fluxo, instale morewrites ou compile com lualatex}%
1809           \@coppemorewritestruefalse
1810         \fi}%
1811   \fi
1812 \fi
1813 % \LoadClass and geometry ran before \ProcessOptions (see above), so the
1814 % two-sided layout can only be applied here. geometry reads left/right as
1815 % inner/outer once twoside is on, so the 3cm/2cm pair set above becomes the
1816 % mirrored gutter with no further arithmetic. \@twoside true is what book's own
1817 % twoside option sets, and is what \cleardoublepage and fancyhdr test.
1818 % In one-sided mode fancyhdr treats every page as odd, so the \fancyhead[R0,LE]
1819 % declarations elsewhere already put the folio in the top right corner of every
1820 % folha, as 2.7 requires.

```

```

1821 \if@coppetwoside
1822 \@twosidettrue
1823 \geometry{twoside}%
1824 \fi
1825 % Built-in language names (used by the loader and the dispatcher below).
1826 \def\coppe@@brazilianstr{brazilian}
1827 \def\coppe@@englishstr{english}
1828 % Load Babel with the foreign language first and the main language last
1829 % (Babel treats the LAST option in the list as the main language). With
1830 % pt-main this produces [english,brazilian]{babel} and with en-main
1831 % [brazilian,english] -- identical to the historical \if@english calls.
1832 \edef\coppe@babelopts{\coppe@foreignlang,\coppe@mainlang}
1833 % Some Babel modules change LaTeX beyond translating captions, and one of
1834 % those changes breaks the norm. With Spanish as the MAIN language,
1835 % babel-spanish redefines \numberline to put a period after the section
1836 % number, so the sumario reads "2.1. SECCION SECUNDARIA" while the section
1837 % heading itself -- which the class formats -- reads "2.1 SECCION SECUNDARIA".
1838 % The 2.6 separates the indicative from the title by a space (NBR 6024) and
1839 % the 3.1.2.1.6 wants the sumario to reproduce the text; the period fails
1840 % both. 'es-nosectiondot' turns off that single adjustment and leaves the rest
1841 % of babel-spanish (quoting, shorthands, roman numerals, captions) alone.
1842 % \coppe@babelextra@<lang> is the general hook: it applies to the MAIN
1843 % language only, which is where these layout modules act, and it is kept out
1844 % of \coppe@babelopts because the language-pack loader below iterates that
1845 % list and would look for coppe-lang-es-nosectiondot.def.
1846 \@namedef{coppe@babelextra@spanish}{,es-nosectiondot}
1847 \let\coppe@babelload\coppe@babelopts
1848 \@ifundefined{coppe@babelextra@\coppe@mainlang}{}{%
1849 \edef\coppe@babelload{\coppe@babelload\@nameuse{coppe@babelextra@\coppe@mainlang}}}
1850 \expandafter\RequirePackage\expandafter[\coppe@babelload]{babel}
1851 % --- Multilingual string dispatcher -----
1852 \copperdefstring{<lang>}{<key>}{<value>} adds one entry to the table for
1853 % language <lang>. The three retrievers below look strings up either at the
1854 % current babel language (\coppestring -- follows \selectlanguage), or pinned
1855 % to the main slot (\coppemainstring) or the foreign slot (\coppeforeignstring).
1856 \providecommand{\copperdefstring}[3]{%
1857 \expandafter\gdef\csname coppe@str@#1@#2\endcsname{#3}}
1858 \providecommand{\coppestring}[1]{\@nameuse{coppe@str@\language @#1}}
1859 \providecommand{\coppemainstring}[1]{\@nameuse{coppe@str@\coppe@mainlang @#1}}
1860 \providecommand{\coppeforeignstring}[1]{\@nameuse{coppe@str@\coppe@foreignlang @#1}}
1861 % Load an extra language pack at any point in the preamble (third-language
1862 % slot for citations or quotations beyond the main+foreign pair).
1863 \providecommand{\usecoppelanguage}[1]{%
1864 \def\coppe@@cur{#1}%
1865 \ifx\coppe@@cur\coppe@@brazilianstr\else
1866 \ifx\coppe@@cur\coppe@@englishstr\else
1867 \InputIfFileExists{coppe-lang-#1.def}{}
1868 {\ClassWarning{coppe}{Language pack coppe-lang-#1.def not found}}%
1869 \fi\fi}
1870 % \titlein{<lang>}{<text>}: record the title in any language. Will be
1871 % consulted by the cover/folha renderers once they are migrated in step 2.
1872 % In step 1 it is just a no-op store -- the historical \title / \foreigntitle
1873 % still drive every visible site, so behavior for pt/en is unchanged.
1874 \providecommand{\titlein}[2]{%

```

```

1875 \expandafter\gdef\csname coppe@title@#1\endcsname{#2}}
1876 % --- Built-in string tables: brazilian + english. The values mirror the
1877 % literals currently scattered through the class; step-1 only populates the
1878 % tables -- no site reads them yet, so the rendered output is unchanged. ----
1879 \copperdefstring{brazilian}{appendix}{Ap\~endice}
1880 \copperdefstring{english} {appendix}{Appendix}
1881 \copperdefstring{brazilian}{annex} {Anexo}
1882 \copperdefstring{english} {annex} {Annex}
1883 \copperdefstring{brazilian}{frame} {Quadro}
1884 \copperdefstring{english} {frame} {Frame}
1885 \copperdefstring{brazilian}{listframe} {Lista de Quadros}
1886 \copperdefstring{english} {listframe} {List of Frames}
1887 \copperdefstring{brazilian}{source} {Fonte}
1888 \copperdefstring{english} {source} {Source}
1889 \copperdefstring{brazilian}{program} {Programa}
1890 \copperdefstring{english} {program} {Program}
1891 \copperdefstring{brazilian}{listprogram} {Lista de Programas}
1892 \copperdefstring{english} {listprogram} {List of Programs}
1893 \copperdefstring{brazilian}{references} {Refer\~encias}
1894 \copperdefstring{english} {references} {References}
1895 \copperdefstring{brazilian}{in:} {em}
1896 \copperdefstring{english} {in:} {in}
1897 \copperdefstring{brazilian}{art} {Ilustra\c{c}\~ao}
1898 \copperdefstring{english} {art} {Art}
1899 \copperdefstring{brazilian}{platform} {Plataforma}
1900 \copperdefstring{english} {platform} {Platform}
1901 \copperdefstring{brazilian}{directedby} {Dire\c{c}\~ao de}
1902 \copperdefstring{english} {directedby} {Directed by}
1903 \copperdefstring{brazilian}{producedby} {Produ\c{c}\~ao de}
1904 \copperdefstring{english} {producedby} {Produced by}
1905 \copperdefstring{brazilian}{listabbreviation}{Lista de Abreviaturas e Siglas}
1906 \copperdefstring{english} {listabbreviation}{List of Abbreviations and Acronyms}
1907 \copperdefstring{brazilian}{listsymbol} {Lista de S{\~i}mbolos}
1908 \copperdefstring{english} {listsymbol} {List of Symbols}
1909 \copperdefstring{brazilian}{glossary} {Gloss{\~a}rio}
1910 \copperdefstring{english} {glossary} {Glossary}
1911 \copperdefstring{brazilian}{advisor} {Orientador}
1912 \copperdefstring{english} {advisor} {Advisor}
1913 \copperdefstring{brazilian}{advisors} {Orientadores}
1914 \copperdefstring{english} {advisors} {Advisors}
1915 \copperdefstring{brazilian}{dept} {Programa}
1916 \copperdefstring{english} {dept} {Department}
1917 \copperdefstring{brazilian}{approvedmale} {Aprovado por}
1918 \copperdefstring{brazilian}{approvedfemale}{Aprovada por}
1919 \copperdefstring{brazilian}{approvedonmale} {Aprovado em}
1920 \copperdefstring{brazilian}{approvedonfemale}{Aprovada em}
1921 \copperdefstring{brazilian}{keywordsname} {Palavras-chave}
1922 \copperdefstring{brazilian}{volumename} {Volume}
1923 \copperdefstring{brazilian}{coadvisor} {Coorientador}
1924 \copperdefstring{brazilian}{coadvisors} {Coorientadores}
1925 \copperdefstring{brazilian}{ofname} {de}
1926 \copperdefstring{brazilian}{researchline}{Linha de pesquisa}
1927 \copperdefstring{brazilian}{tbddate} {a ser determinada}
1928 \copperdefstring{english} {approvedmale} {Approved by}

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```

1929 \copperdefstring{english} {approvedfemale}{Approved by}
1930 \copperdefstring{english} {approvedonmale} {Approved on}
1931 \copperdefstring{english} {approvedonfemale}{Approved on}
1932 \copperdefstring{english} {keywordsname} {Keywords}
1933 \copperdefstring{english} {volumename} {Volume}
1934 \copperdefstring{english} {coadvisor} {Co-advisor}
1935 \copperdefstring{english} {coadvisors} {Co-advisors}
1936 \copperdefstring{english} {ofname} {of}
1937 \copperdefstring{english} {researchline}{Research line}
1938 \copperdefstring{english} {tbddate} {to be determined}
1939 % Institutional names: always Portuguese on the cover/folha (UFRJ template).
1940 \copperdefstring{brazilian}{universityname}{Universidade Federal do Rio de Janeiro}
1941 \copperdefstring{english} {universityname}{Universidade Federal do Rio de Janeiro}
1942 \copperdefstring{brazilian}{cityname} {Rio de Janeiro}
1943 \copperdefstring{english} {cityname} {Rio de Janeiro}
1944 \copperdefstring{brazilian}{statename} {RJ}
1945 \copperdefstring{english} {statename} {RJ}
1946 \copperdefstring{brazilian}{countryname} {Brasil}
1947 \copperdefstring{english} {countryname} {Brasil}
1948 % Month names.
1949 \copperdefstring{brazilian}{monthname1} {Janeiro}
1950 \copperdefstring{brazilian}{monthname2} {Fevereiro}
1951 \copperdefstring{brazilian}{monthname3} {Mar{\c c}o}
1952 \copperdefstring{brazilian}{monthname4} {Abril}
1953 \copperdefstring{brazilian}{monthname5} {Maio}
1954 \copperdefstring{brazilian}{monthname6} {Junho}
1955 \copperdefstring{brazilian}{monthname7} {Julho}
1956 \copperdefstring{brazilian}{monthname8} {Agosto}
1957 \copperdefstring{brazilian}{monthname9} {Setembro}
1958 \copperdefstring{brazilian}{monthname10}{Outubro}
1959 \copperdefstring{brazilian}{monthname11}{Novembro}
1960 \copperdefstring{brazilian}{monthname12}{Dezembro}
1961 \copperdefstring{english}{monthname1} {January}
1962 \copperdefstring{english}{monthname2} {February}
1963 \copperdefstring{english}{monthname3} {March}
1964 \copperdefstring{english}{monthname4} {April}
1965 \copperdefstring{english}{monthname5} {May}
1966 \copperdefstring{english}{monthname6} {June}
1967 \copperdefstring{english}{monthname7} {July}
1968 \copperdefstring{english}{monthname8} {August}
1969 \copperdefstring{english}{monthname9} {September}
1970 \copperdefstring{english}{monthname10}{October}
1971 \copperdefstring{english}{monthname11}{November}
1972 \copperdefstring{english}{monthname12}{December}
1973 % algorithm2e package option for this language (empty -> english default).
1974 \copperdefstring{brazilian}{algorithm2eopt}{portuguese}
1975 \copperdefstring{english} {algorithm2eopt}{ }
1976 % Babel package name for the foreignabstract \begin{otherlanguage}{...}.
1977 \copperdefstring{brazilian}{babelname}{brazilian}
1978 \copperdefstring{english} {babelname}{english}
1979 % Auto-load language packs for any non-built-in language listed in babelopts.
1980 % Built-ins (brazilian, english) ship their string tables inside coppe.cls
1981 % (defined just above) and need no extra file; others (spanish/french/italian
1982 % /...) ship as coppe-lang-<name>.def alongside the class. The dispatcher

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```

1983 % must already be defined when these run, so the loader sits AFTER the tables.
1984 \@for\coppe@@lang:=\coppe@babelopts\do{%
1985   \edef\coppe@@cur{\coppe@@lang}%
1986   \ifx\coppe@@cur\coppe@@brazilianstr\else
1987   \ifx\coppe@@cur\coppe@@englishstr\else
1988     \InputIfFileExists{coppe-lang-\coppe@@cur.def}
1989     {\ClassInfo{coppe}{Loaded language pack coppe-lang-\coppe@@cur.def}}
1990     {\ClassError{coppe}
1991      {Missing language pack 'coppe-lang-\coppe@@cur.def'.}
1992      {Install it next to coppe.cls, or pick a built-in language.}}}%
1993   \fi\fi}
1994 % Quotation marks: csquotes gives the language-aware \enquote; \citacao is its
1995 % Portuguese alias for short, inline quotations (proposta P3).
1996 \RequirePackage{csquotes}
1997 \newcommand{\citacao}[1]{\enquote{#1}}
1998 % --- Code listings & algorithms (TOD03 "tratar listagens"): the listings
1999 %   package gives the "Programa" listing (caption on top, ABNT R46) with
2000 %   \listofprogramas; algorithm2e gives algorithms. Names are babel-aware and
2001 %   the language styles below are based on the user's listagens.tex. -----
2002 \RequirePackage{xcolor}
2003 \RequirePackage{listings}
2004 \RequirePackage{fancyvrb}
2005 \DeclareFixedFont{\ttb}{T1}{txxt}{bx}{n}{10}
2006 \DeclareFixedFont{\ttm}{T1}{txxt}{m}{n}{10}
2007 \lstset{captionpos=t,extendedchars=true,basicstyle=\ttfamily,inputencoding=utf8,%
2008   literate=%
2009   {\~}{\~{a}}1 {\~}{\~{a}}1 {\~}{\~{a}}1 {\~}{\~{a}}1 {\~}{\~{e}}1%
2010   {\~}{\~{e}}1 {\~}{\~{e}}1 {\~}{\~{i}}1 {\~}{\~{i}}1 {\~}{\~{o}}1%
2011   {\~}{\~{o}}1 {\~}{\~{o}}1 {\~}{\~{u}}1 {\~}{\~{u}}1 {\~}{\~{c}}1%
2012   {\~}{\~{c}}1 {\~}{\~{A}}1 {\~}{\~{A}}1 {\~}{\~{A}}1 {\~}{\~{E}}1%
2013   {\~}{\~{E}}1 {\~}{\~{I}}1 {\~}{\~{I}}1 {\~}{\~{O}}1 {\~}{\~{O}}1 {\~}{\~{U}}1%
2014   {\~}{\~{U}}1%
2015 }
2016 % Babel-aware listing names, set after listings' own babel setup.
2017 \AtBeginDocument{%
2018   \renewcommand{\lstlistingname}{\coppestring{program}}%
2019   \renewcommand{\lstlistlistingname}{\coppestring{listprogram}}%
2020 % \listofprogramas in the coppe list style (heading + ToC + running head).
2021 \newcommand{\listofprogramas}%
2022   \chapter*{\lstlistlistingname}%
2023   \coppe@listtoc{\lstlistlistingname}%
2024   \@mkboth{\MakeUppercase\lstlistlistingname}{\MakeUppercase\lstlistlistingname}%
2025   \@starttoc{lol}}
2026 % Per-language styles + environments (python, java, xml, html, prolog).
2027 % The red hook that marks a wrapped line is built ONCE, into a box, and the
2028 % styles below insert the box. Writing the colour inline -- postbreak=\mbox{%
2029 % \textcolor{red}{\hookrightarrow}\space} -- worked for years and broke the
2030 % moment the pdfa option was switched on: pdfx puts xcolor into a converting
2031 % mode (\selectcolormodel plus \convertcolorsU/D), and a colour command run
2032 % from inside listings' line-breaking machinery then fails with "Missing
2033 % number, treated as zero" at the first wrapped line. Every listing in the
2034 % document dies, so the class's own example could not be compiled as PDF/A at
2035 % all. A box is immune: it is built here, in ordinary horizontal mode where
2036 % xcolor works normally, and listings only ever copies it.

```



```

2037 %
2038 % The \sbox waits for \begin{document} because xcolor's colours are not
2039 % defined while the class is still loading.
2040 \newsavebox\coppe@lstarrow
2041 \AtBeginDocument{\sbox\coppe@lstarrow{\textcolor{red}{\hookrightarrow}\space}}
2042 \newcommand\pythonstyle{\lstset{language=Python,basicstyle=\ttm,numbers=left,%
2043   numberstyle=\small\ttfamily,numbersep=0.5em,commentstyle=\color{purple},%
2044   morekeywords={self,as},keywordstyle=\ttb\color{rgb}{0,0,0.5},%
2045   emph={MyClass,__init__},emphstyle=\ttb\color{rgb}{0.6,0,0},%
2046   stringstyle=\color{rgb}{0,0.5,0},frame=tb,showstringspaces=false,%
2047   extendedchars=true,firstnumber=auto,inputencoding=utf8,breaklines=true,%
2048   escapechar=',postbreak=\usebox\coppe@lstarrow}}
2049 \lstnewenvironment{python}[1][\pythonstyle\lstset{#1}%
2050   \csname\@lst @SetFirstNumber\endcsname}{\csname\@lst @SaveFirstNumber\endcsname}
2051 \newcommand\pythoninline[1][\pythonstyle\lstinline!#1!}]
2052 \newcommand\pythonexternal[2][\pythonstyle\lstinputlisting[1]{#2}]}
2053 \newcommand\xmlstyle{\lstset{language=XML,basicstyle=\ttm,numbers=left,%
2054   numberstyle=\small\ttfamily,numbersep=0.5em,%
2055   morekeywords={xs:schema,xs:element,xs:sequence,xs:complexType},%
2056   keywordstyle=\ttb\color{rgb}{0,0,0.5},%
2057   emph={name,type,attributeFormDefault,elementFormDefault,xmlns:xs},%
2058   emphstyle=\ttb\color{rgb}{0.6,0,0},stringstyle=\color{rgb}{0,0.5,0},frame=tb,%
2059   showstringspaces=false,extendedchars=true,inputencoding=utf8,breaklines=true,%
2060   postbreak=\usebox\coppe@lstarrow}}
2061 \lstnewenvironment{xml}[1][\xmlstyle\lstset{#1}]{
2062 \newcommand\htmlstyle{\lstset{language=HTML,basicstyle=\ttm,numbers=left,%
2063   numberstyle=\small\ttfamily,numbersep=0.5em,%
2064   keywordstyle=\ttb\color{rgb}{0,0,0.5},emphstyle=\ttb\color{rgb}{0.6,0,0},%
2065   stringstyle=\color{rgb}{0,0.5,0},frame=tb,showstringspaces=false,%
2066   extendedchars=true,inputencoding=utf8,breaklines=true,%
2067   postbreak=\usebox\coppe@lstarrow}}
2068 \lstnewenvironment{html}[1][\htmlstyle\lstset{#1}]{
2069 \newcommand\prologstyle{\lstset{language=Prolog,basicstyle=\ttm,numbers=left,%
2070   numberstyle=\small\ttfamily,numbersep=0.5em,commentstyle=\color{purple},%
2071   keywordstyle=\ttb\color{rgb}{0,0,0.5},emph={MyClass,__init__},%
2072   emphstyle=\ttb\color{rgb}{0.6,0,0},stringstyle=\color{rgb}{0,0.5,0},frame=tb,%
2073   showstringspaces=false,extendedchars=true,firstnumber=auto,inputencoding=utf8,%
2074   breaklines=true,escapechar=',%
2075   postbreak=\usebox\coppe@lstarrow}}
2076 \lstnewenvironment{prolog}[1][\prologstyle\lstset{#1}%
2077   \csname\@lst @SetFirstNumber\endcsname}{\csname\@lst @SaveFirstNumber\endcsname}
2078 \newcommand\javastyle{\lstset{language=Java,basicstyle=\ttm,numbers=left,%
2079   numberstyle=\small\ttfamily,numbersep=0.5em,commentstyle=\color{purple},%
2080   morekeywords={self,as},keywordstyle=\ttb\color{rgb}{0,0,0.5},%
2081   emphstyle=\ttb\color{rgb}{0.6,0,0},stringstyle=\color{rgb}{0,0.5,0},frame=tb,%
2082   showstringspaces=false,extendedchars=true,firstnumber=auto,inputencoding=utf8,%
2083   breaklines=true,escapechar=',%
2084   postbreak=\usebox\coppe@lstarrow}}
2085 \lstnewenvironment{java}[1][\javastyle\lstset{#1}%
2086   \csname\@lst @SetFirstNumber\endcsname}{\csname\@lst @SaveFirstNumber\endcsname}
2087 \newcommand\javainline[1][\javastyle\lstinline!#1!}]
2088 \newcommand\javaexternal[2][\javastyle\lstinputlisting[1]{#2}]}
2089 % fancyvrb's Verbatim environment is available for program output; the
2090 % \DefineVerbatimEnvironment{gxoutput}/{gxoutputs} aliases that used to live

```

```

2091 % here were author-personal (gx prefix) and were removed in v4.0. Use, for
2092 % example,
2093 % \begin{Verbatim}[frame=lines,numbers=left,numbersep=0.5em]
2094 % ...
2095 % \end{Verbatim}
2096 % (or \DefineVerbatimEnvironment{...}{Verbatim}{...} in the manuscript preamble).
2097 \renewcommand{\theFancyVerbLine}{\small\ttfamily\arabic{FancyVerbLine}}
2098 % Algorithms: algorithm2e, language driven by the main-slot \coppemainstring
2099 % lookup (key "algorithm2eopt"). Empty means use algorithm2e's default
2100 % language (English) -- needed because algorithm2e errors on an empty option.
2101 \edef\coppe@@algotlang{\coppemainstring{algorithm2eopt}}
2102 \ifx\coppe@@algotlang\@empty
2103 \RequirePackage[linesnumbered,lined,algochapter,ruled]{algorithm2e}
2104 \else
2105 \expandafter\RequirePackage\expandafter[\coppe@@algotlang,linesnumbered,lined,algochapter,ruled]{algorithm2e}
2106 \fi
2107 % algorithm2e composes its own caption and therefore ignores the class-wide
2108 % \captionsetup{font=footnotesize,labelsep=endash}. Left alone it printed
2109 % "Algoritmo 6.1:Titulo" in bold at body size, while every other illustration
2110 % printed "Figura 6.1 -- Titulo" in footnotesize -- 2.10 asks for the
2111 % designating word, the number, a TRAVESSAO and the title, uniformly.
2112 %
2113 % The separator and the caption fonts are therefore set to match. The 'ruled'
2114 % layout (rules above and below) is kept: 2.10 prescribes where the caption and
2115 % the source go, not whether the float is delimited, and the rules read here
2116 % like the top-and-bottom delimitation a table carries.
2117 \SetAlgoCaptionSeparator{\textendash}
2118 \SetAlCapFnt{\footnotesize}
2119 \SetAlCapNameFnt{\footnotesize}
2120 \SetAlCapSty{textnormal}
2121 \SetAlCapNameSty{textnormal}
2122 \SetKwFor{Para}{para}{faça}{fim do para}
2123 \SetKwFor{Enquanto}{enquanto}{faça}{fim do enquanto}
2124 % setspace + hyperref + biblatex are loaded at the very end of the class
2125 % (see the block just before %</class>), so their hooks see every macro the
2126 % class defines, and the load order setspace -> hyperref -> biblatex is
2127 % respected.

```

`\department` This macro is used to set the author's affiliation. There are twelve options which correspond to all academic units at COPPE/UFRJ. It defines the current and the foreign names of these units.

```

2128 \newcommand\coppe@deptcode{}
2129 \newcommand\meta@deptname{}
2130 \newcommand\department[1]{%
2131 % The programme CODE is kept alongside the resolved name, and each name is
2132 % declared TWICE. \local@deptname keeps the TeX accent escapes with braces
2133 % (Computa{\c c}{\~ a}o) that the cover and the folha de rosto typeset;
2134 % \meta@deptname is the same name in literal UTF-8, and is the one that goes
2135 % into the XMP. pdfx copies whatever it is given straight through, so feeding it
2136 % the escaped form produced "Computa{c}o" in dc:description -- which is why the
2137 % subject line used to fall back to the bare programme code. Both forms are
2138 % written out here rather than derived from one another because there is no
2139 % reliable way to strip TeX accent syntax at the point where the .xmpdata is
2140 % written, and a table of thirteen entries is cheaper than that machinery.

```

```

2141 \gdef\coppe@deptcode{#1}%
2142 \ifthenelse{\equal{#1}{PEB}}{
2143   {\global\def\local@deptname{Engenharia Biom{\` e}dica}
2144   \global\def\meta@deptname{Engenharia Biomédica}
2145   \global\def\foreign@deptname{Biomedical Engineering}}{}
2146 \ifthenelse{\equal{#1}{PEC}}{
2147   {\global\def\local@deptname{Engenharia Civil}
2148   \global\def\meta@deptname{Engenharia Civil}
2149   \global\def\foreign@deptname{Civil Engineering}}{}
2150 \ifthenelse{\equal{#1}{PEE}}{
2151   {\global\def\local@deptname{Engenharia El{\` e}trica}
2152   \global\def\meta@deptname{Engenharia Elétrica}
2153   \global\def\foreign@deptname{Electrical Engineering}}{}
2154 \ifthenelse{\equal{#1}{PEM}}{
2155   {\global\def\local@deptname{Engenharia Mec{\` a}nica}
2156   \global\def\meta@deptname{Engenharia Mecânica}
2157   \global\def\foreign@deptname{Mechanical Engineering}}{}
2158 \ifthenelse{\equal{#1}{PEMM}}{
2159   {\global\def\local@deptname{Engenharia Metal{\` u}rgica e de Materiais}
2160   \global\def\meta@deptname{Engenharia Metalúrgica e de Materiais}
2161 \global\def\foreign@deptname{Metallurgical and Materials Engineering}}{}
2162 \ifthenelse{\equal{#1}{PEN}}{
2163   {\global\def\local@deptname{Engenharia Nuclear}
2164   \global\def\meta@deptname{Engenharia Nuclear}
2165   \global\def\foreign@deptname{Nuclear Engineering}}{}
2166 \ifthenelse{\equal{#1}{PEN0}}{
2167   {\global\def\local@deptname{Engenharia Oce{\` a}nica}
2168   \global\def\meta@deptname{Engenharia Oceânica}
2169   \global\def\foreign@deptname{Ocean Engineering}}{}
2170 \ifthenelse{\equal{#1}{PPE}}{
2171   {\global\def\local@deptname{Planejamento Energ{\` e}tico}
2172   \global\def\meta@deptname{Planejamento Energético}
2173   \global\def\foreign@deptname{Energy Planning}}{}
2174 \ifthenelse{\equal{#1}{PEP}}{
2175   {\global\def\local@deptname{Engenharia de Produ{\` c}{\` a}o}
2176   \global\def\meta@deptname{Engenharia de Produção}
2177   \global\def\foreign@deptname{Production Engineering}}{}
2178 \ifthenelse{\equal{#1}{PEQ}}{
2179   {\global\def\local@deptname{Engenharia Qu{\` i}mica}
2180   \global\def\meta@deptname{Engenharia Química}
2181   \global\def\foreign@deptname{Chemical Engineering}}{}
2182 \ifthenelse{\equal{#1}{PESC}}{
2183   {\global\def\local@deptname{Engenharia de Sistemas e Computa{\` c}{\` a}o}
2184   \global\def\meta@deptname{Engenharia de Sistemas e Computação}
2185   \global\def\foreign@deptname{Systems Engineering and Computer Science}}{}
2186 \ifthenelse{\equal{#1}{PET}}{
2187   {\global\def\local@deptname{Engenharia de Transportes}
2188   \global\def\meta@deptname{Engenharia de Transportes}
2189   \global\def\foreign@deptname{Transportation Engineering}}{}
2190 \ifthenelse{\equal{#1}{PENT}}{
2191   {\global\def\local@deptname{Engenharia de Nanotecnologia}
2192   \global\def\meta@deptname{Engenharia de Nanotecnologia}
2193   \global\def\foreign@deptname{Nanotechnology Engineering}}{}
2194 }

```

`\title` Used to enter the title in Brazilian Portuguese. Mirrors into `\coppe@title@brazilian` so that the per-language title slot used by the cover/folha renderers (`\coppe@selecttitle`) picks the right text for a pt-main thesis.

```
2195 \renewcommand\title[1]{%
2196   \global\def\local@title{#1}%
2197   \expandafter\gdef\csname coppe@title@brazilian\endcsname{#1}%
2198 }
```

`\foreigntitle` Used to enter the foreign title. Also mirrors into `\coppe@title@english` so that the cover renders the right text for an english-main thesis.

```
2199 \newcommand\foreigntitle[1]{%
2200   \global\def\foreign@title{#1}%
2201   \expandafter\gdef\csname coppe@title@english\endcsname{#1}%
2202 }
```

`\coppe@selecttitle` Expands to the title stored for the current main language. For pt-main this is `\local@title`; for en-main `\foreign@title`; for any other main language the user must call `\titlein{<lang>}{...}` to populate it.

```
2203 \newcommand\coppe@selecttitle{\@nameuse{coppe@title@\coppe@mainlang}}
2204 % Subtitle, mirroring the title machinery slot for slot: \subtitle feeds the
2205 % brazilian slot, \foreignsubtitle the english one, \subtitlein any other.
2206 % 3.1.1 and 3.1.2.1.1(c) require the subtitle on the capa and on the folha de
2207 % rosto, subordinated to the title by a colon.
2208 \newcommand\subtitle[1]{%
2209   \expandafter\gdef\csname coppe@subtitle@brazilian\endcsname{#1}}
2210 \newcommand\foreignsubtitle[1]{%
2211   \expandafter\gdef\csname coppe@subtitle@english\endcsname{#1}}
2212 \newcommand\subtitlein[2]{%
2213   \expandafter\gdef\csname coppe@subtitle@#1\endcsname{#2}}
2214 \newcommand\coppe@selectsubtitle{\@nameuse{coppe@subtitle@\coppe@mainlang}}
2215 % Title as the capa and the folha de rosto print it. Annexes A and B show the
2216 % title in caps and the subtitle in ordinary case after a colon, so the case
2217 % change is applied to the title alone.
2218 \newcommand\coppe@titleblock{%
2219   \MakeUppercase{\coppe@selecttitle}%
2220   \@ifundefined{coppe@subtitle@\coppe@mainlang}{: \coppe@selectsubtitle}}
2221 % Number of volumes (3.1.1 and 3.1.2.1.1(d)): printed only when there is more
2222 % than one, since a single-volume work says nothing about volumes.
2223 % Coorientador (3.1.2.1.1(f)). Same shape as \advisor, including the optional
2224 % treatment; the label switches to the plural on the second call, as the
2225 % advisor label already did.
2226 \newcount\@coadvisor\@coadvisor0
2227 \newcommand\coadvisor[5][1]{%
2228   \global\@namedef{CoppeCoadvisorTitle:\expandafter\the\@coadvisor}{#1}
2229   \global\@namedef{CoppeCoadvisorName:\expandafter\the\@coadvisor}{#2}
2230   \global\@namedef{CoppeCoadvisorSurname:\expandafter\the\@coadvisor}{#3}
2231   \global\@namedef{CoppeCoadvisorDegree:\expandafter\the\@coadvisor}{#4}
2232   \global\@namedef{CoppeCoadvisorInst:\expandafter\the\@coadvisor}{#5}
2233   \global\advance\@coadvisor by 1
2234 }
2235 \newcommand\local@coadvisorstring{%
2236   \ifnum\@coadvisor>1
2237     \coppemainstring{coadvisors}%
2238   }
```

```

2238 \else
2239 \coppemainstring{coadvisor}%
2240 \fi}
2241 % Mesma coisa no idioma estrangeiro, para a pagina de abstract.
2242 \newcommand\foreign@coadvisorstring{%
2243 \ifnum\@coadvisor>1
2244 \coppeforeignstring{coadvisors}%
2245 \else
2246 \coppeforeignstring{coadvisor}%
2247 \fi}
2248 \newcommand\coppe@nvolumes{1}
2249 \newcommand\coppe@thisvolume{1}
2250 \newcommand\volumes[1]{\gdef\coppe@nvolumes{#1}}
2251 \newcommand\volume[1]{\gdef\coppe@thisvolume{#1}}
2252 \newcommand\coppe@volumeline{%
2253 \ifnum\coppe@nvolumes>1
2254 \par\vspace*{4mm}%
2255 \coppemainstring{volumename}\space\coppe@thisvolume\space
2256 \coppemainstring{ofname}\space\coppe@nvolumes
2257 % The closing \par matters: on the capa the next thing is \vspace{\stretch{3}},
2258 % which does not end a paragraph, so without it the city ran on into the volume
2259 % line ("Volume 2 de 3 RIO DE JANEIRO").
2260 \par
2261 \fi}

```

`\advisor` Defines globally the title, name and academic degree of the advisors.

```

2262 \newcount\@advisor\@advisor0
2263 % Argument order since v4.1: the TREATMENT ("Prof.", "Prof.a") became the
2264 % optional slot, and comes empty by default -- no norm asks for it and it only
2265 % clutters the folha de aprovacao. The INSTITUTION, required by 3.1.2.1.3(e),
2266 % became the last mandatory argument; it may be left empty ({}) without error,
2267 % in which case nothing is printed for it. A pre-4.1 call has to be rewritten:
2268 % \advisor[UFRJ]{Prof.}{Nome}{Sobrenome}{D.Sc.} becomes
2269 % \advisor{Nome}{Sobrenome}{D.Sc.}{UFRJ}.
2270 \newcommand\advisor[5][{}]{%
2271 \global\@namedef{CoppeAdvisorTitle:\expandafter\the\@advisor}{#1}
2272 \global\@namedef{CoppeAdvisorName:\expandafter\the\@advisor}{#2}
2273 \global\@namedef{CoppeAdvisorSurname:\expandafter\the\@advisor}{#3}
2274 \global\@namedef{CoppeAdvisorDegree:\expandafter\the\@advisor}{#4}
2275 \global\@namedef{CoppeAdvisorInst:\expandafter\the\@advisor}{#5}
2276 \global\advance\@advisor by 1
2277 \ifnum\@advisor>1
2278 \renewcommand\local@advisorstring{Orientadores}
2279 \renewcommand\foreign@advisorstring{Advisors}
2280 \fi
2281 }

```

`\examiner`

```

2282 \newcount\@examiner\@examiner0
2283 % Until v4.1 this stored only "#1 #2" and DISCARDED the third argument: the
2284 % titulacao every document passed was silently thrown away and never printed,
2285 % though 3.1.2.1.3(e) requires it. Each part is now kept separately, in the same
2286 % order \advisor uses: the optional slot is the TREATMENT, and the institution
2287 % is the last mandatory argument, which may be left empty.

```

```

2288 % CoppeExaminer: is retained with its old meaning for anything reading it.
2289 \newcommand\examiner[4][]{%
2290   \global\@namedef{CoppeExaminerTitle:\expandafter\the\@examiner}{#1}
2291   \global\@namedef{CoppeExaminerName:\expandafter\the\@examiner}{#2}
2292   \global\@namedef{CoppeExaminerDegree:\expandafter\the\@examiner}{#3}
2293   \global\@namedef{CoppeExaminerInst:\expandafter\the\@examiner}{#4}
2294   \global\@namedef{CoppeExaminer:\expandafter\the\@examiner}{#1\ #2}
2295   \global\advance\@examiner by 1
2296 }

```

`\author` It was redefined to allow the identification of the author's first names and surname.

```

2297 \renewcommand\author[2]{%
2298   \global\def\@authname{#1}
2299   \global\def\@authsurn{#2}
2300 }

```

`\date` Sets the *document's* nominal month and year (used by the cover, the folha de rosto, and the catalog). Stores them in private slots (`\coppe@docmonth` / `\coppe@docyear`) rather than mutating TeX's `\month` / `\year` primitives, which would clobber `\today`.

```

2301 \renewcommand\date[2]{%
2302   \def\coppe@docmonth{#1}%
2303   \def\coppe@docyear{#2}}

```

`\local@monthname`

```

2304 \newcommand\local@monthname{\ifcase\coppe@docmonth\or
2305   Janeiro\or Fevereiro\or Mar{\c c}\o\or Abril\or Maio\or Junho\or
2306   Julho\or Agosto\or Setembro\or Outubro\or Novembro\or Dezembro\fi}

```

`\foreign@monthname`

```

2307 \newcommand\foreign@monthname{\ifcase\coppe@docmonth\or
2308   January\or February\or March\or April\or May\or June\or
2309   July\or August\or September\or October\or November\or December\fi}

```

`\keyword`

```

2310 \newcounter{keywords}
2311 \newcommand\keyword[1]{%
2312   \global\@namedef{CoppeKeyword:\expandafter\the\c@keywords}{#1}
2313   \global\addtocounter{keywords}{1}
2314 }
2315 % Keywords for the foreign abstract. Annex F of the manual shows the English
2316 % abstract closing with English keywords, and \keyword alone cannot serve both
2317 % pages. When no \foreignkeyword is given the foreign abstract falls back to
2318 % the \keyword list, which is what a document written before v4.1 expects.
2319 \newcounter{foreignkeywords}
2320 \newcommand\foreignkeyword[1]{%
2321   \global\@namedef{CoppeForeignKeyword:\expandafter\the\c@foreignkeywords}{#1}
2322   \global\addtocounter{foreignkeywords}{1}
2323 }
2324 % Keywords for the optional third abstract (Portuguese, for the banca). Same
2325 % shape and same fallback as \foreignkeyword: without it the Portuguese page
2326 % would close with the main-language keywords under a Portuguese label, which
2327 % is what a Spanish-, French- or Italian-main work would produce.

```

```

2328 \newcounter{braziliankeywords}
2329 \newcommand\braziliankeyword[1]{%
2330   \global\@namedef{CoppeBrazilianKeyword:\expandafter\the\c@braziliankeywords}{#1}
2331   \global\addtocounter{braziliankeywords}{1}
2332 }
2333 % 3.1.2.1.4: the keywords close the resumo, preceded by the term, a colon, and
2334 % separated from one another by semicolons.
2335 % #1 label, #2 csname prefix, #3 how many.
2336 \newcommand\coppe@printkeywords[3]{%
2337   \ifnum#3>0
2338     \par\vspace*{7mm}%
2339     \noindent{\singlespacing\textbf{#1:}\enspace
2340     \count1=0
2341     \@whilenum\count1<#3\do{%
2342       \ifnum\count1>0 ;\space\fi
2343       \csname #2\the\count1\endcsname
2344       \advance\count1 by 1}%
2345     .\par}%
2346   \fi}

```

`\areaconcentracao` Data for the folha adicional (Coleta CAPES) and for the natureza block of the
`\linhapesquisa` folha de rosto and folha de aprovacao. Every one of them defaults to empty, and
`\tipoproducao` each page says what it does with a missing value: the folha adicional leaves the
`\projetovinculado` cell blank, and the folha de aprovacao writes the date as “a ser determinada”.

```

\nomeprojeto 2347 \newcommand\coppe@areaconc{}
\agenciafomento 2348 \newcommand\areaconcentracao[1]{\gdef\coppe@areaconc{#1}}
2349 \newcommand\coppe@dataaprov{}
2350 \newcommand\dataaprovacao[1]{\gdef\coppe@dataaprov{#1}}
2351 \newcommand\coppe@linhapesq{}
2352 \newcommand\linhapesquisa[1]{\gdef\coppe@linhapesq{#1}}
2353 \newcommand\coppe@tipoproducao{}
2354 \newcommand\tipoproducao[1]{\gdef\coppe@tipoproducao{#1}}
2355 \newcommand\coppe@projetovinc{}
2356 \newcommand\projetovinculado[1]{\gdef\coppe@projetovinc{#1}}
2357 \newcommand\coppe@nomeprojeto{}
2358 \newcommand\nomeprojeto[1]{\gdef\coppe@nomeprojeto{#1}}
2359 \newcounter{coppeagencias}
2360 \newcommand\agenciafomento[2]{%
2361   \global\@namedef{CoppeAgencia:\expandafter\the\c@coppeagencias}{#1}%
2362   \global\@namedef{CoppeAgenciaSigla:\expandafter\the\c@coppeagencias}{#2}%
2363   \global\addtocounter{coppeagencias}{1}}

```

`\freeconfig` This command allows easy changing of core class parameters.

```

2364 \newcommand\freeconfig[6]{%
2365   \providecommand\@degree{}
2366   \renewcommand\@degree{#1}
2367   \providecommand\@degreename{}
2368   \renewcommand\@degreename{#2}
2369   \providecommand\local@degname{}
2370   \renewcommand\local@degname{#3}
2371   \providecommand\foreign@degname{}
2372   \renewcommand\foreign@degname{#4}
2373   \providecommand\local@doctype{}
2374   \renewcommand\local@doctype{#5}

```

```

2375 \providecommand\foreign@doctype{}
2376 \renewcommand\foreign@doctype{#6}%
2377 }
2378 % \end{macrocode}
2379 % \end{macro}
2380 %
2381 % \begin{macro}{\frontmatter}
2382 % The number of pages for both frontmatter and mainmatter printed
2383 % in the cataloging details page is computed by means of simple
2384 % \LaTeX\ labels.
2385 % \begin{macrocode}
2386 \renewcommand\frontmatter{%
2387 \cleardoublepage
2388 \@mainmatterfalse
2389 % The count already started at the folha de rosto (see \maketitle) and simply
2390 % continues here: 2.7 requires every folha from the folha de rosto onwards to
2391 % be counted sequentially, printing arabic numerals only from the first textual
2392 % folha. Resetting the counter here -- as this class did until v4.1 -- made the
2393 % Introducao come out as page 1, which the norm does not allow.
2394 %
2395 % The numeraisromanos option keeps the legacy behaviour (lowercase roman in the
2396 % pre-textual part, arabic restarting at 1 in the textual part). That departs
2397 % from 2.7, which has the pre-textual folhas counted but NOT numbered, so the
2398 % option warns.
2399 \if@copperoman
2400 \ClassWarningNoLine{coppe}{Option 'numeraisromanos' prints roman numerals
2401 on the pre-textual folhas and restarts the arabic count at 1 in the
2402 textual part. Section 2.7 of the SiBI manual (9th rev. ed., 2026) has
2403 those folhas counted but not numbered, and the arabic numbering
2404 continuing the count. Drop the option to comply}%
2405 \pagenumbering{roman}%
2406 \fi
2407 % \thispagestyle vale por UMA pagina. Se a folha de aprovacao transbordar --
2408 % uma banca grande com titulo longo consegue --, a pagina seguinte pegava o
2409 % estilo corrente e saia com folio, na parte pre-textual, onde a 2.7 nao
2410 % admite. O grupo com \pagestyle{empty} cobre quantas paginas forem, e o
2411 % \endgroup devolve o estilo anterior.
2412 \begingroup
2413 \pagestyle{empty}\thispagestyle{empty}
2414 \singlespacing
2415 \makefrontpage
2416 \clearpage
2417 \endgroup
2418 \if@copperoman
2419 \fancypagestyle{plain}{\fancyhf{} \fancyhead[R0,LE]{\coppe@folio}%
2420 \renewcommand{\headrulewidth}{0pt}}%
2421 \pagestyle{coppe}%
2422 \else
2423 \fancypagestyle{plain}{\fancyhf{} \renewcommand{\headrulewidth}{0pt}}%
2424 \pagestyle{empty}%
2425 \fi
2426 }

```

\mainmatter


```

2427 \renewcommand\mainmatter{%
2428   \coppe@mainBegin
2429   \cleardoublepage
2430   \@mainmattertrue
2431   \fancypagestyle{plain}{\fancyhf{}}\fancyhead[R0,LE]{\coppe@folio}%
2432   \renewcommand{\headrulewidth}{0pt}}%
2433   \pagestyle{coppe}%
2434 % No \pagenumbering{arabic} in the default path: it would reset the counter to
2435 % 1 and make the Introducao folha 1, contrary to 2.7. The counter has been
2436 % arabic and running since the folha de rosto; all that changes here is that
2437 % the folio starts being PRINTED. Only the legacy numeraisromanos path, which
2438 % switched to roman in the front matter, has to switch back -- and there the
2439 % restart at 1 is intrinsic to that (non-conforming) scheme.
2440 \ifcopperoman\pagenumbering{arabic}\fi

```

`\backmatter`

```

2441 \renewcommand\backmatter{%
2442   \ifopenright
2443     \cleardoublepage
2444   \else
2445     \clearpage
2446   \fi}
2447 %

```

`\coppetexfinalpage` Optional colophon-style closing page intended to face the rear cover. The macro forces a verso (even) page so that, in a two-sided print, the reader sees the colophon on the left and the inside of the rear cover on the right; pushes the text to the bottom of the page with `\vfill`; and types it in Portuguese unconditionally (the colophon is institutional metadata, not body content, so it is rendered in pt-BR regardless of the thesis main language). Three slot macros are user-overridable: `\coppefinalengine` (engine name; auto-detected via `iftex`), `\coppefinalfont` (text body font; defaults to Computer Modern, the L^AT_EX default), and `\coppefinalmanual` (the reference manual title; defaults to the current COPPE/UFRJ norm). Override them before `\coppetexfinalpage` with `\renewcommand` if the document uses a non-default font or follows an earlier edition of the norm.

```

2448 \providecommand{\coppefinalengine}{%
2449   \ifPDFTeX\hologo{pdfLaTeX}\fi
2450   \ifLuaTeX\hologo{LuaLaTeX}\fi
2451   \ifXeTeX\hologo{XeLaTeX}\fi}
2452 % A classe carrega lmodern (as bitmap da Computer Modern saiam em Type 3 e
2453 % inviabilizavam o PDF/A), entao o colofao diz Latin Modern -- quem trocar a
2454 % fonte redefine este comando.
2455 \providecommand{\coppefinalfont}{%
2456   \ifcoppesansserif
2457     Latin Modern Sans (\texttt{lmodern})%
2458   \else
2459     Latin Modern (\texttt{lmodern})%
2460   \fi}
2461 \providecommand{\coppefinalmanual}{%
2462   \emph{Norma para a Elaborac~ao Gr'afica de Teses e Disserta~oes
2463     da COPPE/UFRJ}}
2464 \newcommand{\coppetexfinalpage}{%

```

```

2465 \clearpage
2466 % Force the colophon onto a verso (even) page so it faces the rear cover.
2467 % If the current page is odd (recto), fill it with an empty verso first.
2468 \if@twoside
2469   \ifodd\c@page
2470     \hbox{}\thispagestyle{empty}\newpage
2471   \fi
2472 \fi
2473 \thispagestyle{empty}%
2474 \null\vfill
2475 \begin{otherlanguage}{brazilian}%
2476   \begin{center}
2477     {\footnotesize\setstretch{1.0}%
2478       \textsc{Colof}\~ao}\par
2479     \vspace{0.7em}
2480     \begin{minipage}{0.75\textwidth}\centering
2481       Este documento foi composto em \LaTeX{} com \coppefinalengine{}
2482       em \today, na fonte \coppefinalfont. A bibliografia foi processada
2483       por Bib\LaTeX{} e Biber. Foi utilizada a classe \CoppeTeX,
2484       vers\~ao\~ao\coppe@version, dispon\~ai vel no reposit\~orio GitHub
2485       \mbox{\url{https://github.com/COPPE-UFRJ/CoppeTeX}}, em
2486       conformidade com a \coppefinalmanual.
2487     \end{minipage}\par
2488   \end{center}
2489 \end{otherlanguage}%
2490 \vspace*{2cm}%
2491 \clearpage}

```

`\makecover` The ABNT cover (capa, NBR 14724) suggested in the manual's Anexo A. It is the institution block — the university, the full COPPE name, and the graduate program taken from `\department` — followed by the author, the title, and the city and year, all centred and uppercased. The optional UFRJ and COPPE logos head the page. The capa is not counted: it precedes the (roman) front matter and is produced before the `titlepage` (folha de rosto).

```

2492 \newcommand\makecover{%
2493   \begin{titlepage}%
2494   % A capa e um modelo institucional, e sai em espaco simples qualquer que seja o
2495   % espacamento do corpo. Sob a opcao doublespacing ela transbordava para uma
2496   % segunda e ate uma terceira folha -- que ainda por cima levavam folio, porque
2497   % o \thispagestyle do titlepage vale por uma pagina so. 0 \pagestyle{empty} e
2498   % local ao ambiente e cobre quantas paginas forem.
2499   \pagestyle{empty}%
2500   \singlespacing
2501   \centering
2502   \makebox[\linewidth]{%
2503     \includegraphics[height=2cm]{ufrj-logo}\hfill
2504     \includegraphics[height=2cm]{coppe-logo}}\par
2505   \vspace{1.5cm}
2506   {\singlespacing
2507     \MakeUppercase{\local@universityname}\par
2508     \nohyphens{\MakeUppercase{Instituto Alberto Luiz Coimbra de \
2509       P{\~o}s-Gradua{\c c}{\~a}o e Pesquisa de Engenharia}}\par
2510     \MakeUppercase{Programa de \local@deptname}\par}

```

```

2511 \vspace{\stretch{2}}
2512 \textbf{\MakeUppercase{\textbf{\@authname\ \@authsur}}}\par
2513 \vspace{\stretch{2}}
2514 \nohyphens{\coppe@titleblock}\par
2515 \coppe@volumeline
2516 \vspace{\stretch{3}}
2517 \MakeUppercase{\local@cityname}\par
2518 \coppe@docyear
2519 \end{titlepage}%
2520 }

\maketitle

2521 \renewcommand\maketitle{%
2522 % The capa is neither counted nor numbered (2.7), so it lives in its own
2523 % throwaway numbering scheme; \pagenumbering{arabic} right after it restarts
2524 % the counter, making the FOLHA DE ROSTO page 1 -- which is where 2.7 says the
2525 % count begins. Nothing is printed here: \frontmatter keeps \pagestyle{empty}
2526 % until the textual part starts.
2527 \pagenumbering{alph}
2528 \ltx@ifpackageloaded{hyperref}{\coppe@hypersetup}{}%
2529 \makecover
2530 \pagenumbering{arabic}
2531 \begin{titlepage}
2532 % Mesma razao da capa: modelo institucional, espaco simples, e sem folio ainda
2533 % que transborde.
2534 \pagestyle{empty}%
2535 \singlespacing
2536 \begin{flushleft}
2537 \vspace*{1.5mm}
2538 \setlength\baselineskip{0pt}
2539 \setlength\parskip{1mm}
2540 \makebox[\linewidth]{%
2541 \includegraphics[height=2cm]{ufrj-logo}\hfill
2542 \includegraphics[height=2cm]{coppe-logo}}
2543 \end{flushleft}
2544 \vspace{1.05cm}
2545 \begin{center}
2546 \nohyphens{\@authname\ \@authsur}\par
2547 \vspace*{3cm}
2548 \nohyphens{\coppe@titleblock}\par
2549 \coppe@volumeline
2550 \end{center}
2551 \vspace*{2.1cm}
2552 \begin{flushright}
2553 % 2.4: on the folha de rosto the natureza/objetivo/instituicao block is set
2554 % SINGLE spaced, and aligned from the middle of the mancha grafica to the right
2555 % margin. 0.5\textwidth tracks the mancha, so the block follows automatically if
2556 % the margins ever change -- the old fixed 8.45cm did not.
2557 \begin{minipage}{0.5\textwidth}
2558 \singlespacing\frontcover@maintext
2559 \end{minipage}\par
2560 \end{flushright}
2561 % Only the natureza block is indented to the right (2.4). The linha de pesquisa
2562 % and the orientadores return to the LEFT MARGIN, as the CAPES model of the

```

```

2563 % folha de rosto and Annex B of the manual show. Until v4.1 they sat inside the
2564 % right-hand block, which is what a recent review flagged.
2565 \vspace*{12mm}
2566 {\singlespacing
2567 \ifthenelse{\equal{\coppe@linhapesq}{}}{\}%
2568 \noindent\coppemainstring{researchline}: \coppe@linhapesq\par
2569 \vspace*{5mm}}%
2570 \nohyphens{%
2571 \noindent\begin{tabularx}{\textwidth}[b]{@{}l@{ }>{\raggedright\arraybackslash}X@{}}
2572 \local@advisorstring: &
2573 \count1=0
2574 \toks@={}
2575 \@whilenum \count1<\@advisor \do{%
2576 \ifcase\count1 % same as \ifnum0=\count1
2577 \toks@=\expandafter{\csname CoppeAdvisorName:\the\count1%
2578 \expandafter\endcsname\expandafter\space%
2579 \csname CoppeAdvisorSurname:\the\count1\endcsname\\}
2580 \else
2581 \toks@=\expandafter\expandafter\expandafter{%
2582 \expandafter\the\expandafter\toks@%
2583 \expandafter&\expandafter\space%
2584 \csname CoppeAdvisorName:\the\count1\expandafter\endcsname%
2585 \expandafter\space\csname CoppeAdvisorSurname:\the\count1\endcsname\\
2586 }%
2587 \fi
2588 \advance\count1 by 1}
2589 \the\toks@
2590 \end{tabularx}}%
2591 \ifnum\@coadvisor>0
2592 \par\vspace*{2mm}%
2593 \nohyphens{%
2594 \noindent\begin{tabularx}{\textwidth}[b]{@{}l@{ }>{\raggedright\arraybackslash}X@{}}
2595 \local@coadvisorstring: &
2596 \count1=0
2597 \toks@={}
2598 \@whilenum \count1<\@coadvisor \do{%
2599 \ifcase\count1
2600 \toks@=\expandafter{\csname CoppeCoadvisorName:\the\count1%
2601 \expandafter\endcsname\expandafter\space%
2602 \csname CoppeCoadvisorSurname:\the\count1\endcsname\\}
2603 \else
2604 \toks@=\expandafter\expandafter\expandafter{%
2605 \expandafter\the\expandafter\toks@%
2606 \expandafter&\expandafter\space%
2607 \csname CoppeCoadvisorName:\the\count1\expandafter\endcsname%
2608 \expandafter\space\csname CoppeCoadvisorSurname:\the\count1\endcsname\\
2609 }%
2610 \fi
2611 \advance\count1 by 1}
2612 \the\toks@
2613 \end{tabularx}}%
2614 \fi
2615 \par}
2616 \vspace*{\fill}

```

```

2617 % 3.1.2.1.1(g): the folha de rosto carries the ANO DE DEPOSITO alone -- the year
2618 % the revised final version is handed to the Programa. The month was never asked
2619 % for here (the capa, Annex A, also shows only the year), and printing it
2620 % invited confusion with the defence date. \date still takes month and year:
2621 % the month is what the abstract pages and the colophon use.
2622 \begin{center}
2623 \local@cityname\par
2624 \coppe@docyear
2625 \end{center}
2626 \end{titlepage}
2627 % Folha adicional (3.1.2.1.2): a sheet of its own right after the folha de
2628 % rosto, ahead of the approval sheet produced by \frontmatter. Until the 2026
2629 % edition this content lived on the VERSO of the folha de rosto (NBR 14724,
2630 % R57); the verso is gone with the printed copy.
2631 % LaTeX's titlepage environment ends with \setcounter{page}{1} whenever the
2632 % document is one-sided (see \endtitlepage in classes.dtx). That silently ate
2633 % the folha de rosto from the count, putting the folha de aprovacao at 1 instead
2634 % of 2. The count is therefore restated here: the folha de rosto WAS folha 1, so
2635 % whatever follows it is folha 2.
2636 \setcounter{page}{2}%
2637 % The page carrying the ficha catalografica is explicitly excluded from the
2638 % count -- 2.7 of the 2026 manual ("a folha adicional que contem a ficha
2639 % catalografica nao pode ser contada ou numerada"), and the 2025 edition said
2640 % the same of the verso of the folha de rosto. Hence the counter is stepped
2641 % back after it.
2642 % An exame de qualificacao is not deposited with the library: no ficha
2643 % catalografica, and no Coleta CAPES sheet. Everything else gets both.
2644 \if@coppeexame\else
2645 \makefolhaadicional\clearpage\addtocounter{page}{-1}%
2646 \fi
2647 \global\let\maketitle\relax%
2648 \global\let\and\relax}

2649 % 3.1.2.1.3 lists the elements of the folha de aprovacao in order: (a) the
2650 % AUTHOR, (b) the title in full, (c) the natureza/objetivo/instituicao/area de
2651 % concentracao, (d) the date of approval, (e) the examining board. Until v4.1
2652 % this page opened with the title in caps and the author under it, and set the
2653 % natureza as a wall of uppercase (\frontpage@maintext). Annex D shows neither.
2654 % One board member. \coppe@membroid is the identification line required by
2655 % 3.1.2.1.3(e): name, titulacao and institution. Since v4.1 the TREATMENT
2656 % ("Prof.", "Prof.a") is the optional slot and comes empty by default -- no norm
2657 % asks for it -- while titulacao and institution are what actually identify the
2658 % member and are shown whenever they are given. Each of the three is dropped
2659 % silently when empty, so nothing breaks if the author leaves one blank.
2660 \newcommand\coppe@membroid[4]{%
2661 \ifthenelse{\equal{#1}{} }{\ #1\ }{#2}%
2662 \ifthenelse{\equal{#3}{} }{\ , #3}%
2663 \ifthenelse{\equal{#4}{} }{\ , #4}%
2664 % Until v4.1 the class could draw a rule above each name, for a hand signature,
2665 % and \coppe@membrogap held the gap above that rule, shrinking as the board grew
2666 % so seven members still fitted on one sheet. The deposit has been digital only
2667 % since Resolucao CEPG n. 246/2023, so there is nothing left to sign by hand and
2668 % the rules are gone. The length stays declared because the seven-member sizing
2669 % below still uses it as the gap between entries.

```

```

2670 \newlength\coppe@membrogap
2671 \setlength\coppe@membrogap{9mm minus 2mm}
2672 \newcount\coppe@bancasize
2673 \newcommand\coppe@membro[4]{%
2674   \noindent\hspace*{8mm}%
2675   \begin{minipage}{\dimexpr\textwidth-8mm\relax}
2676     \coppe@membroid{#1}{#2}{#3}{#4}
2677   \end{minipage}\par\vspace{2mm}}
2678 \newcommand\makefrontpage{%
2679   \begin{center}
2680     \nohyphens{\@authname\ \@authsur}\par
2681     \vspace*{7mm}
2682     \sloppy\nohyphens{\coppe@selecttitle
2683       \@ifundefined{coppe@subtitle@\coppe@mainlang}{:
2684         \coppe@selectsubtitle}}\par
2685   \end{center}\par
2686   \vspace*{8mm}
2687   % Same block, same indentation and same single spacing as the folha de rosto:
2688   % 2.4 names the two pages together.
2689   \begin{flushright}
2690     \begin{minipage}{0.5\textwidth}
2691       \singlespacing\frontcover@maintext
2692     \end{minipage}
2693   \end{flushright}
2694   \vspace*{12mm}
2695   % (d) date of approval, from \dataaprovacao. With no date given the page used to
2696   % print a rule to fill in by hand, which the digital-only deposit made pointless
2697   % -- nobody writes on a PDF. It now says so in words, in the main language:
2698   % "a ser determinada". The line is never blank and never a rule.
2699   \noindent\local@approvedonname:\enspace
2700   \ifthenelse{\equal{\coppe@dataaprov}{}}{
2701     {\coppemainstring{tbddate}}{\coppe@dataaprov}\par
2702   \vspace*{12mm}
2703   % (e) the examining board. By DEFAULT only the examiners are listed: the folha
2704   % de aprovacao records who examined the work, and the advisor conducted it. With
2705   % the 'orientadorexamina' option the advisors and coadvisors go back in, ahead
2706   % of the examiners, as president of the board (3.1.2.1.3(e)).
2707   %
2708   % The list is assembled into \coppe@bancalist with \g@addto@macro instead of
2709   % being built inside a tabularx: each entry now carries four fields (treatment,
2710   % name, titulacao, institution) and the old \toks@/\expandafter accumulation
2711   % could not carry them without becoming unreadable. \protected@edef keeps
2712   % accents and other protected commands in the names intact.
2713   % Size of the board decides the spacing above. The thresholds are measured, not
2714   % guessed: with the 9mm gap a seven-member board pushed three members onto a
2715   % second page, 7mm brought six back onto one sheet, 5mm did the same for seven,
2716   % and eight needed 3.5mm -- eight came up in the adversarial suite, in a
2717   % Spanish thesis whose longer natureza block leaves less room than the
2718   % Portuguese one. Boards of five or fewer keep the original 9mm, so every
2719   % document written before this change renders exactly as it did.
2720   % The count is of what actually gets PRINTED, so a document that leaves the
2721   % advisors out does not get the tight spacing of a board it does not show.
2722   \coppe@bancasize=\@examiner
2723   \if@coppeorientexam

```

```

2724 \advance\coppe@bancasize by \@advisor
2725 \advance\coppe@bancasize by \@coadvisor
2726 \fi
2727 \ifnum\coppe@bancasize>7
2728 \setlength\coppe@membrogap{3.5mm minus 1.5mm}%
2729 \else\ifnum\coppe@bancasize>6
2730 \setlength\coppe@membrogap{5mm minus 2mm}%
2731 \else\ifnum\coppe@bancasize>5
2732 \setlength\coppe@membrogap{7mm minus 2mm}%
2733 \fi\fi\fi
2734 \gdef\coppe@bancalist{%
2735 % Orientadores e coorientadores so entram sob 'orientadorexamina'. Fora dela a
2736 % folha lista apenas quem examinou.
2737 \if@coppeorientexam
2738 \count1=0
2739 \@whilenum\count1<\@advisor\do{%
2740 \protected@edef\coppe@tmp{\noexpand\coppe@membro
2741 {\csname CoppeAdvisorTitle:\the\count1\endcsname}%
2742 {\csname CoppeAdvisorName:\the\count1\endcsname}\space
2743 {\csname CoppeAdvisorSurname:\the\count1\endcsname}%
2744 {\csname CoppeAdvisorDegree:\the\count1\endcsname}%
2745 {\csname CoppeAdvisorInst:\the\count1\endcsname}}%
2746 \expandafter\g@addto@macro\expandafter\coppe@bancalist
2747 \expandafter{\coppe@tmp}%
2748 \advance\count1 by 1}%
2749 % Coorientadores come right after the orientadores and before the rest of the
2750 % board, keeping the order 3.1.2.1.3(e) prescribes.
2751 \count1=0
2752 \@whilenum\count1<\@coadvisor\do{%
2753 \protected@edef\coppe@tmp{\noexpand\coppe@membro
2754 {\csname CoppeCoadvisorTitle:\the\count1\endcsname}%
2755 {\csname CoppeCoadvisorName:\the\count1\endcsname}\space
2756 {\csname CoppeCoadvisorSurname:\the\count1\endcsname}%
2757 {\csname CoppeCoadvisorDegree:\the\count1\endcsname}%
2758 {\csname CoppeCoadvisorInst:\the\count1\endcsname}}%
2759 \expandafter\g@addto@macro\expandafter\coppe@bancalist
2760 \expandafter{\coppe@tmp}%
2761 \advance\count1 by 1}%
2762 \fi
2763 \count1=0
2764 \@whilenum\count1<\@examiner\do{%
2765 \protected@edef\coppe@tmp{\noexpand\coppe@membro
2766 {\csname CoppeExaminerTitle:\the\count1\endcsname}%
2767 {\csname CoppeExaminerName:\the\count1\endcsname}%
2768 {\csname CoppeExaminerDegree:\the\count1\endcsname}%
2769 {\csname CoppeExaminerInst:\the\count1\endcsname}}%
2770 \expandafter\g@addto@macro\expandafter\coppe@bancalist
2771 \expandafter{\coppe@tmp}%
2772 \advance\count1 by 1}%
2773 \noindent\local@approvedname:\par
2774 \nohyphens{\singlespacing\coppe@bancalist}\par
2775 % No city/date block at the foot: 3.1.2.1.3 ends the folha de aprovacao at the
2776 % examining board, and Annex D shows nothing below the signature lines. The date
2777 % that belongs on this page is the date of APPROVAL, printed above.

```

```

2778 % \frontpage@bottomtext stays defined for anyone calling it directly.
2779 \vspace*{\fill}}

2780 \newcommand\coppe@hypersetup{%
2781 \begin{group}
2782 % changes to \toks@ and \count@ are kept local;
2783 % it's not necessary for them, but it is usually the case
2784 % for \count1, because the first ten counters are written
2785 % to the DVI file, thus you got lucky because of PDF output
2786 \toks@={}% in this special case not necessary
2787 \count@=0 %
2788 \@whilenum\count@<\value{keywords}\do{%
2789 % * a keyword separator is not necessary,
2790 % if there is just one keyword
2791 % * \csname CoppeKeyword:\the\count@\endcsname must be expanded
2792 % at least once, to get rid of the loop depended \count@
2793 \ifcase\count@ % same as \ifnum0=\count@
2794 \toks@=\expandafter{\csname CoppeKeyword:\the\count@\endcsname}%
2795 \else
2796 \toks@=\expandafter\expandafter\expandafter{%
2797 \expandafter\the\expandafter\toks@
2798 \expandafter;\expandafter\space
2799 \csname CoppeKeyword:\the\count@\endcsname
2800 }%
2801 \fi
2802 \advance\count@ by 1 %
2803 }%
2804 \edef\x{\endgroup
2805 \noexpand\hypersetup{%
2806 pdfkeywords={\the\toks@}%
2807 }%
2808 }%
2809 \x
2810 \hypersetup{%
2811 pdfauthor={\@authname\ \@authsur},
2812 pdftitle={\local@title},
2813 pdfsubject={\local@doctype\ de \@degree\ em \local@deptname\ da COPPE/UFRJ},
2814 pdfcreator={LaTeX with CoppeTeX toolkit},
2815 breaklinks={true},
2816 raiselinks={true},
2817 pageanchor={true},
2818 }}

```

`\makecatalog` When the document has illustrations, it is required to insert “: il.” between the number of pages of the textual part and the page dimension. We have created a label to flag the existence of lists of figures. It is checked to be undefined using the plain `TeX` command `\@isundefined` (cf. the `TeX` FAQ).

```

2819 \newcommand\makecatalog{%
2820 \vspace*{\fill}
2821 \begin{center}
2822 \setlength{\fboxsep}{5mm}
2823 \framebox[120mm][c]{\makebox[5mm][c]{}}%
2824 \begin{minipage}[c]{105mm}
2825 \setlength{\parindent}{5mm}

```



```

2826 \noindent\sloppy\nohyphens\@authsurn,
2827 \nohyphens\@authname\par
2828 \nohyphens{%
2829 \coppe@selecttitle/\@authname\ \@authsurn. -- \local@cityname:
2830 UFRJ/COPPE, \coppe@docyear.}\par
2831 \pageref{front:pageno},
2832 \pageref{LastPage}
2833 p.\@ifundefined{r@cat:lofflag}{\pageref{cat:lofflag}} $29,7$cm.\par
2834 % There is an issue here. When the last entry must be split between lines,
2835 % the spacing between it and the next paragraph becomes smaller.
2836 % Should we manually introduce a fixed space? But how could we know that
2837 % a name was split? Is this happening yet?
2838 \nohyphens{%
2839 \begin{tabularx}{100mm}[b]{@{}l@{ }>{\raggedright\arraybackslash}X@{}}
2840 \local@advisorstring: &
2841 \count1=0
2842 \toks@={}
2843 \@whilenum \count1<\@advisor \do{%
2844 \ifcase\count1 % same as \ifnum0=\count1
2845 \toks@=\expandafter{\csname CoppeAdvisorName:\the\count1%
2846 \expandafter\endcsname\expandafter\space%
2847 \csname CoppeAdvisorSurname:\the\count1\endcsname\\}
2848 \else
2849 \toks@=\expandafter\expandafter\expandafter{%
2850 \expandafter\the\expandafter\toks@
2851 \expandafter&\expandafter\space
2852 \csname CoppeAdvisorName:\the\count1\expandafter\endcsname%
2853 \expandafter\space\csname CoppeAdvisorSurname:\the\count1\endcsname\\}
2854 }%
2855 \fi
2856 \advance\count1 by 1}
2857 \the\toks@
2858 \end{tabularx}}\par
2859 \nohyphens{\local@doctype\ ({\MakeLowercase\@degreename}) --
2860 UFRJ/COPPE/Programa de \local@deptname, \coppe@docyear.}\par
2861 Refer{\^ e}ncias Bibliogr{\` a}ficas: p. \pageref{bib:begin} -- \pageref{bib:end}.\par
2862 \count1=0
2863 \count2=1
2864 \nohyphens{\@whilenum \count1<\value{keywords} \do {%
2865 \number\count2. \csname CoppeKeyword:\the\count1 \endcsname.
2866 \advance\count1 by 1
2867 \advance\count2 by 1}
2868 I. \csname CoppeAdvisorSurname:0\endcsname,%
2869 \ \csname CoppeAdvisorName:0\endcsname%
2870 \ifthenelse{\@advisor>1}{\ \emph{et~al.}}{}}.
2871 II. \local@universityname, COPPE, Programa de \local@deptname.
2872 III. T{\` i}tulo.}
2873 \end{minipage}}
2874 \end{center}
2875 \vspace*{\fill}}

```

\fichacatalografica The folha adicional required from August 2026 (SiBI manual, 9th rev. ed.,
\makefolhaadicional 3.1.2.1.2 and Annex H): the CAPES collection fields plus the ficha cata-
lografica, sitting immediately after the folha de rosto and counted by no-

body. `\fichacatalografica` takes the PDF produced by the SiBI generator at <http://fichacatalografica.sibi.ufrj.br/>, optionally with a page number.

The sheet is institutional, so it is always in Portuguese whatever the main language of the work – and deliberately without `\selectlanguage`, since the brazilian babel language need not be loaded in, say, a Spanish-main document.

```

2876 \newcommand\coppe@fichafila{}
2877 \newcommand\coppe@fichapage{1}
2878 \newcommand\fichacatalografica[2][1]{%
2879   \gdef\coppe@fichapage{#1}\gdef\coppe@fichafila{#2}}
2880 \newcommand\coppe@marcatipo[1]{%
2881   \ifthenelse{\equal{\coppe@tipoproducao}{#1}}{$\times$}{\phantom{$\times$}}}
2882 \newcommand\coppe@marcavinc[1]{%
2883   \ifthenelse{\equal{\coppe@projetoavinc}{#1}}{$\times$}{\phantom{$\times$}}}
2884 % Annex H prints the five fields inside a CLOSED FRAME, one cell each, with a
2885 % horizontal rule between cells and NOTHING to write on: the space to fill in is
2886 % the blank space of the cell. Until v4.1 the class laid the fields out as loose
2887 % paragraphs and drew a dotted \hrulefill to write on, which is not what the
2888 % model shows. \coppe@campo now prints the value when there is one and leaves
2889 % empty vertical space when there is not.
2890 \newlength\coppe@capespap
2891 \setlength\coppe@capespap{11mm}
2892 \newcommand\coppe@campo[1]{%
2893   \ifthenelse{\equal{#1}{}}{\vspace*{\coppe@capespap}}{\par#1}}
2894 % One cell of the frame: the label, then whatever the field holds.
2895 \newcommand\coppe@capescelula[2]{%
2896   \begin{minipage}[t]{\dimexpr\linewidth-2\tabcolsep\relax}
2897     \vspace{1mm}\par#2\vspace{1.5mm}
2898   \end{minipage}\hrule
2899 \newcommand\coppe@checativo{%
2900   \ifthenelse{\equal{\coppe@tipoproducao}{}}{}{%
2901     \ifthenelse{\equal{\coppe@tipoproducao}{bibliografica}}{OR
2902       \equal{\coppe@tipoproducao}{artistica}}{OR
2903       \equal{\coppe@tipoproducao}{tecnologica}}{OR
2904       \equal{\coppe@tipoproducao}{tecnica}}{}{%
2905       \ClassWarning{coppe}{Unknown \string\coppe@tipoproducao{} value
2906         '\coppe@tipoproducao'. Annex H of the SiBI manual allows only
2907         bibliografica, artistica, tecnologica or tecnica; no box will be
2908         ticked}}}}
2909 \newcommand\makefolhaadiciona{%
2910   \clearpage
2911   \thispagestyle{empty}%
2912   \coppe@checativo
2913   {\singlespacing
2914     \begin{center}
2915       {\large\bfseries Informa\c c\~oes Coleta CAPES}
2916     \end{center}
2917     \vspace{7mm}
2918     \noindent\begin{tabular}{|@{\hspace{\tabcolsep}}p{\dimexpr\textwidth-2\tabcolsep-2\arrayru
2919       \hline
2920       \coppe@capescelula{Tipo de produ\c c\~ao intelectual:}{%
2921         \vspace{1mm}
2922         \hspace*{10mm}(\coppe@marcatipo{bibliografica})\enspace
2923         Bibliogr\~africa\par

```

```

2924 \hspace*{10mm}(\coppe@marcatipo{artistica})\enspace Art'\{i}stica\par
2925 \hspace*{10mm}(\coppe@marcatipo{tecnologica})\enspace Tecnol\'ogica\par
2926 \hspace*{10mm}(\coppe@marcatipo{tecnica})\enspace T\'ecnica}
2927 \coppe@capescelula{Projeto de pesquisa vinculado \'a produco
2928 intelectual?}{%
2929 \vspace{1mm}
2930 \hspace*{10mm}(\coppe@marcavinc{sim})\enspace Sim\par
2931 \hspace*{10mm}(\coppe@marcavinc{nao})\enspace No}
2932 \coppe@capescelula{Nome do projeto de pesquisa:}{%
2933 \coppe@campo{\coppe@nomeprojeto}}
2934 \coppe@capescelula{\'Area de concentrao da produco
2935 intelectual:}{\coppe@campo{\coppe@areaconc}}
2936 \coppe@capescelula{Agncia(s) de fomento financiadora(s) da produco
2937 intelectual:}{%
2938 \ifthenelse{value{coppeagencias}=0}{\vspace*{\coppe@capesgap}}{%
2939 \count1=0
2940 \@whilenum\count1<\value{coppeagencias}\do{%
2941 \hspace*{10mm}\csname CoppeAgencia:\the\count1\endcsname
2942 \ (\csname CoppeAgenciaSigla:\the\count1\endcsname)\par
2943 \advance\count1 by 1}}
2944 \end{tabular}\par
2945 \vspace{9mm}
2946 \begin{center}
2947 {\large\bfseries Ficha catalogr\'afica}\par
2948 \vspace{4mm}
2949 \ifthenelse{equal{\coppe@fichafile}}{%
2950 \if@copperascunho
2951 \makecatalog
2952 \else
2953 \framebox[125mm][c]{\parbox[c][62mm][c]{115mm}{\centering\footnotesize
2954 Espao reservado \'a ficha catalogr\'afica.\[1ex]
2955 Gere-a em \texttt{fichacatalografica.sibi.ufrj.br} e inclua o
2956 arquivo com \texttt{\string\fichacatalografica\{arquivo.pdf\}},
2957 ou solicite-a \'a biblioteca do seu Programa.}}%
2958 \fi}{%
2959 \includegraphics[page=\coppe@fichapage,width=125mm]{\coppe@fichafile}}%
2960 \end{center}
2961 \par}}

```

\dedication

```

2962 \newcommand\dedication[1]{
2963 \gdef\@dedic{#1}
2964 \cleardoublepage
2965 \vspace*{\fill}
2966 \begin{flushright}
2967 \begin{minipage}{60mm}
2968 \raggedleft \it \normalsize \@dedic
2969 \end{minipage}
2970 \end{flushright}}

```

\coppe@coadvisorrows The coadvisor block of the abstract pages, in the shape the folha de rosto already uses. It prints nothing at all unless the `coorientador` class option is given AND the document actually declared a `\coadvisor`, so the default output is byte for byte what it was. The argument is the label macro, which differs per page: the

main-language pages pass \local@coadvisorstring and the foreign one passes \foreign@coadvisorstring.

```

2971 \newcommand\coppe@coadvisorrows[1]{%
2972   \if@coppecoadvresumo
2973     \ifnum\@coadvisor>0
2974       \par\vspace*{2mm}%
2975       \noindent\begin{tabularx}{\textwidth}[b]{0{}}10{ }>{\raggedright\arraybackslash}X0{}}
2976         #1: &
2977         \count1=0
2978         \toks@={}
2979         \@whilenum \count1<\@coadvisor \do{%
2980           \ifcase\count1
2981             \toks@=\expandafter{\csname CoppeCoadvisorName:\the\count1%
2982               \expandafter\endcsname\expandafter\space%
2983               \csname CoppeCoadvisorSurname:\the\count1\endcsname\\}
2984           \else
2985             \toks@=\expandafter\expandafter\expandafter{%
2986               \expandafter\the\expandafter\toks@%
2987               \expandafter&\expandafter\space%
2988               \csname CoppeCoadvisorName:\the\count1\expandafter\endcsname%
2989               \expandafter\space\csname CoppeCoadvisorSurname:\the\count1\endcsname\\}
2990             }%
2991           \fi
2992           \advance\count1 by 1}
2993         \the\toks@
2994       \end{tabularx}%
2995     \fi
2996   \fi}

```

abstract (*env.*) This is a specialization of the abstract in the article standard class.

```

2997 \newenvironment{abstract}{%
2998   \cleardoublepage
2999   \thispagestyle{plain}
3000   \abstract@top\text\par
3001   \vspace*{8.6mm}
3002   \begin{center}
3003     \sloppy\nohyphens{\MakeUppercase\local@title}\par
3004     \vspace*{13.2mm}
3005     \@authname\ \@authsurn \par
3006     \vspace*{7mm}
3007     \local@monthname/\coppe@docyear
3008   \end{center}\par
3009   \vspace*{1.5cm}
3010   \noindent%
3011   \begin{tabularx}{\textwidth}[b]{0{}}10{ }>{\raggedright\arraybackslash}X0{}}
3012     \local@advisorstring: &
3013     \count1=0
3014     \toks@={}
3015     \@whilenum \count1<\@advisor \do{%
3016       \ifcase\count1 % same as \ifnum0=\count1
3017         \toks@=\expandafter{\csname CoppeAdvisorName:\the\count1%
3018           \expandafter\endcsname\expandafter\space%
3019           \csname CoppeAdvisorSurname:\the\count1\endcsname\\}
3020       \else

```

```

3021      \toks@=\expandafter\expandafter\expandafter{%
3022      \expandafter\the\expandafter\toks@
3023      \expandafter&\expandafter\space
3024      \csname CoppeAdvisorName:\the\count1\expandafter\endcsname%
3025      \expandafter\space\csname CoppeAdvisorSurname:\the\count1\endcsname\\
3026      }%
3027      \fi
3028      \advance\count1 by 1}
3029      \the\toks@
3030 \end{tabularx}%
3031 \coppe@coadvisorrows{\local@coadvisorstring}\par
3032 \vspace*{2mm}
3033 \noindent\local@deptstring: \local@deptname\par
3034 \vspace*{7mm}}}%
3035 \coppe@printkeywords{\coppemainstring{keywordsname}}%
3036 {CoppeKeyword:}{\value{keywords}}}%
3037 \vspace*{\fill}}

```

foreignabstract (*env.*)

```

3038 \newenvironment{foreignabstract}{%
3039 \cleardoublepage
3040 \thispagestyle{plain}
3041 \begin{otherlanguage}{\coppe@foreignlang}
3042 \foreignabstract@toptext\par
3043 \vspace*{8.6mm}
3044 \begin{center}
3045 \sloppy\nohyphens{\MakeUppercase\foreign@title}\par
3046 \vspace*{13.2mm}
3047 \@authname\ \@authsurn \par
3048 \vspace*{7mm}
3049 \foreign@monthname/\coppe@docyear
3050 \end{center}\par
3051 \vspace*{1.5cm}
3052 \noindent%
3053 \begin{tabularx}{\textwidth}[b]{@{}l@{ }>{\raggedright\arraybackslash}X@{}}
3054 \foreign@advisorstring: &
3055 \count1=0
3056 \toks@={}
3057 \@whilenum \count1<\@advisor \do{%
3058 \ifcase\count1 % same as \ifnum0=\count1
3059 \toks@=\expandafter{\csname CoppeAdvisorName:\the\count1%
3060 \expandafter\endcsname\expandafter\space%
3061 \csname CoppeAdvisorSurname:\the\count1\endcsname\\}
3062 \else
3063 \toks@=\expandafter\expandafter\expandafter{%
3064 \expandafter\the\expandafter\toks@
3065 \expandafter&\expandafter\space
3066 \csname CoppeAdvisorName:\the\count1\expandafter\endcsname%
3067 \expandafter\space\csname CoppeAdvisorSurname:\the\count1\endcsname\\
3068 }%
3069 \fi
3070 \advance\count1 by 1}
3071 \the\toks@
3072 \end{tabularx}%

```

```

3073 \coppe@coadvisorrows{\foreign@coadvisorstring}\par
3074 \vspace*{2mm}
3075 \noindent\foreign@deptstring: \foreign@deptname\par
3076 \vspace*{7mm}}{%
3077 \ifnum\value{foreignkeywords}>0
3078   \coppe@printkeywords{\coppeforeignstring{keywordsname}}%
3079   {CoppeForeignKeyword:}{\value{foreignkeywords}}%
3080 \else
3081   \coppe@printkeywords{\coppeforeignstring{keywordsname}}%
3082   {CoppeKeyword:}{\value{keywords}}%
3083 \fi
3084 \end{otherlanguage}
3085 \vspace*{\fill}
3086 \global\let\@author\@empty
3087 \global\let\@date\@empty
3088 \global\let\foreign@title\@empty
3089 \global\let\foreign@title\relax
3090 \global\let\local@title\@empty
3091 \global\let\local@title\relax
3092 \global\let\author\relax
3093 \global\let\author\relax
3094 \global\let\date\relax}

```

brazilianabstract (*env.*) Optional third abstract for non-pt main languages: a Brazilian Portuguese abstract printed in addition to **abstract** (main language) and **foreignabstract** (English by default). Wraps its body in `\begin{otherlanguage}{brazilian}` so the babel typography (hyphenation, quotation marks) is correct regardless of the current main language. Uses the same `\local@...` Portuguese strings the **abstract** env uses today. Meaningful only when main is neither **brazilian** nor **english**; harmless otherwise (the document just won't include it).

```

3095 \newenvironment{brazilianabstract}{%
3096   \cleardoublepage
3097   \thispagestyle{plain}
3098   \begin{otherlanguage}{brazilian}
3099   \abstract@toptext\par
3100   \vspace*{8.6mm}
3101   \begin{center}
3102     \sloppy\nohyphens{\MakeUppercase\local@title}\par
3103     \vspace*{13.2mm}
3104     \@authname\ \@authsurv \par
3105     \vspace*{7mm}
3106     \local@monthname/\coppe@docyear
3107   \end{center}\par
3108   \vspace*{1.5cm}
3109   \noindent%
3110   \begin{tabularx}{\textwidth}[b]{@{}l@{ }>\raggedright\arraybackslash}X@{}}
3111     \local@advisorstring: &
3112     \count1=0
3113     \toks@={}
3114     \@whilenum \count1<\@advisor \do{%
3115       \ifcase\count1
3116         \toks@=\expandafter{\csname CoppeAdvisorName:\the\count1%
3117         \expandafter\endcsname\expandafter\space%

```

```

3118     \csname CoppeAdvisorSurname:\the\count1\endcsname\\}
3119   \else
3120     \toks@=\expandafter\expandafter\expandafter{%
3121       \expandafter\the\expandafter\toks@
3122       \expandafter&\expandafter\space
3123       \csname CoppeAdvisorName:\the\count1\expandafter\endcsname%
3124       \expandafter\space\csname CoppeAdvisorSurname:\the\count1\endcsname\\
3125     }%
3126   \fi
3127   \advance\count1 by 1}
3128   \the\toks@
3129 \end{tabularx}%
3130 \coppe@coadvisorrows{\local@coadvisorstring}\par
3131 \vspace*{2mm}
3132 \noindent\local@deptstring: \local@deptname\par
3133 \vspace*{7mm}}{%
3134 % \coppestring takes ONE argument and resolves against the current language;
3135 % this runs before \end{otherlanguage}, so \language is already brazilian.
3136 \ifnum\value{braziliankeywords}>0
3137   \coppe@printkeywords{\coppestring{keywordsname}}%
3138   {CoppeBrazilianKeyword:}{\value{braziliankeywords}}%
3139 \else
3140   \coppe@printkeywords{\coppestring{keywordsname}}%
3141   {CoppeKeyword:}{\value{keywords}}%
3142 \fi
3143 \end{otherlanguage}
3144 \vspace*{\fill}}

```

\listoffigures

```

3145 \renewcommand\listoffigures{%
3146   \coppe@hasLof
3147   \if@twocolumn
3148     \@restonecoltrue\onecolumn
3149   \else
3150     \@restonecolfalse
3151   \fi
3152   \chapter*{\listfigurename}%
3153   \coppe@listtoc{\listfigurename}%
3154   \@mkboth{\MakeUppercase\listfigurename}%
3155           {\MakeUppercase\listfigurename}%
3156   \@starttoc{lof}%
3157   \if@restonecol\twocolumn\fi
3158 }

```

\listoftables

```

3159 \renewcommand\listoftables{%
3160   \if@twocolumn
3161     \@restonecoltrue\onecolumn
3162   \else
3163     \@restonecolfalse
3164   \fi
3165   \chapter*{\listtablename}%
3166   \coppe@listtoc{\listtablename}%
3167   \@mkboth{%

```

```

3168      \MakeUppercase\listtablename}%
3169      {\MakeUppercase\listtablename}%
3170      \@starttoc{lot}%
3171      \if@restonecol\twocolumn\fi
3172      }

```

\printlosymbols

```

3173 % 0 \renewcommand ficava FORA do grupo e vazava: depois de imprimir a lista de
3174 % simbolos, \glossaryname continuava valendo "Lista de Simbolos" pelo resto do
3175 % documento, e o Glossario pos-textual da 3.1.4.2 saia com o titulo errado --
3176 % no sumario tambem. Dentro do grupo, a troca dura o que tem de durar.
3177 \newcommand\printlosymbols{%
3178 \begingroup\@coppeglosspretrue
3179 \renewcommand\glossaryname{\listsymbolname}%
3180 \@input@{\jobname.los}\endgroup}

```

\makelosymbols

```

3181 \def\makelosymbols{%
3182   \newwrite\@losfile
3183   \immediate\openout\@losfile=\jobname.syx
3184   \newcommand\syml[3][\@bsphack\begingroup
3185     \ifstrempy{##1}{\def\@tempsymb1{##2=}}{\def\@tempsymb1{##1=}}}%
3186     \@sanitize%
3187     \@wrls{\@tempsymb1}{##2}{##3}}\typeout%
3188   {Writing index of symbols file \jobname.syx}%
3189   \let\makelosymbols\@empty%
3190 }%
3191 \@onlypreamble\makelosymbols

3192 \AtBeginDocument{%
3193 \ifpackageloaded{hyperref}{%
3194   \newcommand\@wrls[3]{%
3195     \protected@write\@losfile{%
3196       {\string\indexentry{#1[#2] #3|hyperpage}{\thepage}}}%
3197     \endgroup%
3198     \@esphack}}{%
3199   \newcommand\@wrls[3]{%
3200     \protected@write\@losfile{%
3201       {\string\indexentry{#1[#2] #3}{\thepage}}}%
3202     \endgroup%
3203     \@esphack}}}%

```

\printloabbreviations

```

3204 \newcommand\printloabbreviations{%
3205 \begingroup\@coppeglosspretrue
3206 \renewcommand\glossaryname{\listabbreviationname}%
3207 \@input@{\jobname.lab}\endgroup}

```

\makeloabbreviations

```

3208 \def\makeloabbreviations{%
3209   \newwrite\@labfile
3210   \immediate\openout\@labfile=\jobname.abx
3211   \newcommand\abbrev[3][\@bsphack\begingroup
3212     \ifstrempy{##1}{\def\@tempsymb1{##2=}}{\def\@tempsymb1{##1=}}

```



```

3213             \@sanitize
3214             \@wrlab{\@tempsymb1}{##2}{##3}\typeout
3215   {Writing index of abbreviations file \jobname.abx}%
3216   \let\makeloabbreviations\@empty
3217 }
3218 \@onlypreamble\makeloabbreviations

3219 \AtBeginDocument{%
3220 \ifpackageloaded{hyperref}{%
3221   \newcommand\@wrlab[3]{%
3222     \protected@write\@labfile{%
3223       {\string\indexentry{#1[#2] #3|hyperpage}{\thepage}}%
3224     \endgroup%
3225     \@esphack}}{%
3226   \newcommand\@wrlab[3]{%
3227     \protected@write\@labfile{%
3228       {\string\indexentry{#1[#2] #3}{\thepage}}%
3229     \endgroup%
3230     \@esphack}}}%

\newsigla Acronyms (siglas), manual2024 2.8 / R43. \newsigla{key}{SIGLA}{full form}
\sigla declares one; \sigla{key} prints “full form (SIGLA)” on first use and registers it in
the list of abbreviations and acronyms (through \abbrev, so \makeloabbreviations
must be active), then just “SIGLA” on later uses. \sigla*{key} forces the full
form. The first-use flag is set globally so it survives groups (floats, footnotes).

3231 \newcommand\newsigla[3]{%
3232   \@namedef{coppe@sigla@s@#1}{#2}%
3233   \@namedef{coppe@sigla@l@#1}{#3}%
3234   \newcommand\coppe@sigla@reg[1]{%
3235     \ifundefined{abbrev}{}%
3236     {\abbrev{\@nameuse{coppe@sigla@s@#1}}{\@nameuse{coppe@sigla@l@#1}}}%
3237   \newcommand\coppe@sigla@full[1]{%
3238     \@nameuse{coppe@sigla@l@#1}\space(\@nameuse{coppe@sigla@s@#1})%
3239   \newcommand\coppe@sigla@undef[1]{%
3240     \PackageWarning{coppe}{Sigla '#1' nao declarada com \string\newsigla}%
3241     \textbf{??#1??}}
3242   \newcommand\coppe@sigla@first[1]{%
3243     \global\@namedef{coppe@sigla@done@#1}{\coppe@sigla@reg{#1}\coppe@sigla@full{#1}}
3244   \newcommand\coppe@sigla@norm[1]{%
3245     \ifundefined{coppe@sigla@s@#1}{\coppe@sigla@undef{#1}}%
3246     {\@ifundefined{coppe@sigla@done@#1}{\coppe@sigla@first{#1}}%
3247     {\@nameuse{coppe@sigla@s@#1}}}%
3248   \newcommand\coppe@sigla@star[1]{%
3249     \ifundefined{coppe@sigla@s@#1}{\coppe@sigla@undef{#1}}%
3250     {\global\@namedef{coppe@sigla@done@#1}{\coppe@sigla@full{#1}}}%
3251   \newcommand\sigla{\@ifstar\coppe@sigla@star\coppe@sigla@norm}

3252 %%% \AtBeginDocument{%
3253 %%% \ifpackageloaded{hyperref}{%
3254 %%%   \def\@wrlab#1#2{%
3255 %%%     \protected@write\@labfile{%
3256 %%%       {\string\indexentry{#1} #2|hyperpage}{\thepage}}%
3257 %%%     \endgroup
3258 %%%     \@esphack}}{%

```

```

3259 %%%% \def\@wrlab#1#2{%
3260 %%%% \protected@write\@labfile{%
3261 %%%% {\string\indexentry{[#1] #2}{\arabic{page}}}%
3262 %%%% \endgroup
3263 %%%% \@esphack}}

3264 % Catalog-label writers. Two language-related traps must be avoided here:
3265 % 1. The label NAMES contain ':' and '.', which babel makes active in
3266 % some languages (e.g. Brazilian's "shorthand" set). If the write
3267 % argument is tokenised while one of those is active, pdfTeX aborts
3268 % with "<inserted text> }\endwrite" because the shorthand expansion
3269 % injects unbalanced braces. \detokenize emits its argument as
3270 % catcode-12 characters regardless of the current activation, so the
3271 % label NAMES survive whatever language is current at writer time.
3272 % 2. Spanish babel redefines \@roman (and therefore \roman{...}) to
3273 % produce small-caps roman numerals via \es@scroman, whose expansion
3274 % contains an \uppercase group that is unbalanced inside \write. The
3275 % same crash signature appears. The fix is to bypass LaTeX's
3276 % \roman/\arabic counter wrappers entirely and use the raw e-TeX
3277 % primitives \romannumeral and \number directly on the \c@page count
3278 % register --- both are fully expandable, language-agnostic, and
3279 % always produce plain ASCII digits or lowercase roman letters.
3280 % \Roman{page} is left in place because babel-spanish only redefines
3281 % \@Roman under the (non-default) ucroman option.
3282 \AtBeginDocument{%
3283 \ltx@ifpackageloaded{hyperref}{%
3284 \def\coppe@bibEnd{%
3285 \immediate\write\@auxout{%
3286 \string\newlabel{\detokenize{bib:end}}%
3287 {}{\number\c@page}{\detokenize{page.}\number\c@page}{}}}%
3288 \def\coppe@bibBegin{%
3289 \immediate\write\@auxout{%
3290 \string\newlabel{\detokenize{bib:begin}}%
3291 {}{\number\c@page}{\detokenize{page.}\number\c@page}{}}}%
3292 \def\coppe@mainBegin{%
3293 \immediate\write\@auxout{%
3294 \string\newlabel{\detokenize{front:pageno}}%
3295 {}{\Roman{page}}{\detokenize{page.}\romannumeral\c@page}{}}}%
3296 \def\coppe@hasLof{%
3297 \immediate\write\@auxout{%
3298 \string\newlabel{\detokenize{cat:lofflag}}%
3299 {}{\detokenize{:~il.}}{\detokenize{page.}\romannumeral\c@page}{}}}%
3300 }{%
3301 \def\coppe@bibEnd{%
3302 \immediate\write\@auxout{%
3303 \string\newlabel{\detokenize{bib:end}}{\number\c@page}{}}}%
3304 \def\coppe@bibBegin{%
3305 \immediate\write\@auxout{%
3306 \string\newlabel{\detokenize{bib:begin}}{\number\c@page}{}}}%
3307 \def\coppe@mainBegin{%
3308 \immediate\write\@auxout{%
3309 \string\newlabel{\detokenize{front:pageno}}{\Roman{page}}{}}}%
3310 \def\coppe@hasLof{%
3311 \immediate\write\@auxout{%
3312 \string\newlabel{\detokenize{cat:lofflag}}{\detokenize{:~il.}}{}}}%

```

```

3313 }%
3314 }
3315 % The reference list is now produced by biblatex's \printbibliography (see the
3316 % bibliography setup after babel, above). The former thebibliography
3317 % redefinition and its \bibindent are no longer used and have been removed; the
3318 % bib:begin/bib:end catalog labels are now written by \AtBeginBibliography /
3319 % \AtEndBibliography via \coppe@bibBegin and \coppe@bibEnd, defined just above.

```

longquote (*env.*)

```

3320 \newlength{\recuolongquote}
3321 \setlength{\recuolongquote}{4cm}
3322
3323 \newenvironment*{longquote}[1][default]{%
3324 \begin{list}{}{%
3325 \setlength{\leftmargin}{\recuolongquote}%
3326 \setlength{\rightmargin}{0pt}%
3327 \setlength{\itemindent}{0pt}%
3328 \setlength{\listparindent}{0pt}%
3329 \setlength{\parsep}{0pt}%
3330 \setlength{\topsep}{\baselineskip}%
3331 }%
3332 \item\relax
3333 \footnotesize
3334 \singlespacing
3335 \ifthenelse{\not\equal{#1}{default}}{%
3336 {\itshape\selectlanguage{#1}}%
3337 }%
3338 \ignorespaces
3339 }{%
3340 \end{list}%
3341 \ignorespacesafterend
3342 }

3343 \newenvironment{theglossary}{%
3344 \if@twocolumn%
3345 \@restonecoltrue\onecolumn%
3346 \else%
3347 \@restonecolfalse%
3348 \fi%
3349 \@mkboth{\MakeUppercase\glossaryname}%
3350 {\MakeUppercase\glossaryname}%
3351 \chapter*{\glossaryname}%
3352 \if@coppeglosspre
3353 \coppe@listtoc{\glossaryname}%
3354 \else
3355 \addcontentsline{toc}{chapter}{\glossaryname}%
3356 \fi
3357 \list{}
3358 {\setlength{\listparindent}{0in}%
3359 \setlength{\labelwidth}{1.0in}%
3360 \setlength{\leftmargin}{1.5in}%
3361 \setlength{\labelsep}{0.5in}%
3362 \setlength{\itemindent}{0in}}%
3363 \sloppy}%

```

```

3364 {\if@restonecol\twocolumn\fi%
3365 \endlist}
3366 %
3367 \renewenvironment{theindex}{%
3368 \if@twocolumn
3369 \@restonecolfalse
3370 \else
3371 \@restonecoltrue
3372 \fi
3373 \twocolumn[\@makeschapterhead{\indexname}]{%
3374 \mkboth{\MakeUppercase\indexname}%
3375 {\MakeUppercase\indexname}%
3376 \thispagestyle{plain}\parindent\z@
3377 \addcontentsline{toc}{chapter}{\indexname}
3378 \parskip\z@ \@plus .3\p@\relax
3379 \columnseprule \z@
3380 \columnsep 35\p@
3381 \let\item\@idxitem}
3382 {\if@restonecol\onecolumn\else\clearpage\fi}
3383 \newcommand\listabbreviationname{\coppemainstring{listabbreviation}}
3384 \newcommand\listsymbolname{\coppemainstring{listsymbol}}
3385 \newcommand\glossaryname{\coppemainstring{glossary}}
3386 %
3387 \newcommand\local@advisorstring{Orientador}
3388 \newcommand\foreign@advisorstring{Advisor}
3389 % Fed by the string dispatcher instead of a hard-coded Portuguese literal.
3390 % Expansion is lazy (it happens inside \makefrontpage), so \coppe@mainlang is
3391 % already set by the time this is used.
3392 \ifthenelse{\boolean{maledoc}}{%
3393 \newcommand\local@approvedname{\coppemainstring{approvedmale}}%
3394 }{%
3395 \newcommand\local@approvedname{\coppemainstring{approvedfemale}}%
3396 }
3397 \ifthenelse{\boolean{maledoc}}{%
3398 \newcommand\local@approvedonname{\coppemainstring{approvedonmale}}%
3399 }{%
3400 \newcommand\local@approvedonname{\coppemainstring{approvedonfemale}}%
3401 }
3402 \newcommand\local@universityname{Universidade Federal do Rio de Janeiro}
3403 \newcommand\local@deptstring{Programa}
3404 \newcommand\foreign@deptstring{Department}
3405 \newcommand\local@cityname{Rio de Janeiro}
3406 \newcommand\local@statename{RJ}
3407 \newcommand\local@countryname{Brasil}
3408 %
3409 % 3.1.2.1.1(e) of the 2026 manual requires the natureza, the objetivo, the
3410 % name of the institution AND the area de concentracao in this block. The area
3411 % was missing; it is appended as its own sentence when \areaconcentracao was
3412 % given, and simply omitted otherwise.
3413 \newcommand\frontcover@maintext{
3414 \sloppy\nohyphens{\local@doctype\ de \@degree\name\
3415 \ifthenelse{\boolean{maledoc}}{apresentado}{apresentada}
3416 ao Programa de P{\` o}s-gradua{\c c}{\~ a}o em \local@deptname,
3417 COPPE, da \local@universityname, como parte dos requisitos

```

```

3418 necess{' a}rios {' a} obten{c c}{~ a}o do t{' i}tulo de
3419 \local@degname\ em \local@deptname.}%
3420 \ifthenelse{equal{\coppe@areaconc}{}}{ }{%
3421   \par\vspace{1ex}%
3422   \sloppy\nohyphens{' Area de concentra{c c}{~ a}o: \coppe@areaconc.}}%
3423 }
3424 %
3425 \newcommand\frontpage@maintext{
3426   \noindent {\MakeUppercase\local@doctype}
3427   \ifthenelse{\boolean{maledoc}}{SUBMETIDO}{SUBMETIDA}
3428   \sloppy\nohyphens{AO CORPO DOCENTE DO INSTITUTO ALBERTO LUIZ COIMBRA
3429   DE P{' O}S-GRADUA{c c}{~ A}O E PESQUISA DE ENGENHARIA DA
3430   UNIVERSIDADE FEDERAL DO RIO DE JANEIRO COMO PARTE DOS REQUISITOS
3431   NECESS{' A}RIOS PARA A OBTEN{c c}{~ A}O DO GRAU DE
3432   {\MakeUppercase\local@degname} EM CI{\^E}NCIAS EM
3433   {\MakeUppercase\local@deptname.\par}}%
3434 }
3435 %
3436 \newcommand\frontpage@bottomtext{%
3437   \begin{center}
3438     {\MakeUppercase{\local@cityname, \local@statename\ -- \local@countryname}}\par
3439     {\MakeUppercase\local@monthname\ DE \coppe@docyear}
3440   \end{center}%
3441 }
3442 %
3443 \newcommand\abstract@toptext{%
3444   \noindent Resumo \ifthenelse{\boolean{maledoc}}{do}{da}
3445   \local@doctype\ \ifthenelse{\boolean{maledoc}}{apresentado}{apresentada}
3446   \sloppy\nohyphens{' a} COPPE/UFRJ como parte dos requisitos
3447   necess{' a}rios para a obten{c c}{~ a}o do grau de
3448   \local@degname\ em Ci{\^e}ncias (\@degree)}
3449 }
3450 \newcommand\foreignabstract@toptext{%
3451   \noindent \sloppy\nohyphens{Abstract of \foreign@doctype\ presented to
3452   COPPE/UFRJ as a partial fulfillment of the requirements for the
3453   degree of \foreign@degname\ of Science (\@degree)}
3454 }
3455 %

```

Late-load packages: setspace, hyperref, biblatex

These three packages *must* be the last things the class loads, and in this exact order:

- **setspace** – supplies `\singlespacing`, `\onehalfspacing` and friends. It is loaded here so the line-spacing state it carries around is settled before **hyperref** patches all the cross-reference and TOC plumbing.
- **hyperref** – patches an enormous number of LaTeX internals (`\section`, `\label`, `\ref`, `\bibcite`, list-of-* commands, ...). Loading it as late as possible is the standard rule of thumb so it sees the final version of those macros, including the ones the **coppe** class redefines (e.g. `\@chapter`, `\listofquadros`).

- `biblatex` – must come after `hyperref` so its citation links and back-references hook into `hyperref` correctly; otherwise the `\cite` commands print as plain (non-clickable) text.

```

3456 \RequirePackage{setspace}
3457 % Apply the default line spacing now that setspace is available. The flag was
3458 % set above by \DeclareOption{doublespacing}; absent the option, one-and-a-half.
3459 \if@coppepdoublespace\doublespacing\else\onehalfspacing\fi
3460 % pdfx MUST precede hyperref -- it loads hyperref itself, with the options
3461 % PDF/A needs, and cannot fix up an already-loaded one. Level a-2b rather than
3462 % a-1b (the "ISO 19005-1" of Annex I): a-1b forbids transparency, which
3463 % tcolorbox and several graphics packages produce, so a-1b would fail on
3464 % documents that are otherwise perfectly fine. a-2b is ISO 19005-2, accepted by
3465 % Pantheon and by the SiBI's own instructions, which also list PDF/A-2b.
3466 \if@coppepdfa
3467   \RequirePackage[a-2b]{pdfx}
3468 % pdfTeX stamps /PTEX.Fullbanner into the document information dictionary.
3469 % It is not one of the keys ISO 19005-2 6.6.2.3.2 maps to an XMP property, and
3470 % no extension schema declares it, so it is a DocInfo entry with no XMP
3471 % counterpart -- a finding a PDF/A checker has every reason to raise, and one
3472 % that costs nothing to remove. -1 sets every suppression bit.
3473 %
3474 % The two engines spell this differently, and not just in the name. pdfTeX has
3475 % \pdfsuppressptexinfo, whose bits only ever govern PTEX.* entries, so -1 (all
3476 % of them) is safe. LuaTeX moved the pdfTeX integer parameters behind
3477 % \pdfvariable in 0.85 AND renamed this one to suppressoptionalinfo, whose bits
3478 % reach further: besides the four PTEX.* entries (1, 2, 4, 8) it can also
3479 % suppress Creator, the dates, Producer, Trapped and -- the dangerous one --
3480 % the trailer /ID, which PDF/A requires. So -1 is exactly what must NOT be
3481 % written there; 15 is the four PTEX.* bits and nothing else. Checked on a real
3482 % LuaTeX run: /ID survives.
3483 %
3484 % Getting the name wrong is not a quiet failure. \pdfvariable with an unknown
3485 % variable name makes LuaTeX typeset the name, and the document stops with
3486 % "Missing \begin{document}" -- which is how the first attempt at this was
3487 % caught.
3488 \ifdefined\pdfsuppressptexinfo
3489   \pdfsuppressptexinfo=-1
3490 \else
3491   \ifdefined\pdfvariable
3492     \pdfvariable suppressoptionalinfo=15
3493   \fi
3494 \fi
3495 % pdfx takes the XMP metadata from \jobname.xmpdata, which it reads when it is
3496 % LOADED -- before the preamble has run, so before \title, \author and
3497 % \keyword exist. The file is therefore WRITTEN by the class and picked up on
3498 % the NEXT run. The build already runs pdflatex three times for cross-
3499 % references, so the deposited PDF always carries current metadata; only a
3500 % first-ever compilation of a brand-new file emits pdfx's "no \jobname.xmpdata"
3501 % warning and lacks dc:title.
3502 %
3503 % The write happens at the END of the document, not at \begin{document}. This
3504 % class has always documented -- and every example follows it -- that \title,
3505 % \author, \department and \keyword go AFTER \begin{document}, next to

```

```

3506 % \maketitle. A write at \begin{document} therefore ran before any of them
3507 % existed and died with "Undefined control sequence" on \@authname, so the
3508 % pdfa option was unusable in exactly the style the manual teaches. It only
3509 % ever worked for a document that set its metadata in the preamble. At the end
3510 % everything is set whichever style the author chose, and \immediate still puts
3511 % the file on disk in time for the next run.
3512 %
3513 % The hook is enddocument/afteraux, and the stream is \@mainaux rather than one
3514 % of our own. TeX has sixteen output streams and example.tex was already using
3515 % all of them (toc, lof, lot, loa, lol, loq, los, idx, glo, aux, bcf, ...), so
3516 % \newwrite here failed with "No room for a new \write" on precisely the
3517 % document that most needs validating. enddocument/afteraux fires after LaTeX
3518 % has closed the .aux, which frees that stream for the one \immediate\openout
3519 % we need and costs the document nothing. Kernels older than 2020-10-01 have no
3520 % such hook; there the class falls back to \AtEndDocument with a stream of its
3521 % own, which is the pre-existing behaviour.
3522 %
3523 % Every field is guarded, and the guards are not cosmetic. \coppe@selecttitle
3524 % is \@nameuse{coppe@title@<lang>}, and \csname on a name that does not exist
3525 % yet MAKES it \relax: unexpandable, so \write puts the control sequence NAME
3526 % into the file instead of the title. The .xmpdata then carried the literal
3527 % \coppe@title@brazilian, pdfx read it back on the following run, and the
3528 % document stopped compiling at all -- a single bad run bricked the file until
3529 % someone deleted the .xmpdata by hand. A field that is not set is now simply
3530 % omitted, which pdfx handles, instead of poisoning the next run.
3531 %
3532 % The writes must be \immediate. \protected@write defers to the shipout, and
3533 % the stream is closed here and now, so deferred writes would land nowhere --
3534 % which is exactly what happened first time round: a 0-byte .xmpdata.
3535 % \let\protect\string inside the group gives the protection that
3536 % \protected@write would have given: accents and other robust commands are
3537 % written literally instead of being expanded. pdfx reads the .xmpdata as
3538 % LaTeX, so \a arrives there intact.
3539 \ifdefined\AddToHook
3540   \let\coppe@xmpout\@mainaux
3541   \def\coppe@xmphook{\AddToHook{enddocument/afteraux}}%
3542 \else
3543   \newwrite\coppe@xmpout
3544   \let\coppe@xmphook\AtEndDocument
3545 \fi
3546 \coppe@xmphook{%
3547   \gdef\coppe@xmpkeys{%
3548     \count@=0
3549     \@whilenum\count@<\value{keywords}\do{%
3550       \ifnum\count@>0
3551         \g@addto@macro\coppe@xmpkeys{\string\sep\space}%
3552       \fi
3553       \protected@edef\coppe@tmp{\csname CoppeKeyword:\the\count@\endcsname}%
3554       \expandafter\g@addto@macro\expandafter\coppe@xmpkeys
3555       \expandafter{\coppe@tmp}%
3556       \advance\count@ by 1}%
3557   \immediate\openout\coppe@xmpout=\jobname.xmpdata
3558   \begingroup
3559   \let\protect\string

```

```

3560 \ifundefined{coppe@title@coppe@mainlang}{}%
3561 \immediate\write\coppe@xmpout{\string\Title{\coppe@selecttitle}}}%
3562 \ifundefined{coppe@authname}{}%
3563 \immediate\write\coppe@xmpout{%
3564 \string\Author{\coppe@authname\space\coppe@authsur}}}%
3565 \immediate\write\coppe@xmpout{\string\Keywords{\coppe@xmpkeys}}}%
3566 \ifthenelse{equal{\coppe@deptcode}{}}{ }{%
3567 \immediate\write\coppe@xmpout{\string\Subject{\local@doctype\space de
3568 \coppe@deptname. Programa de \meta@deptname\space(\coppe@deptcode),
3569 COPPE/UFRJ}}}%
3570 \immediate\write\coppe@xmpout{\string\Publisher{\local@universityname}}}%
3571 \ifundefined{coppe@docyear}{}%
3572 \immediate\write\coppe@xmpout{\string\Date{\coppe@docyear}}}%
3573 \endgroup
3574 \immediate\closeout\coppe@xmpout}%
3575 \fi
3576 \RequirePackage{hyperref}
3577 % Bibliography engine: the full coppe BibLaTeX/biber ABNT style, replacing the
3578 % former natbib+bibtex setup. The numbers class option selects the numeric
3579 % citation style; otherwise the author-date default is used. natbib=true keeps
3580 % \citet/\citep available for users migrating from natbib.
3581 \if@coppenumeric
3582 \RequirePackage[backend=biber,datamodel=coppe,%
3583 bibstyle=coppe-numeric,citestyle=coppe-numeric,natbib=true]{biblatex}
3584 \else
3585 \RequirePackage[backend=biber,datamodel=coppe,%
3586 bibstyle=coppe,citestyle=coppe,natbib=true]{biblatex}
3587 \fi
3588 % Reference-list heading "Referencias" (pt) / "References" (en) -- manual2024
3589 % 3.1.4.1 requires the post-textual list to be titled "Referencias", not
3590 % "Bibliografia" (babel's default \bibname). The title is rendered through
3591 % \coppestring so it follows the current main language in BOTH the chapter
3592 % heading and the .toc entry (biblatex's \bibstring proved unreliable across
3593 % the .toc write, and babel's \bibname caption override was equally fragile).
3594 \defbibheading{bibliography}{%
3595 \chapter*{\coppestring{references}}%
3596 \addcontentsline{toc}{chapter}{\coppestring{references}}}
3597 % The catalog page range (bib:begin/bib:end) was written by the old
3598 % thebibliography redefinition, which biblatex's \printbibliography does not
3599 % use. biblatex offers \AtBeginBibliography but no end counterpart, so the
3600 % begin label uses that hook and the end label is emitted by wrapping
3601 % \printbibliography to write it just after the reference list.
3602 \AtBeginBibliography{\coppe@bibBegin}
3603 \let\coppe@orig@printbibliography\printbibliography
3604 \renewcommand\printbibliography[1][]{%
3605 \coppe@orig@printbibliography[#1]}%
3606 \coppe@bibEnd}
3607 \endclass
3608 \glossary
3609 actual '='
3610 quote '!'
3611 level '>'
3612 %%% delim_0 " , p. "

```



```

3613 delim_0 "\\dotfill "
3614 lethead_flag 0
3615 headings_flag 0
3616 preamble
3617 "\n\\begin{theglossary}\n  \\makeatletter"
3618 postamble
3619 "\n  \\end{theglossary}\n"
3620 </glossary>

3621 <*class>
3622 \\newcommand{\\annex}{
3623   \\renewcommand{\\appendixname}{\\coppestring{annex}}%
3624   \\appendix
3625   \\@coppeannextrue
3626 % \\appendix has already re-issued the centred chapter format, but it baked in
3627 % the \\appendixname of that moment. Re-issue it now that the name is "Anexo".
3628   \\coppe@appendixheadings
3629 }
3630 </class>

```

BibLaTeX style files

Generated from this .dtx by coppe.ins (one docstrip module each).

```

3631 <*dbx>
3632 \\ProvidesFile{coppe.dbx}[2026/09/09 v4.1 CoppeTeX ABNT datamodel]
3633 % Custom entry types beyond biblatex's standard datamodel (games, TV, maps).
3634 % map = cartographic document (NBR 6023 4.2.12); biblatex has no native @map.
3635 % The remaining ABNT "exotic" types (legislation, jurisdiction, legal, patent,
3636 % standard, music, audio, image, artwork, video, dataset, letter, periodical,
3637 % performance) are already in biblatex's standard datamodel.
3638 \\DeclareDatamodelEntrytypes{game,videogame,boardgame,tvshow,map}
3639 % Custom fields.
3640 \\DeclareDatamodelFields[type=field,datatype=literal]{course}      % "(... em <course>)"
3641 \\DeclareDatamodelFields[type=field,datatype=literal]{coppedegree} % degree word
3642 \\DeclareDatamodelFields[type=field,datatype=literal]{platform}    % videogame platform
3643 \\DeclareDatamodelFields[type=field,datatype=literal]{scale}       % map scale (escala)
3644 \\DeclareDatamodelFields[type=list,datatype=literal]{artist}      % game/illustration art
3645 \\DeclareDatamodelFields[type=list,datatype=literal]{director}    % movie/tvshow/video
3646 \\DeclareDatamodelFields[type=list,datatype=literal]{producer}
3647 \\DeclareDatamodelFields[type=field,datatype=literal]{caderno}     % newspaper section
3648 \\DeclareDatamodelFields[type=field,datatype=literal]{relator}     % jurisprudence rapporteur
3649 \\DeclareDatamodelFields[type=field,datatype=literal]{filingdate}  % patente: deposito
3650 \\DeclareDatamodelFields[type=field,datatype=literal]{grantdate}  % patente: concessao
3651 \\DeclareDatamodelEntryfields{course,coppedegree}
3652 \\DeclareDatamodelEntryfields[game,videogame,boardgame,software,movie,tvshow,video]%
3653   {artist,platform,director,producer,version}
3654 \\DeclareDatamodelEntryfields[map]{scale}
3655 \\DeclareDatamodelEntryfields[article]{caderno}
3656 \\DeclareDatamodelEntryfields[jurisdiction]{relator}
3657 \\DeclareDatamodelEntryfields[patent]{filingdate,grantdate}
3658 </dbx>
3659 <*bbx>
3660 \\ProvidesFile{coppe.bbx}[2026/09/09 v4.1 CoppeTeX ABNT bibliography style]
3661 % The reference list is single-spaced (ABNT 2.4) via \\singlespacing in

```

```

3662 % \AtBeginBibliography. coppe.cls already loads setspace, but the style must work
3663 % standalone too (e.g. \usepackage[bibstyle=coppe]{biblatex} in a plain article),
3664 % so load setspace here as well (idempotent).
3665 \RequirePackage{setspace}
3666 % ABNT reference list keeps the date at the end (imprenta), same for author-date
3667 % and numeric modes, so we derive from the standard style, not authoryear.
3668 \RequireBibliographyStyle{standard}
3669
3670 % Note: standard style never dashes repeated authors, which already matches the
3671 % post-2020 ABNT rule (repeat the entry in full), so no 'dashed' option is needed.
3672 \ExecuteBibliographyOptions{%
3673   maxbibnames=99,      % ABNT 4.3.1.3: list all authors by default
3674   giveninits=false,
3675   uniquename=false,
3676   uniquelist=false,
3677   labeldateparts=true, % needed for author-date extradate (1999a/1999b)
3678   sorting=nyt,
3679 }
3680
3681 % Language strings (Disponível em / Acesso em / In, etc.)
3682 \DeclareLanguageMapping{brazilian}{brazilian-coppe}
3683 \DeclareLanguageMapping{english}{english-coppe}
3684
3685 % --- Names: all reversed "FAMILY, Given", separated by ";" in the list ---
3686 \DeclareNameAlias{author}{family-given}
3687 \DeclareNameAlias{editor}{family-given}
3688 \DeclareNameAlias{translator}{family-given}
3689 \DeclareNameAlias{default}{family-given}
3690 \DeclareDelimFormat[bib,biblist]{multinamedelim}{\addsemicolon\space}
3691 \DeclareDelimFormat[bib,biblist]{finalnamedelim}{\addsemicolon\space}
3692
3693 % ABNT (NBR 6023 8.1.1.4): a collection entered under the name of its
3694 % organizer prints the kind of responsibility in parentheses after the
3695 % name, abbreviated, lowercase and in the singular even for several
3696 % organizers -- e.g. "SOUSA, A. M. de; OLIVEIRA, L. da C. (org.)". The
3697 % standard style prints ", Org." (comma, no parentheses) when the editor
3698 % takes the author slot, so override editor+others to wrap the role string
3699 % in "(...)". The host position ("In: NAME (org.)") already parenthesises
3700 % the role via coppe:bookeditor below. The role word itself is the
3701 % lowercase, singular "org." (and "ed.", "cur.", ...) set in the language
3702 % packs (brazilian-coppe.lbx et al.).
3703 \renewbibmacro*{editor+others}{%
3704   \ifbool{expr{ test \ifuseeditor and not test {\ifnameundef{editor}}} }
3705     {\printnames{editor}%
3706      \setunit*\addspace}%
3707     \printtext{\mkbibparens{\usebibmacro{editor+othersstrg}}}%
3708     \clearname{editor}}
3709   {}
3710
3711 % Same parenthesised role for drivers that take the editor through the plain
3712 % editor macro instead of editor+others (e.g. @manual).
3713 \renewbibmacro*{editor}{%
3714   \ifbool{expr{ test \ifuseeditor and not test {\ifnameundef{editor}}} }
3715     {\printnames{editor}%
3716      \setunit*\addspace}%

```

```

3716 \printtext{\mkbibparens{\usebibmacro{editorstrg}}}%
3717 \clearname{editor}}
3718 {}
3719
3720 % Uppercase the surname (family + prefix) only inside the bibliography list;
3721 % citations keep normal caps (post-2020 rule).
3722 % ABNT date (NBR 6023 §4.3.5.5.2): "dia mês. ano" -- without the brazilian "de"
3723 % between the parts (e.g. "4 fev. 1992", "abr. 1996"). #1=year, #2=month, #3=day
3724 % field names; the month stays abbreviated via \mkbibmonth.
3725 % Autor-entidade com subordinacao: a 4.3.2.13 poe em caixa alta a jurisdicao ou
3726 % a entidade superior, e deixa em caixa normal o que vem depois -- o orgao
3727 % subordinado ("BRASIL. Supremo Tribunal Federal.") e o qualificador entre
3728 % parenteses ("SAO PAULO (Estado)."). \mkbibnamefamily so conhecia nome de
3729 % pessoa e punha tudo em caixa alta. Corta-se no primeiro " (" e, dentro do que
3730 % sobra a esquerda, no primeiro ". ": o que estiver antes vai para caixa alta,
3731 % o resto sai como foi digitado.
3732 % O biber entrega o nome literal ENTRE CHAVES ("{Brasil. Supremo Tribunal
3733 % Federal}"), e as chaves escondem o ponto do casamento de padrao abaixo: sem
3734 % desembrulhar, o nome inteiro caia no \MakeUppercase. \coppe@ucstrip tira a
3735 % chave externa -- um argumento delimitado que seja um grupo unico perde as
3736 % chaves na captura.
3737 % Autor-entidade: a 4.3.2.13 poe em caixa alta a jurisdicao ou a entidade
3738 % superior e deixa o orgao subordinado em caixa normal -- "BRASIL. Supremo
3739 % Tribunal Federal." Qual das duas coisas o nome e, so quem escreve sabe:
3740 % "Associacao Brasileira de Normas Tecnicas" e uma entidade so, e vai inteira
3741 % para versal; "Brasil. Supremo Tribunal Federal" tem subordinacao, e a parte
3742 % depois do ponto fica como esta.
3743 %
3744 % A regra, entao, e mecanica e previsivel: um nome que TENHA ponto sai
3745 % exatamente como o autor digitou; um nome sem ponto vai para caixa alta. O
3746 % teste e feito sobre a forma detokenizada, porque entre as palavras de um
3747 % nome o biblatex nao poe espaco -- poe \bibnamedelim e \bibnamedelimi --, e
3748 % procurar ". " no fluxo de tokens nunca encontra nada.
3749 % O parentese conta junto com o ponto: "Sao Paulo (Estado)" leva o
3750 % qualificador em caixa normal, e so quem escreve sabe disso.
3751 % O argumento chega como \namepartfamily -- uma sequencia de controle, nao o
3752 % texto. Detokenizar sem expandir antes devolve a string "\namepartfamily",
3753 % que nao tem ponto nenhum, e por isso TODO nome caia no \MakeUppercase.
3754 \protected\def\coppe@ucfamily#1{%
3755 \protected@edef\coppe@uctmp{#1}%
3756 \edef\coppe@ucstr{\detokenize\expandafter{\coppe@uctmp}}%
3757 \expandafter\coppe@ucscan\coppe@ucstr.\coppe@ucstop{#1}}
3758 \protected\def\coppe@ucscan#1.#2\coppe@ucstop#3{%
3759 \ifblank{#2}
3760 {\expandafter\coppe@ucscanp\coppe@ucstr(\coppe@ucstop{#3}}
3761 {#3}}
3762 \protected\def\coppe@ucscanp#1(#2\coppe@ucstop#3{%
3763 \ifblank{#2}{\MakeUppercase{#3}}{#3}}
3764 % A 4.3 separa os meses de um intervalo por barra -- "jan./jun. 1991" --, e o
3765 % biblatex usa travessao.
3766 \renewcommand*{\bibdaterangesep}{/}
3767 \protected\def\coppe@abntdate#1#2#3{%
3768 \iffieldundef{#3}{\stripzeros{\thefield{#3}}\nobreakspace}%
3769 \iffieldundef{#2}{\mkbibmonth{\thefield{#2}}\iffieldundef{#1}{\space}}%

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```

3770 \iffielddundef{#1}{\stripzeros{\thefield{#1}}}}
3771 \AtBeginBibliography{%
3772 \singlespacing % entries single-spaced internally (2.4)
3773 \setlength{\bibitemsep}{\baselineskip}% one blank line between entries (2.4)
3774 \renewcommand*{\mkbibnamefamily}[1]{\coppe@ucfamily{#1}}%
3775 \renewcommand*{\mkbibnameprefix}[1]{\MakeUppercase{#1}}%
3776 \let\mkbibdatelong\coppe@abntdate % ABNT date format (drop the "de")
3777 \let\mkbibdateshort\coppe@abntdate
3778 }
3779
3780 % --- Titles: bold for whole-document types; plain for parts; entry-by-title
3781 % (no author/editor) -> not bold, first word in CAPS (ABNT 4.2). ---
3782 % \coppeUCtitle uppercases the first SIGNIFICANT word of the title (ABNT 4.2).
3783 % If the title starts with an article (o/a/os/as/um/uma/uns/umas, the/an), the
3784 % article AND the next word are uppercased; otherwise just the first word. The
3785 % delimited macros split off words; \coppeUCkeep strips the trailing-space
3786 % sentinel so no spurious space is left before the following punctuation.
3787 \protected\def\coppeUCtitle#1{\coppeUCtitleA#1 \coppeUCstop}
3788 \protected\def\coppeUCtitleA#1 #2\coppeUCstop{%
3789 \coppeUCifart{#1}
3790 {\MakeUppercase{#1}\ifblank{#2}{\space\coppeUCtitleB#2\coppeUCstop}}%
3791 {\MakeUppercase{#1}\ifblank{#2}{\space\coppeUCkeep#2\coppeUCstop}}%
3792 \protected\def\coppeUCtitleB#1 #2\coppeUCstop{%
3793 \MakeUppercase{#1}\ifblank{#2}{\space\coppeUCkeep#2\coppeUCstop}}
3794 \protected\def\coppeUCkeep#1 \coppeUCstop{#1}
3795 % \coppeUCifart{word}{if-article}{if-not}: case-insensitive membership test.
3796 \protected\def\coppeUCifart#1{%
3797 \begingroup
3798 \lowercase{\def\coppeUC@w{#1}}%
3799 \edef\coppeUC@w{\coppeUC@w}%
3800 \expandafter\in@\expandafter{\coppeUC@w}{,a,o,as,os,um,uma,uns,umas,the,an,}%
3801 \ifin@\aftergroup\@firstoftwo\else\aftergroup\@secondoftwo\fi
3802 \endgroup}
3803 \DeclareFieldFormat{title}{%
3804 \ifboolexpr{ test {\ifnameundef{labelname}} }
3805 {#1}%
3806 {\mkbibbold{#1}}}
3807 % First word of the title in CAPS for entry-by-title (ABNT 4.2). The titlecase
3808 % format receives the RAW title text (the title format above only sees the
3809 % field-printing commands), so \coppeUCtitle can split off and uppercase the
3810 % first word here.
3811 \DeclareFieldFormat{titlecase}{%
3812 \ifboolexpr{ test {\ifnameundef{labelname}} }
3813 {\coppeUCtitle{#1}}%
3814 {#1}}
3815 \DeclareFieldFormat[article,incollection,inbook,inproceedings,suppperiodical,review]{title}{%
3816 \DeclareFieldFormat[thesis]{title}{\mkbibbold{#1}}
3817 % biblatex's standard style quotes patent (and unpublished) titles; ABNT bolds
3818 % the title of a whole document, so override those back to bold.
3819 \DeclareFieldFormat[patent]{title}{\mkbibbold{#1}}
3820 % Legal documents (NBR 6023 4.2.6): the law's identification/ementa is NOT
3821 % highlighted; the entry leads with the jurisdiction (entity author, CAPS).
3822 \DeclareFieldFormat[legislation,jurisdiction,legal]{title}{#1}
3823 \DeclareFieldFormat[legislation,jurisdiction,legal]{volume}{v\adddot\adnbspace#1}

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3824 \DeclareFieldFormat[legislation,jurisdiction,legal]{number}{n\addot\addnbspace#1}
3825 \DeclareFieldFormat[journaltitle]{\mkbibbold{#1}}
3826 \DeclareFieldFormat{booktitle}{\mkbibbold{#1}}
3827 \DeclareFieldFormat{maintitle}{\mkbibbold{#1}}
3828
3829 % --- Pages: ABNT uses "p." (also for ranges); theses use "f." for pagetotal ---
3830 \DeclareFieldFormat[pagetotal]{#1\addnbspace p\addot}
3831 \DeclareFieldFormat[thesis]{pagetotal}{#1\addnbspace f\addot}
3832 \DeclareFieldFormat{pages}{p\addot\addnbspace#1}
3833
3834 % --- Periodical volume / number prefixes ---
3835 \DeclareFieldFormat[article]{volume}{v\addot\addnbspace#1}
3836 \DeclareFieldFormat[article]{number}{n\addot\addnbspace#1}
3837
3838 % --- "In:" before host title (incollection/inproceedings/inbook) ---
3839 \renewbibmacro*{in:}{\printtext{\bibstring{in}\addcolon\space}}
3840
3841 % --- Online: "Disponível em: <url>." + "Acesso em: <date>." via lbx strings ---
3842 \DeclareFieldFormat[url]{\bibstring{availablefrom}\addcolon\space\url{#1}}
3843 % O 4.2 do manual fecha toda entrada eletrônica assim, sem exceção:
3844 %   Disponível em: <url>. Acesso em: <data>.
3845 % O biblatex, por padrão, põe a data de acesso entre parênteses e sem
3846 % dois-pontos -- "<url> (Acesso em <data>)" --, que é a convenção dele, não a
3847 % da norma.
3848 \DeclareFieldFormat[urldate]{\bibstring{urlseen}\addcolon\space#1}
3849 % A descrição física ("1 carta", "3 fotografias", "1 partitura") vem ANTES do
3850 % bloco eletrônico em todos os exemplos da 4.2. Os drivers padrão do biblatex
3851 % a imprimem DEPOIS, então ela é impressa aqui e o campo é limpo, para não
3852 % sair duas vezes.
3853 \renewbibmacro*{url+urldate}{%
3854   \iffielddundef{addendum}
3855     {}
3856     {\printfield{addendum}\clearfield{addendum}%
3857       \setunit{\addot\space}}%
3858   \iffielddundef{url}
3859     {}
3860     {\printfield{url}%
3861       \iffielddundef{urlyear}{\setunit{\addot\space}\printurldate}}
3862
3863 % --- ABNT article driver: no "In:"; Journal, Local, v., n., p., date, DOI/URL.
3864 %   Newspapers(entrystubtype=newspaper, set by the @artigodejornal source map)
3865 %   differ: the newspaper name (journaltitle) and city, then either
3866 %     "<date>. <caderno>, p. <pages>" (when a caderno/section is given), or
3867 %     "p. <pages>, <date>" (page before date, when none),
3868 %   per NBR 6023 §4.2.3.6. ---
3869 \DeclareBibliographyDriver{article}{%
3870   \usebibmacro{bibindex}%
3871   \usebibmacro{begentry}%
3872   \usebibmacro{author/translator+others}%
3873   \setunit{\labelnamepunct}\newblock
3874   \usebibmacro{title}%
3875   \newunit\newblock
3876   \printfield{journaltitle}%
3877 % "J. Manag. Stud., Oxford": a vírgula depois de um título abreviado terminado

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3878 % em ponto e engolida pelo rastreador de pontuacao, e a 4.2.3.4 a exige.
3879 \iflistundef{location}{\printtext{,}\setunit{\space}\printlist{location}}%
3880 \iffieldequalstr{entrysubtype}{newspaper}
3881   {\iffieldundef{caderno}
3882     {\setunit{\addcomma\space}\printfield{pages}%
3883       \setunit{\addcomma\space}\printdate}%
3884     {\setunit{\addcomma\space}\printdate%
3885       \setunit{\adddot\space}\printfield{caderno}%
3886       \setunit{\addcomma\space}\printfield{pages}}}%
3887   {\setunit{\addcomma\space}\printfield{volume}%
3888     \setunit{\addcomma\space}\printfield{number}%
3889     \setunit{\addcomma\space}\printfield{pages}%
3890     \setunit{\addcomma\space}\printdate}%
3891 \newunit\newblock
3892 \printfield{doi}%
3893 \newunit\newblock
3894 \usebibmacro{url+urldate}%
3895 \usebibmacro{finentry}}
3896
3897 % --- incollection: ABNT puts the host organizer before the booktitle, as
3898 % "AUTOR. Parte. In: ORGANIZADOR (Org.). Booktitle. Local: Editora, ano,
3899 % p. X-Y." The organizer role string is "Org./Orgs." (see the .lbx). ---
3900 \newbibmacro*{coppe:bookeditor}{%
3901   \ifnameundef{editor}
3902     {}
3903     {\printnames{editor}%
3904       \setunit*{\addspace}\printtext{\mkbibparens{\usebibmacro{editorstrg}}}%
3905       \clearname{editor}%
3906       \setunit{\adddot\space}}}%
3907 \newbibmacro*{coppe:partdriver}{%
3908   \usebibmacro{bibindex}\usebibmacro{begentry}%
3909   \usebibmacro{author/translator+others}%
3910   \setunit{\labelnamepunct}\newblock
3911   \usebibmacro{title}%
3912   \newunit\newblock
3913   \usebibmacro{in:}%
3914   \usebibmacro{coppe:bookeditor}%
3915   \printfield{booktitle}%
3916   \newunit\newblock
3917   \printfield{edition}%
3918   \newunit\newblock
3919   \printlist{location}%
3920   \setunit*{\addcolon\space}\printlist{publisher}%
3921   \setunit*{\addcomma\space}\printdate%
3922   \newunit\newblock
3923   \printfield{pages}%
3924   \setunit{\addcomma\space}\printfield{chapter}%
3925   \newunit\newblock
3926   \usebibmacro{url+urldate}%
3927   \usebibmacro{finentry}}
3928 \DeclareBibliographyDriver{incollection}{\usebibmacro{coppe:partdriver}}
3929 \DeclareBibliographyDriver{inbook}{\usebibmacro{coppe:partdriver}}
3930
3931 % --- type field: resolve a known bibliography string (babel-aware), else literal ---

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```

3932 \DeclareFieldFormat{type}{\ifbibstring{#1}{\bibstring{#1}}{#1}}
3933
3934 % --- Thesis: "<type> (em <course>)" using the custom course field (ABNT). ---
3935 % "(Mestrado em Memoria Social e Documento)": a 4.2.1.1 poe o grau dentro dos
3936 % parenteses, antes do curso, e deixa o substantivo do documento fora deles --
3937 % "Dissertacao (Mestrado em ...)". O grau vem do campo coppedegree, que o
3938 % sourcemap preenche junto com o tipo.
3939 % O grau sai do proprio tipo do trabalho, que ja e uma chave de string; passar
3940 % a chave por um campo e depois resolve-la com \ifbibstring nao funciona,
3941 % porque \thefield nao se expande onde \ifbibstring precisa. Um campo
3942 % coppedegree preenchido a mao pelo autor tem precedencia.
3943 \newcommand*\coppe@degword{%
3944   \iffieldequalstr{type}{mscdiss}{\bibstring{coppemscdeg}}{%
3945     \iffieldequalstr{type}{dscsthesis}{\bibstring{coppedscdeg}}{%
3946       \iffieldequalstr{type}{phdthesis}{\bibstring{coppephddeg}}{%
3947         \iffieldequalstr{type}{tcc}{\bibstring{coppetccdeg}}{}}}}
3948   {\printfield{coppedegree}}}
3949
3950 \DeclareFieldFormat{course}{%
3951   \mkbibparens{\coppe@degword\addspace\coppestring{in:}\addspace#1}}
3952 % A ordem da 4.2.1.1 para tese e dissertacao:
3953 %   AUTOR. Titulo. ano. NNN f. Tipo (Grau em Curso) -- Instituicao, Local, ano.
3954 % A data aparece duas vezes de proposito, como nos exemplos do manual: uma
3955 % depois do titulo, outra fechando a nota de tese.
3956 \DeclareBibliographyDriver{thesis}{%
3957   \usebibmacro{bibindex}\usebibmacro{begentry}%
3958   \usebibmacro{author}%
3959   \setunit{\labelnamepunct}\newblock
3960   \usebibmacro{title}%
3961   \newunit\newblock
3962   \printdate%
3963   \newunit\newblock
3964   \printfield{pagetotal}%
3965   \newunit\newblock
3966   \printfield{type}\setunit*{\addspace}\printfield{course}%
3967   \iflistundef{institution}
3968     {}
3969     {\setunit*{\addspace\textendash\addspace}\printlist{institution}}%
3970   \setunit*{\addcomma\space}\printlist{location}%
3971   \setunit*{\addcomma\space}\printdate%
3972   \newunit\newblock
3973   \printfield{note}%
3974   \newunit\newblock
3975   \usebibmacro{url+urldate}%
3976   \usebibmacro{finentry}}
3977 % --- Games / audiovisual: render the custom fields (artist, platform,
3978 %   director, producer, version) the generic misc driver ignores. ---
3979 \newbibmacro*{coppe:media}{%
3980   \iflistundef{artist}{ }\setunit{\addperiod\space}%
3981   \printtext{\coppestring{art}\addcolon\space}\printlist{artist}}%
3982   \iffieldequalstr{platform}{ }\setunit{\addperiod\space}%
3983   \printtext{\coppestring{platform}\addcolon\space}\printfield{platform}}%
3984   \iflistundef{director}{ }\setunit{\addperiod\space}%
3985   \printtext{\coppestring{directedby}\addspace}\printlist{director}}%

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```

3986 \iflistundef{producer}{\setunit{\addperiod\space}%
3987 \printtext{\coppestring{producedby}\addspace}\printlist{producer}}
3988 \newbibmacro*{coppe:mediadriv}{%
3989 \usebibmacro{bibindex}\usebibmacro{begentry}%
3990 \usebibmacro{author/translator+others}%
3991 \setunit{\labelnamepunct}\newblock
3992 \usebibmacro{title}%
3993 \newunit\newblock
3994 \usebibmacro{coppe:media}%
3995 \newunit\newblock
3996 \usebibmacro{coppe:impreta}%
3997 \newunit\newblock
3998 \printfield{version}%
3999 \newunit\newblock
4000 \printfield{note}%
4001 \newunit\newblock
4002 \usebibmacro{url+urldate}%
4003 \usebibmacro{finentry}}
4004 % --- Publicacao periodica (4.2.3.1 a 4.2.3.3) -----
4005 % O titulo do periodico vai INTEIRO em caixa alta, e nao so a primeira palavra
4006 % como numa obra sem autor: e a forma dos exemplos do manual. Sem driver
4007 % proprio saia o default do biblatex -- "SAO Paulo Medical Journal (1941)." --,
4008 % sem local, sem editora e com o ano entre parenteses.
4009 \DeclareFieldFormat[periodical,suppperiodical]{title}{\MakeUppercase{#1}}
4010 \DeclareFieldFormat[periodical,suppperiodical]{volume}{v\adddot\addnbspace#1}
4011 \DeclareFieldFormat[periodical,suppperiodical]{number}{n\adddot\addnbspace#1}
4012 \DeclareBibliographyDriver{periodical}{%
4013 \usebibmacro{bibindex}\usebibmacro{begentry}%
4014 % Pelo campo, e nao pelo bibmacro: o bibmacro 'title' passa pelo formato
4015 % 'titlecase', e num periodico -- que nao tem autor -- ele nao chegava a
4016 % imprimir coisa nenhuma. A entrada saia comecando por um ponto solto.
4017 \printfield{title}%
4018 \newunit\newblock
4019 \printlist{location}%
4020 \iflistundef{publisher}{\setunit*{\addcolon\space}\printlist{publisher}}%
4021 \setunit*{\addcomma\space}\printfield{volume}%
4022 \setunit*{\addcomma\space}\printfield{number}%
4023 \setunit*{\addcomma\space}\printdate%
4024 \newunit\newblock
4025 \printfield{note}%
4026 \newunit\newblock
4027 \usebibmacro{url+urldate}%
4028 \usebibmacro{finentry}}
4029 \DeclareBibliographyDriver{suppperiodical}{%
4030 \usedriver{}{periodical}\usebibmacro{finentry}}
4031 % --- Eventos (4.2.4) -----
4032 % O nome do evento vai em caixa alta, seguido do numero da edicao, do ano e da
4033 % cidade -- "ENCONTRO ANUAL DA ANPAD, 14., 1982, Florianopolis." --, e so
4034 % depois vem o titulo dos anais, que e o elemento em destaque. Por isso o nome
4035 % do evento tem campo proprio (|eventtitle|, sinonimo |evento|) em vez de vir
4036 % empacotado no titulo: nada mais na entrada vai para caixa alta.
4037 \DeclareFieldFormat{eventtitle}{\MakeUppercase{#1}}
4038 % "Anais [...]" nao e uma obra entrada pelo titulo: quem faz as vezes de
4039 % cabeca da entrada e o nome do evento, ja em caixa alta. Sem esta declaracao

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4040 % o titulo dos anais saia "ANAIS [...]" e ainda perdia o ponto que o separa
4041 % da impressa.
4042 \DeclareFieldFormat[proceedings]{title}{#1}
4043 \DeclareFieldFormat[proceedings,inproceedings]{number}{#1\adddot}
4044 \DeclareFieldFormat[proceedings,inproceedings]{volume}{v\adddot\addnbspace#1}
4045 \newbibmacro*{coppe:evento}{%
4046   \printfield{eventtitle}%
4047   \setunit*{\addcomma\space}\printfield{number}%
4048   \setunit*{\addcomma\space}\printeventdate%
4049   \setunit*{\addcomma\space}\printfield{venue}}
4050 \DeclareBibliographyDriver{proceedings}{%
4051   \usebibmacro{bibindex}\usebibmacro{begentry}%
4052   \usebibmacro{coppe:evento}%
4053   \newunit\newblock
4054 % O ponto que separa o titulo dos anais da impressa e literal: o rastreador de
4055 % pontuacao do biblatex nao reconhece o fim de um titulo terminado em "]", e
4056 % nem \newunit nem \adddot produzia alguma coisa ali.
4057   \iffielddundef{title}{\printfield{title}\printtext{.}}%
4058   \setunit{\space}\newblock
4059   \usebibmacro{coppe:impressa}%
4060   \newunit\newblock
4061   \usebibmacro{url+urldate}%
4062   \usebibmacro{finentry}}
4063 \DeclareBibliographyDriver{inproceedings}{%
4064   \usebibmacro{bibindex}\usebibmacro{begentry}%
4065   \usebibmacro{author/translator+others}%
4066   \setunit{\labelnamepunct}\newblock
4067   \usebibmacro{title}%
4068   \newunit\newblock
4069   \usebibmacro{in:}%
4070   \usebibmacro{coppe:evento}%
4071   \newunit\newblock
4072   \iffielddundef{booktitle}{\printfield{booktitle}\printtext{.}}%
4073   \setunit{\space}\newblock
4074   \usebibmacro{coppe:impressa}%
4075   \newunit\newblock
4076   \printfield{volume}%
4077   \setunit*{\addcomma\space}\printfield{pages}%
4078   \newunit\newblock
4079   \usebibmacro{url+urldate}%
4080   \usebibmacro{finentry}}
4081 \DeclareBibliographyDriver{game}{\usebibmacro{coppe:mediadriver}}
4082 \DeclareBibliographyDriver{videogame}{\usebibmacro{coppe:mediadriver}}
4083 \DeclareBibliographyDriver{boardgame}{\usebibmacro{coppe:mediadriver}}
4084 \DeclareBibliographyDriver{movie}{\usebibmacro{coppe:mediadriver}}
4085 \DeclareBibliographyDriver{tvshow}{\usebibmacro{coppe:mediadriver}}
4086 % Cartographic document (map, NBR 6023 4.2.12): like a book (author/title/
4087 % impressa) plus the mandatory scale ("Escala 1:N").
4088 \DeclareFieldFormat{scale}{\bibstring{coppescale}\addnbspace#1}
4089 % O dois-pontos separa local e editora; sem editora ele vazava para a data
4090 % ("Recife: 1976"), e a 4.2.12 quer "Recife, 1976". Por isso a guarda.
4091 \newbibmacro*{coppe:impressa}{%
4092   \printlist{location}%
4093   \iflistundef{publisher}

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4094     {}
4095     {\setunit*{\addcolon\space}\printlist{publisher}}%
4096     \setunit*{\addcomma\space}\printdate}
4097 \DeclareBibliographyDriver{map}{%
4098   \usebibmacro{bibindex}\usebibmacro{begentry}%
4099   \usebibmacro{author/translator+others}%
4100   \setunit{\labelnamepunct}\newblock
4101   \usebibmacro{title}%
4102   \newunit\newblock
4103   \usebibmacro{coppe:imprenta}%
4104   \newunit\newblock
4105   % "Recife, 1976. 71 fotografias. Escala 1: 20.000." -- a descricao fisica vem
4106   % ANTES da escala, e por isso ela nao pode esperar pelo bloco eletronico.
4107   \iffielddundef{addendum}{\printfield{addendum}\clearfield{addendum}}%
4108   \newunit\newblock
4109   \printfield{scale}%
4110   \newunit\newblock
4111   \printfield{note}%
4112   \newunit\newblock
4113   \usebibmacro{url+urldate}%
4114   \usebibmacro{finentry}}
4115 % Patent (NBR 6023 4.2.5): inventor. \textbf{title}. Depositante: holder. type
4116 % number. date. The holder (depositante) is printed in normal case, not the
4117 % ALL-CAPS family treatment the bibliography gives the leading author.
4118 \newbibmacro*{coppe:depositor}{%
4119   \ifnameundef{holder}
4120     {}
4121     {\printtext{\bibstring{depositor}\addcolon\space}%
4122       \begingroup
4123         \def\mkbibnamefamily##1{##1}%
4124         \def\mkbibnameprefix##1{##1}%
4125         \printnames{holder}%
4126       \endgroup}}
4127 \DeclareBibliographyDriver{patent}{%
4128   \usebibmacro{bibindex}%
4129   \usebibmacro{begentry}%
4130   \usebibmacro{author/translator+others}%
4131   \setunit{\labelnamepunct}\newblock
4132   \usebibmacro{title}%
4133   \newunit\newblock
4134   \usebibmacro{coppe:depositor}%
4135   \newunit\newblock
4136   \printfield{type}%
4137   \setunit*{\addspace}\printfield{number}%
4138   \newunit\newblock
4139   % A 4.2.5 fecha a patente com as duas datas, nomeadas: "Deposito: 27 jan. 2006.
4140   % Concessao: 25 mar. 2008." Sem elas so saia o ano.
4141   \iffielddundef{filingdate}
4142     {}
4143     {\printtext{\bibstring{coppefiling}\addcolon\space}\printfield{filingdate}%
4144       \setunit{\addot\space}}%
4145   \iffielddundef{grantdate}
4146     {}
4147     {\printtext{\bibstring{coppegrant}\addcolon\space}\printfield{grantdate}}%

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```

4148 \ifboolexpr{ test {\iffieldundef{filingdate}} and test {\iffieldundef{grantdate}} }
4149   {\printdate}
4150   }{%
4151 \newunit\newblock
4152 \usebibmacro{url+urldate}%
4153 \usebibmacro{finentry}}
4154 % Exotic ABNT types that biblatex's standard style leaves without a driver:
4155 % give each a baseline (book-like for published docs: norma/partitura; misc for
4156 % the rest) so they print. Per-type ABNT refinements can follow later.
4157 % \usedriver desliga o \finentry do driver chamado, porque quem chama e que
4158 % deve fechar a entrada -- e aqui ninguem fechava. Resultado: norma, partitura,
4159 % audiovisual, iconografico, obra de arte, tridimensional e conjunto de dados
4160 % saiam sem o ponto final, que a 4.2 poe em toda referencia.
4161 \DeclareBibliographyDriver{standard}{\usedriver{}{book}\usebibmacro{finentry}}
4162 \DeclareBibliographyDriver{music}{\usedriver{}{book}\usebibmacro{finentry}}
4163 % Legal documents (NBR 6023 4.2.6-4.2.8): jurisdiction/entity (CAPS author),
4164 % the law id + ementa (plain, not bold), then location and date.
4165 \newbibmacro*{coppe:legaldriver}{%
4166   \usebibmacro{bibindex}\usebibmacro{begentry}%
4167   \usebibmacro{author/translator+others}%
4168   \setunit{\labelnamepunct}\newblock
4169   \usebibmacro{title}%
4170   \newunit\newblock
4171   % Jurisprudência: "Relator: <nome>." after the decision title (NBR 6023).
4172   \iffieldundef{relator}{\printtext{Relator\addcolon\space}\printfield{relator}\newunit\ne
4173   \printfield{edition}%
4174   \newunit\newblock
4175   \printlist{location}%
4176   % A 4.2.6 traz o veiculo de publicacao com volume, numero e pagina -- "Sao
4177   % Paulo, v. 62, n. 3, p. 217-220, 1998." --, que se perdiam por completo.
4178   \iflistundef{publisher}{\setunit*{\addcolon\space}\printlist{publisher}}%
4179   \setunit*{\addcomma\space}\printfield{volume}%
4180   \setunit*{\addcomma\space}\printfield{number}%
4181   \setunit*{\addcomma\space}\printfield{pages}%
4182   \setunit{\addcomma\space}\printdate%
4183   \newunit\newblock
4184   \printfield{note}%
4185   \newunit\newblock
4186   \usebibmacro{url+urldate}%
4187   \usebibmacro{finentry}}
4188 % 4.2.14: "UNIVERSIDADE ... Rio de Janeiro, 1998. Disponivel em: ..." -- o
4189 % driver padrao do biblatex para @online nao imprime o local.
4190 \DeclareBibliographyDriver{online}{%
4191   \usebibmacro{bibindex}\usebibmacro{begentry}%
4192   \usebibmacro{author/translator+others}%
4193   \setunit{\labelnamepunct}\newblock
4194   \usebibmacro{title}%
4195   \newunit\newblock
4196   \usebibmacro{coppe:imprenta}%
4197   \newunit\newblock
4198   \printfield{note}%
4199   \newunit\newblock
4200   \usebibmacro{url+urldate}%
4201   \usebibmacro{finentry}}

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4202 \DeclareBibliographyDriver{legislation}{\usebibmacro{coppe:legaldriver}}
4203 \DeclareBibliographyDriver{jurisdiction}{\usebibmacro{coppe:legaldriver}}
4204 \DeclareBibliographyDriver{legal}{\usebibmacro{coppe:legaldriver}}
4205 % Audio e video vao pelo driver de midia, que imprime diretor e produtor --
4206 % "Producao de Jorge Ramos de Andrade." da 4.2.9. O driver misc os ignorava.
4207 \DeclareBibliographyDriver{audio}{\usebibmacro{coppe:mediadriver}}
4208 \DeclareBibliographyDriver{video}{\usebibmacro{coppe:mediadriver}}
4209 \DeclareBibliographyDriver{software}{\usedriver{}{misc}\usebibmacro{finentry}}
4210 \DeclareBibliographyDriver{image}{\usedriver{}{misc}\usebibmacro{finentry}}
4211 \DeclareBibliographyDriver{artwork}{\usedriver{}{misc}\usebibmacro{finentry}}
4212 \DeclareBibliographyDriver{performance}{\usedriver{}{misc}\usebibmacro{finentry}}
4213 \DeclareBibliographyDriver{dataset}{\usedriver{}{misc}\usebibmacro{finentry}}
4214
4215 % --- biber source map: Portuguese synonyms (entry types + fields) + thesis family ---
4216 \DeclareSourcemap{%
4217   \maps[datatype=bibtex]{%
4218     % entry-type PT synonyms -> canonical English
4219     \map{step[typesource=livro,typetarget=book]}
4220     \map{step[typesource=artigo,typetarget=article]}
4221     \map{step[typesource=artigodejornal,typetarget=article,final]\step[fieldset=entrysubtyp
4222     \map{step[typesource=partedelivro,typetarget=incollection]}
4223     \map{step[typesource=emlivro,typetarget=inbook]}
4224     \map{step[typesource=verbete,typetarget=inreference]}
4225     \map{step[typesource=anais,typetarget=proceedings]}
4226     \map{step[typesource=trabalhodeevento,typetarget=inproceedings]}
4227     \map{step[typesource=periodico,typetarget=periodical]}
4228     \map{step[typesource=relatorio,typetarget=report]}
4229     \map{step[typesource=relatoriotecnico,typetarget=report]}
4230     \map{step[typesource>manual,typetarget>manual]}
4231     \map{step[typesource=correspondencia,typetarget=letter]}
4232     \map{step[typesource=patente,typetarget=patent]}
4233     \map{step[typesource=diverso,typetarget=misc]}
4234     \map{step[typesource=jogo,typetarget=game]}
4235     \map{step[typesource=jogoeletronico,typetarget=videogame]}
4236     \map{step[typesource=jogodetabuleiro,typetarget=boardgame]}
4237     \map{step[typesource=filme,typetarget=movie]}
4238     \map{step[typesource=serietv,typetarget=tvshow]}
4239     \map{step[typesource=programadetelevisao,typetarget=tvshow]}
4240     % Exotic ABNT types (NBR 6023 4.2.5-4.2.14); targets are biblatex-native
4241     % except map (declared in coppe.dbx).
4242     \map{step[typesource=legislacao,typetarget=legislation]}
4243     \map{step[typesource=atonomativo,typetarget=legislation]}
4244     \map{step[typesource=jurisprudencia,typetarget=jurisdiction]}
4245     \map{step[typesource=documentojuridico,typetarget=legal]}
4246     \map{step[typesource=documentocivil,typetarget=legal]}
4247     \map{step[typesource=norma,typetarget=standard]}
4248     \map{step[typesource=partitura,typetarget=music]}
4249     \map{step[typesource=iconografico,typetarget=image]}
4250     \map{step[typesource=obradearte,typetarget=artwork]}
4251     \map{step[typesource=tridimensional,typetarget=artwork]}
4252     \map{step[typesource=cartografico,typetarget=map]}
4253     \map{step[typesource=mapa,typetarget=map]}
4254     \map{step[typesource=audiovisual,typetarget=video]}
4255     \map{step[typesource=conjuntodedados,typetarget=dataset]}

```

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4256 \map{\step[tources=fasciculo,typetarget=supperperiodical]}
4257 \map{\step[tources=trabalhoapresentado,typetarget=unpublished]}
4258 % thesis family: retype to thesis and preset the (babel-aware) type word
4259 \map{\step[tources=tese,typetarget=thesis]}
4260 \map{\step[tources=dissertacao,typetarget=thesis]}
4261 \map{\step[tources=masterdissertation,typetarget=thesis,final]\step[fieldset=type,fiel
4262 \map{\step[tources=mscdissertation,typetarget=thesis,final]\step[fieldset=type,fiel
4263 \map{\step[tources=dissertacaomestrado,typetarget=thesis,final]\step[fieldset=type,fi
4264 \map{\step[tources=dscthesis,typetarget=thesis,final]\step[fieldset=type,fieldvalue=d
4265 \map{\step[tources=tesedoutorado,typetarget=thesis,final]\step[fieldset=type,fieldval
4266 \map{\step[tources=phdthesis,typetarget=thesis,final]\step[fieldset=type,fieldvalue=p
4267 \map{\step[tources=teseph,typetarget=thesis,final]\step[fieldset=type,fieldvalue=phd
4268 \map{\step[tources=tcc,typetarget=thesis,final]\step[fieldset=type,fieldvalue=tcc]}
4269 % field PT synonyms (unaccented) -> canonical English
4270 \map{\step[fieldsource=periodico,fieldtarget=journaltitle]}
4271 \map{\step[fieldsource=revista,fieldtarget=journaltitle]}
4272 \map{\step[fieldsource=jornal,fieldtarget=journaltitle]}
4273 \map{\step[fieldsource=titulo,fieldtarget=title]}
4274 \map{\step[fieldsource=subtitulo,fieldtarget=subtitle]}
4275 \map{\step[fieldsource=autor,fieldtarget=author]}
4276 \map{\step[fieldsource=editora,fieldtarget=publisher]}
4277 \map{\step[fieldsource=local,fieldtarget=location]}
4278 \map{\step[fieldsource=ano,fieldtarget=year]}
4279 \map{\step[fieldsource=data,fieldtarget=date]}
4280 \map{\step[fieldsource=edicao,fieldtarget=edition]}
4281 \map{\step[fieldsource=paginas,fieldtarget=pages]}
4282 \map{\step[fieldsource=numero,fieldtarget=number]}
4283 \map{\step[fieldsource=tradutor,fieldtarget=translator]}
4284 \map{\step[fieldsource=organizador,fieldtarget=editor]}
4285 \map{\step[fieldsource=instituicao,fieldtarget=institution]}
4286 \map{\step[fieldsource=tipo,fieldtarget=type]}
4287 \map{\step[fieldsource=nota,fieldtarget=note]}
4288 \map{\step[fieldsource=dataacesso,fieldtarget=urldate]}
4289 \map{\step[fieldsource=curso,fieldtarget=course]}
4290 \map{\step[fieldsource=grau,fieldtarget=coppedegree]}
4291 \map{\step[fieldsource=artista,fieldtarget=artist]}
4292 \map{\step[fieldsource=plataforma,fieldtarget=platform]}
4293 \map{\step[fieldsource=diretor,fieldtarget=director]}
4294 \map{\step[fieldsource=produtor,fieldtarget=producer]}
4295 \map{\step[fieldsource=versao,fieldtarget=version]}
4296 \map{\step[fieldsource=escala,fieldtarget=scale]}
4297 \map{\step[fieldsource=evento,fieldtarget=eventtitle]}
4298 \map{\step[fieldsource=dataevento,fieldtarget=eventdate]}
4299 \map{\step[fieldsource=localevento,fieldtarget=venue]}
4300 \map{\step[fieldsource=deposito,fieldtarget=filingdate]}
4301 \map{\step[fieldsource=concessao,fieldtarget=grantdate]}
4302 }%
4303 }
4304
4305 % --- ABNT list layout: flush left, single spacing, blank line between entries ---
4306 \setlength{\bibhang}{0pt}
4307 \setlength{\bibitemsep}{\baselineskip}
4308
4309 </bbx>

```

```

4310 <*cbx>
4311 \ProvidesFile{coppe.cbx}[2026/09/09 v4.1 CoppeTeX ABNT author-date citations]
4312 % Author-date is the CoppeTeX 4.0 default. authoryear-comp gives compressed
4313 % author-year cites with normal-caps names (post-2020 rule) out of the box.
4314 \RequireCitationStyle{authoryear-comp}
4315
4316 % --- NBR 10520:2023 author separator -----
4317 % Parenthetical citations separate authors with ";" -> (Silva; Souza, 2020).
4318 % Narrative citations (\textcite, natbib \citet) keep comma + "e/and" ->
4319 % Silva, Souza e Oliveira (2020). The reference list keeps its own ";" via
4320 % coppe.bbx ([bib,biblist]); these contextless formats cover the cite side.
4321 \DeclareDelimFormat{multinamedelim}{\addsemicolon\space}
4322 \DeclareDelimFormat{finalnamedelim}{\addsemicolon\space}
4323 \DeclareDelimFormat[textcite]{multinamedelim}{\addcomma\space}
4324 \DeclareDelimFormat[textcite]{finalnamedelim}{\addspace\bibstring{and}\space}
4325
4326 % --- apud: citation of a citation (ABNT NBR 6023 §4.1.2.2 / NBR 10520) -----
4327 % ABNT lists only the work actually CONSULTED. So only the consulted work (#4, a
4328 % .bib key) is cited and enters the reference list; the ORIGINAL work (#2 author,
4329 % #3 year) was not consulted and is given as plain text -- it is never a cite key,
4330 % so it is never added to the bibliography.
4331 % \citetapud[<page>]{<orig author>}{<orig year>}{<consulted key>}
4332 % -> Orig Author (year) apud Consulted (yr[, page]) [narrative]
4333 % \citepapud[<page>]{<orig author>}{<orig year>}{<consulted key>}
4334 % -> (Orig Author, year apud CONSULTED, yr[, page]) [parenthetical]
4335 % #1 = optional postnote (page/loc) on the consulted work; "apud" is latin -> italic.
4336 \DeclareRobustCommand{\citetapud}[4][]{%
4337   #2\space(#3)\space\mkbibemph{apud}\space\textcite{#1}{#4}}
4338 \DeclareRobustCommand{\citepapud}[4][]{%
4339   \mkbibparens{#2\addcomma\space#3\space\mkbibemph{apud}\space
4340     \citeauthor{#4}\addcomma\space\citeyear{#4}%
4341     \ifblank{#1}{ }\{\addcomma\space\mkpageprefix[pagination]{#1}\}}
4342 </cbx>
4343 <*lbr>
4344 \ProvidesFile{brazilian-coppe.lbx}[2026/09/09 v4.1 CoppeTeX brazilian strings]
4345 \InheritBibliographyExtras{brazilian}
4346 \NewBibliographyString{availablefrom,mscdiss,dscthes, tcc,depositor,coppescale,
4347   coppemscdeg,coppedscdeg,coppephddeg,coppetccdeg,coppefiling,coppegrant}
4348 \DeclareBibliographyStrings{%
4349   inherit = {brazilian},
4350   availablefrom = {{Dispon\'}{\i}vel em}{Dispon\'}{\i}vel em}},
4351   coppemscdeg = {{Mestrado}{Mestrado}},
4352   coppedscdeg = {{Doutorado}{Doutorado}},
4353   coppephddeg = {{Doutorado}{Doutorado}},
4354   coppetccdeg = {{Gradua\c{c}\~{a}o}{Gradua\c{c}\~{a}o}},
4355   coppefiling = {{Dep\'}osito}{Dep\'}osito}},
4356   coppegrant = {{Concess\~{a}o}{Concess\~{a}o}},
4357   urlseen = {{Acesso em}{Acesso em}},
4358 % A 4.3 abrevia os meses nas tres primeiras letras seguidas de ponto -- menos
4359 % maio, que se escreve por extenso. O biblalex-brazilian abrevia "mai.".
4360   may = {{maio}{maio}},
4361   in = {{In}{In}},
4362   editor = {{organizador}{org\addot}},
4363   editors = {{organizadores}{org\addot}},

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4364 depositor      = {{Depositante}{Depositante}},
4365 coppescale      = {{Escala}{Escala}},
4366 msdiss          = {{Disserta\c{c}\~{a}o}{Disserta\c{c}\~{a}o}},
4367 dscthesi        = {{Tese}{Tese}},
4368 phdthesis       = {{Tese}{Tese}},
4369 tcc             = {{Trabalho de Conclus\~{a}o de Curso}{Trabalho de Conclus\~{a}o de Curso}}
4370 }
4371 </lbr>
4372 <*lbr>
4373 \ProvidesFile{english-coppe.lbx}[2026/09/09 v4.1 CoppeTeX english strings]
4374 \InheritBibliographyExtras{english}
4375 \NewBibliographyString{availablefrom,msdiss,dscthesi,tcc,depositor,coppescale,
4376   coppemscdeg,coppedscdeg,coppephddeg,coppetccdeg,coppefiling,coppegrant}
4377 \DeclareBibliographyStrings{%
4378   inherit        = {english},
4379   availablefrom = {{Available from}{Available from}},
4380   coppemscdeg   = {{M\addot Sc\addot}{M\addot Sc\addot}},
4381   coppedscdeg   = {{D\addot Sc\addot}{D\addot Sc\addot}},
4382   coppephddeg   = {{Ph\addot D\addot}{Ph\addot D\addot}},
4383   coppetccdeg   = {{B\addot Sc\addot}{B\addot Sc\addot}},
4384   coppefiling   = {{Filed}{Filed}},
4385   coppegrant    = {{Granted}{Granted}},
4386   urlseen       = {{Access on}{Access on}},
4387   in            = {{In}{In}},
4388   editor        = {{editor}{ed\addot}},
4389   editors       = {{editors}{ed\addot}},
4390   depositor     = {{Depositor}{Depositor}},
4391   coppescale    = {{Scale}{Scale}},
4392   msdiss        = {{M\addot Sc\addot\ Dissertation}{M\addot Sc\addot\ Diss\addot}},
4393   dscthesi      = {{D\addot Sc\addot\ Thesis}{D\addot Sc\addot\ Thesis}},
4394   phdthesis     = {{Ph\addot D\addot\ Thesis}{Ph\addot D\addot\ Thesis}},
4395   tcc           = {{Undergraduate Thesis}{Undergraduate Thesis}},
4396 }
4397 </lbr>
4398 <*lbr>
4399 \ProvidesFile{spanish-coppe.lbx}[2026/09/09 v4.1 CoppeTeX spanish strings]
4400 \InheritBibliographyExtras{spanish}
4401 \NewBibliographyString{availablefrom,msdiss,dscthesi,tcc,depositor,coppescale,
4402   coppemscdeg,coppedscdeg,coppephddeg,coppetccdeg,coppefiling,coppegrant}
4403 \DeclareBibliographyStrings{%
4404   inherit        = {spanish},
4405   availablefrom = {{Disponible en}{Disponible en}},
4406   coppemscdeg   = {{Maestr'\{i}a}{Maestr'\{i}a}},
4407   coppedscdeg   = {{Doctorado}{Doctorado}},
4408   coppephddeg   = {{Doctorado}{Doctorado}},
4409   coppetccdeg   = {{Grado}{Grado}},
4410   coppefiling   = {{Dep\ 'osito}{Dep\ 'osito}},
4411   coppegrant    = {{Concesi\ 'on}{Concesi\ 'on}},
4412   urlseen       = {{Consultado el}{Consultado el}},
4413   in            = {{En}{En}},
4414   editor        = {{editor}{ed\addot}},
4415   editors       = {{editores}{ed\addot}},
4416   depositor     = {{Solicitante}{Solicitante}},
4417   coppescale    = {{Escala}{Escala}},

```

```

4418 msdiss      = {{Tesis de Maestr'\{i\}a}{Tesis de Maestr'\{i\}a}},
4419 dscthesi s   = {{Tesis Doctoral}{Tesis Doctoral}},
4420 phdthesis    = {{Tesis Doctoral}{Tesis Doctoral}},
4421 tcc          = {{Trabajo de Fin de Carrera}{Trabajo de Fin de Carrera}},
4422 }
4423 </l b x e s>
4424 <*l b x f r>
4425 \ProvidesFile{french-coppe.lbx}[2026/09/09 v4.1 CoppeTeX french strings]
4426 \InheritBibliographyExtras{french}
4427 \NewBibliographyString{availablefrom,msdiss,dscthesi s,tcc,depositor,coppescale,
4428   coppemscdeg,coppedscdeg,coppephddeg,coppetccdeg,coppefiling,coppegrant}
4429 \DeclareBibliographyStrings{%
4430   inherit      = {french},
4431   availablefrom = {{Disponible \'a}{Disponible \'a}},
4432   coppemscdeg  = {{Master}{Master}},
4433   coppedscdeg  = {{Doctorat}{Doctorat}},
4434   coppephddeg  = {{Doctorat}{Doctorat}},
4435   coppetccdeg  = {{Licence}{Licence}},
4436   coppefiling  = {{D\'ep\'ot}{D\'ep\'ot}},
4437   coppegrant   = {{D\'elivrance}{D\'elivrance}},
4438   urlseen      = {{Consult\'e le}{Consult\'e le}},
4439   in           = {{Dans}{Dans}},
4440   editor        = {{\'editeur}{\'ed\adddot}},
4441   editors       = {{\'editeurs}{\'ed\adddot}},
4442   depositor    = {{D\'eposant}{D\'eposant}},
4443   coppescale   = {{\'Echelle}{\'Echelle}},
4444   msdiss        = {{M\'emoire de Master}{M\'emoire de Master}},
4445   dscthesi s    = {{Th\'ese de Doctorat}{Th\'ese de Doctorat}},
4446   phdthesis     = {{Th\'ese de Doctorat}{Th\'ese de Doctorat}},
4447   tcc          = {{M\'emoire de fin d\'\'etudes}{M\'emoire de fin d\'\'etudes}},
4448 }
4449 </l b x f r>
4450 <*l b x i t>
4451 \ProvidesFile{italian-coppe.lbx}[2026/09/09 v4.1 CoppeTeX italian strings]
4452 \InheritBibliographyExtras{italian}
4453 \NewBibliographyString{availablefrom,msdiss,dscthesi s,tcc,depositor,coppescale,
4454   coppemscdeg,coppedscdeg,coppephddeg,coppetccdeg,coppefiling,coppegrant}
4455 \DeclareBibliographyStrings{%
4456   inherit      = {italian},
4457   availablefrom = {{Disponibile da}{Disponibile da}},
4458   urlseen      = {{Consultato il}{Consultato il}},
4459   in           = {{In}{In}},
4460   editor        = {{curatore}{cur\adddot}},
4461   editors       = {{curatori}{cur\adddot}},
4462   depositor    = {{Depositante}{Depositante}},
4463   coppescale   = {{Scala}{Scala}},
4464   msdiss        = {{Tesi di Laurea Magistrale}{Tesi di Laurea Magistrale}},
4465   dscthesi s    = {{Tesi di Dottorato}{Tesi di Dottorato}},
4466   phdthesis     = {{Tesi di Dottorato}{Tesi di Dottorato}},
4467   tcc          = {{Tesi di Laurea Triennale}{Tesi di Laurea Triennale}},
4468 }
4469 </l b x i t>
4470 <*l a n g e s>
4471 %% coppe-lang-spanish.def -- CoppeTeX 4.x Spanish class-string pack.

```



```

4472 %% Loaded by coppe.cls when \documentclass[spanish]{coppe} (or any other
4473 %% spanish slot) is active. Populates the dispatcher table for the
4474 %% spanish language and registers the biblatex language mapping.
4475 %% Institutional names (universityname/cityname/statename/countryname)
4476 %% are deliberately kept Portuguese: the COPPE/UFRJ cover and folha de
4477 %% rosto are a Brazilian institutional template that does not translate.
4478 \ProvidesFile{coppe-lang-spanish.def}[2026/09/09 v4.1 CoppeTeX spanish class strings]
4479 \copperdefstring{spanish}{appendix}      {Ap\'endice}
4480 \copperdefstring{spanish}{annex}        {Anexo}
4481 \copperdefstring{spanish}{frame}        {Cuadro}
4482 \copperdefstring{spanish}{listframe}     {Lista de Cuadros}
4483 \copperdefstring{spanish}{source}        {Fuente}
4484 \copperdefstring{spanish}{program}       {Programa}
4485 \copperdefstring{spanish}{listprogram}   {Lista de Programas}
4486 \copperdefstring{spanish}{references}    {Referencias}
4487 \copperdefstring{spanish}{in:}          {en}
4488 \copperdefstring{spanish}{art}          {Arte}
4489 \copperdefstring{spanish}{platform}      {Plataforma}
4490 \copperdefstring{spanish}{directedby}    {Direcci\'on}
4491 \copperdefstring{spanish}{producedby}    {Producci\'on}
4492 \copperdefstring{spanish}{listabbreviation}{Lista de Abreviaturas y Siglas}
4493 \copperdefstring{spanish}{listsymbol}    {Lista de S\'imbolos}
4494 \copperdefstring{spanish}{glossary}      {Glosario}
4495 \copperdefstring{spanish}{advisor}       {Director}
4496 \copperdefstring{spanish}{advisors}      {Directores}
4497 \copperdefstring{spanish}{dept}          {Programa}
4498 \copperdefstring{spanish}{approvedmale}  {Aprobado por}
4499 \copperdefstring{spanish}{approvedfemale}{Aprobada por}
4500 \copperdefstring{spanish}{approvedonmale}{Aprobado el}
4501 \copperdefstring{spanish}{approvedonfemale}{Aprobada el}
4502 \copperdefstring{spanish}{keywordsname} {Palabras clave}
4503 \copperdefstring{spanish}{volumename}    {Volumen}
4504 \copperdefstring{spanish}{coadvisor}     {Codirector}
4505 \copperdefstring{spanish}{coadvisors}    {Codirectores}
4506 \copperdefstring{spanish}{ofname}        {de}
4507 \copperdefstring{spanish}{researchline}{L\'inea de investigaci\'on}
4508 \copperdefstring{spanish}{tbddate}       {por determinar}
4509 \copperdefstring{spanish}{universityname}{Universidade Federal do Rio de Janeiro}
4510 \copperdefstring{spanish}{cityname}      {Rio de Janeiro}
4511 \copperdefstring{spanish}{statename}     {RJ}
4512 \copperdefstring{spanish}{countryname}   {Brasil}
4513 \copperdefstring{spanish}{monthname1}    {enero}
4514 \copperdefstring{spanish}{monthname2}    {febrero}
4515 \copperdefstring{spanish}{monthname3}    {marzo}
4516 \copperdefstring{spanish}{monthname4}    {abril}
4517 \copperdefstring{spanish}{monthname5}    {mayo}
4518 \copperdefstring{spanish}{monthname6}    {junio}
4519 \copperdefstring{spanish}{monthname7}    {julio}
4520 \copperdefstring{spanish}{monthname8}    {agosto}
4521 \copperdefstring{spanish}{monthname9}    {septiembre}
4522 \copperdefstring{spanish}{monthname10}   {octubre}
4523 \copperdefstring{spanish}{monthname11}   {noviembre}
4524 \copperdefstring{spanish}{monthname12}   {diciembre}
4525 \copperdefstring{spanish}{algorithm2eopt}{spanish}

```

```

4526 \copperdefstring{spanish}{babelname} {spanish}
4527 %% biblatex is loaded by coppe.cls AFTER this .def file is read; defer the
4528 %% language-mapping registration until \begin{document} so biblatex's
4529 %% \DeclareLanguageMapping is available.
4530 \AtBeginDocument{\DeclareLanguageMapping{spanish}{spanish-coppe}}
4531 \langues
4532 \*langfr
4533 %% coppe-lang-french.def -- CoppeTeX 4.x French class-string pack.
4534 %% See coppe-lang-spanish.def for the structure. Institutional names
4535 %% remain Portuguese; only document-content strings translate.
4536 \ProvidesFile{coppe-lang-french.def}[2026/09/09 v4.1 CoppeTeX french class strings]
4537 \copperdefstring{french}{appendix} {Annexe}
4538 \copperdefstring{french}{annex} {Annexe}
4539 \copperdefstring{french}{frame} {Encadr\'e}
4540 \copperdefstring{french}{listframe} {Liste des encadr\'es}
4541 \copperdefstring{french}{source} {Source}
4542 \copperdefstring{french}{program} {Programme}
4543 \copperdefstring{french}{listprogram} {Liste des programmes}
4544 \copperdefstring{french}{references} {R\'eferences}
4545 \copperdefstring{french}{in:} {dans}
4546 \copperdefstring{french}{art} {Illustration}
4547 \copperdefstring{french}{platform} {Plateforme}
4548 \copperdefstring{french}{directedby} {R\'ealis\'e par}
4549 \copperdefstring{french}{producedby} {Produit par}
4550 \copperdefstring{french}{listabbreviation} {Liste des abr\'eviations et acronymes}
4551 \copperdefstring{french}{listsymbol} {Liste des symboles}
4552 \copperdefstring{french}{glossary} {Glossaire}
4553 \copperdefstring{french}{advisor} {Directeur}
4554 \copperdefstring{french}{advisors} {Directeurs}
4555 \copperdefstring{french}{dept} {Programme}
4556 \copperdefstring{french}{approvedmale} {Approuv\'e par}
4557 \copperdefstring{french}{approvedfemale} {Approuv\'ee par}
4558 \copperdefstring{french}{approvedonmale} {Approuv\'e le}
4559 \copperdefstring{french}{approvedonfemale} {Approuv\'ee le}
4560 \copperdefstring{french}{keywordsname} {Mots-cl\'es}
4561 \copperdefstring{french}{volumename} {Volume}
4562 \copperdefstring{french}{coadvisor} {Co-directeur}
4563 \copperdefstring{french}{coadvisors} {Co-directeurs}
4564 \copperdefstring{french}{ofname} {de}
4565 \copperdefstring{french}{researchline} {Axe de recherche}
4566 \copperdefstring{french}{tbddate} {\`a d\'eterminer}
4567 \copperdefstring{french}{universityname} {Universidade Federal do Rio de Janeiro}
4568 \copperdefstring{french}{cityname} {Rio de Janeiro}
4569 \copperdefstring{french}{statename} {RJ}
4570 \copperdefstring{french}{countryname} {Brasil}
4571 \copperdefstring{french}{monthname1} {janvier}
4572 \copperdefstring{french}{monthname2} {f\'evrier}
4573 \copperdefstring{french}{monthname3} {mars}
4574 \copperdefstring{french}{monthname4} {avril}
4575 \copperdefstring{french}{monthname5} {mai}
4576 \copperdefstring{french}{monthname6} {juin}
4577 \copperdefstring{french}{monthname7} {juillet}
4578 \copperdefstring{french}{monthname8} {ao\~ut}
4579 \copperdefstring{french}{monthname9} {septembre}

```

```

4580 \copperdefstring{french}{monthname10}{octobre}
4581 \copperdefstring{french}{monthname11}{novembre}
4582 \copperdefstring{french}{monthname12}{d\'ecembre}
4583 \copperdefstring{french}{algorithm2eopt}{french}
4584 \copperdefstring{french}{babelname} {french}
4585 \AtBeginDocument{\DeclareLanguageMapping{french}{french-coppe}}
4586 \langfr
4587 \*langit
4588 %% coppe-lang-italian.def -- CoppeTeX 4.x Italian class-string pack.
4589 %% See coppe-lang-spanish.def for the structure. The algorithm2e
4590 %% language option for Italian is spelled "italiano" -- that's the
4591 %% spelling algorithm2e ships, even though babel calls the language
4592 %% "italian".
4593 \ProvidesFile{coppe-lang-italian.def}[2026/09/09 v4.1 CoppeTeX italian class strings]
4594 \copperdefstring{italian}{appendix} {Appendice}
4595 \copperdefstring{italian}{annex} {Allegato}
4596 \copperdefstring{italian}{frame} {Riquadro}
4597 \copperdefstring{italian}{listframe} {Elenco dei riquadri}
4598 \copperdefstring{italian}{source} {Fonte}
4599 \copperdefstring{italian}{program} {Programma}
4600 \copperdefstring{italian}{listprogram} {Elenco dei programmi}
4601 \copperdefstring{italian}{references} {Riferimenti}
4602 \copperdefstring{italian}{in:} {in}
4603 \copperdefstring{italian}{art} {Illustrazione}
4604 \copperdefstring{italian}{platform} {Piattaforma}
4605 \copperdefstring{italian}{directedby} {Regia di}
4606 \copperdefstring{italian}{producedby} {Prodotto da}
4607 \copperdefstring{italian}{listabbreviation}{Elenco di abbreviazioni e sigle}
4608 \copperdefstring{italian}{listsymbol} {Elenco dei simboli}
4609 \copperdefstring{italian}{glossary} {Glossario}
4610 \copperdefstring{italian}{advisor} {Relatore}
4611 \copperdefstring{italian}{advisors} {Relatori}
4612 \copperdefstring{italian}{dept} {Programma}
4613 \copperdefstring{italian}{approvedmale} {Approvato da}
4614 \copperdefstring{italian}{approvedfemale} {Approvata da}
4615 \copperdefstring{italian}{approvedonmale} {Approvato il}
4616 \copperdefstring{italian}{approvedonfemale}{Approvata il}
4617 \copperdefstring{italian}{keywordsname} {Parole chiave}
4618 \copperdefstring{italian}{volumename} {Volume}
4619 \copperdefstring{italian}{coadvisor} {Correlatore}
4620 \copperdefstring{italian}{coadvisors} {Correlatori}
4621 \copperdefstring{italian}{ofname} {di}
4622 \copperdefstring{italian}{researchline}{Linea di ricerca}
4623 \copperdefstring{italian}{tbddate} {da determinare}
4624 \copperdefstring{italian}{universityname}{Universidade Federal do Rio de Janeiro}
4625 \copperdefstring{italian}{cityname} {Rio de Janeiro}
4626 \copperdefstring{italian}{statename} {RJ}
4627 \copperdefstring{italian}{countryname} {Brasil}
4628 \copperdefstring{italian}{monthname1} {gennaio}
4629 \copperdefstring{italian}{monthname2} {febbraio}
4630 \copperdefstring{italian}{monthname3} {marzo}
4631 \copperdefstring{italian}{monthname4} {aprile}
4632 \copperdefstring{italian}{monthname5} {maggio}
4633 \copperdefstring{italian}{monthname6} {giugno}

```

```

4634 \copperdefstring{italian}{monthname7} {luglio}
4635 \copperdefstring{italian}{monthname8} {agosto}
4636 \copperdefstring{italian}{monthname9} {settembre}
4637 \copperdefstring{italian}{monthname10}{ottobre}
4638 \copperdefstring{italian}{monthname11}{novembre}
4639 \copperdefstring{italian}{monthname12}{dicembre}
4640 \copperdefstring{italian}{algorithm2eopt}{italiano}
4641 \copperdefstring{italian}{babelname} {italian}
4642 \AtBeginDocument{\DeclareLanguageMapping{italian}{italian-coppe}}
4643 \</langit>
4644 <*numbbx>
4645 \ProvidesFile{coppe-numeric.bbx}[2026/09/09 v4.1 CoppeTeX ABNT numeric bibliography]
4646 % Numeric reference list for the ABNT numeric system (NBR 10520). It reuses the
4647 % full coppe ABNT style (drivers, CAPS surnames, bold titles, "In:", p./f.,
4648 % the PT->EN source map and the language mappings) and only adds the [n] list
4649 % label and the numeric sorting on top.
4650 \RequireBibliographyStyle{coppe}
4651
4652 % ABNT numeric system lists references in the order they are first cited
4653 % (NBR 10520), i.e. unsorted. Override with sorting=nty for an alphabetical
4654 % numbered list. 'labelnumber' generates the [n] field.
4655 \ExecuteBibliographyOptions{labelnumber,sorting=none}
4656
4657 % Bracketed labels, both in the list and for any shorthand entries.
4658 \DeclareFieldFormat{labelnumberwidth}{\mkbibbrackets{#1}}
4659 \DeclareFieldFormat{shorthandwidth}{\mkbibbrackets{#1}}
4660
4661 % Reference list driven by the [n] label (copied from biblatex's numeric.bbx),
4662 % keeping the ABNT blank line between entries via \bibitemsep set in coppe.bbx.
4663 \defbibenvironment{bibliography}
4664 {
4665   \list
4666     {\printtext[labelnumberwidth]{%
4667       \printfield{labelprefix}%
4668       \printfield{labelnumber}}}
4669     {\setlength{\labelwidth}{\labelnumberwidth}%
4670      \setlength{\leftmargin}{\labelwidth}%
4671      \setlength{\labelsep}{\biblabelsep}%
4672      \addtolength{\leftmargin}{\labelsep}%
4673      \setlength{\itemsep}{\bibitemsep}%
4674      \setlength{\parsep}{\bibparsep}}%
4675     {\renewcommand*{\makelabel}[1]{\hss#1}}
4676   {\endlist}
4677   {\item}
4678 </numbbx>
4679 <*numcbx>
4680 \ProvidesFile{coppe-numeric.cbx}[2026/09/09 v4.1 CoppeTeX ABNT numeric citations]
4681 % Numeric citation system (ABNT NBR 10520): in-text references are bracketed
4682 % numbers [1], compressed and sorted within a single cite group. This is the
4683 % opt-in alternative to the author-date default (coppe.cbx); it pairs with
4684 % bibstyle=coppe-numeric. With the biblatex natbib option, \cite -> [1] and
4685 % \citet -> Author [1].
4686 \RequireCitationStyle{numeric-comp}
4687 </numcbx>

```

10.2 The bibliographic databases

The two BibLaTeX/biber databases shipped with the class — `example.bib` (the sample database cited by the example monograph) and `coppe.bib` (the references of this manual) — are generated from this source too, so the whole distribution still has a single editable origin. They live inside an `\iffalse` block below: the documentation driver never typesets them, and only `coppe.ins` extracts them, through the `examplebib` and `coppebib` docstrip targets. Edit the entries here, never the generated `.bib` files.

10.3 The generated companion files

Everything the distribution ships is extracted from this file: the class, the BibLaTeX styles, the language packs, the databases, the `.ist`, the demonstration documents and the `latexmkrc`. One `pdflatex coppe.ins` rebuilds all of it and there is no second place to edit. Each one sits in an `\iffalse` block: the documentation driver never typesets them, and only `coppe.ins` extracts them.

The line is drawn at what is DISTRIBUTED. What exists only to prove the class works stays outside, as ordinary files under version control:

`tests/` the regression suite. Nobody installing CoppeTeX needs it, and generating it from here would put proof and product in the same box.

`adversativa/` twelve documents – four kinds of work by three languages – each firing everything the class offers at once.

`tools/*.ps1` the build harness itself.

`NORMA_COPPE_2026.tex`, `futuremanual2026.tex` documents about the norm that live in this repository with their own life cycle.

Both suites are run by one command, `tools\prova.ps1`, and that is what has to come out clean before a version is tagged.

10.3.1 `example_pt.tex` -- pt-main demo

`example_pt.tex` – CoppeTeX 4.0 multilingual demo, main language: brazilian. Short demonstration document (cover + folha + abstracts + 1 chapter) used by the CPGP submission package. The full pt-main example with bibliography and lists is the historical `example.tex`.

10.3.2 `example_en.tex` -- en-main demo

`example_en.tex` – CoppeTeX 4.0 multilingual demo, main language: english.

10.3.3 `example_es.tex` -- es-main demo

`example_es.tex` – CoppeTeX 4.0 multilingual demo, main language: spanish. Requires the language pack files `coppe-lang-spanish.def` and `spanish-coppe.lbx` to be installed alongside `coppe.cls`.

10.3.4 `example_fr.tex` -- fr-main demo

`example_fr.tex` – CoppeTeX 4.0 multilingual demo, main language: french. Requires the language pack files `coppe-lang-french.def` and `french-coppe.lbx` to be installed alongside `coppe.cls`.

10.3.5 `example_it.tex` -- it-main demo

`example_it.tex` – CoppeTeX 4.0 multilingual demo, main language: italian. Requires the language pack files `coppe-lang-italian.def` and `italian-coppe.lbx` to be installed alongside `coppe.cls`.

10.3.6 `example_pdfa.tex` -- the full example as PDF/A

`example_pdfa.tex` – o `example.tex` completo, compilado com a opção ‘pdfa’.

Existe para dar ao veraPDF o documento mais exigente que a classe produz: o exemplo inteiro, com bibliografia, listas, figuras, tabelas, algoritmos e as caixas do `tcolorbox` – ou seja, transparência, que é justamente o que o `a-1b` proibiria e o `a-2b` admite. Validar só um documento curto não diria nada sobre o que vai de fato para o depósito.

`\PassOptionsToClass` tem de vir ANTES do `\documentclass` que o `example.tex` traz dentro de si; é o único jeito de acrescentar uma opção de classe a um arquivo gerado pelo `docstrip` sem duplicá-lo aqui e deixar as duas cópias divergirem com o tempo.

10.3.7 `covers_5languages.tex` -- the five covers side by side

`covers_5languages.tex` – One-page visual demonstration of the CoppeTeX 4.0 multilingual model: the five built-in main languages side by side, all sharing the same Portuguese institutional template, differing only in the title (and the body language, not visible on the cover).

This file uses the article class deliberately – it is a meta-document about the `coppe` class, not a thesis. It expects the five cover PNGs `cover_pt.png`, `cover_en.png`, `cover_es.png`, `cover_fr.png`, `cover_it.png` to live in the same directory (generated by `pdftoppm` from the `example_<lang>.pdf` demos).

10.3.8 `latexmkrc` -- latexmk/Overleaf build configuration

`latexmkrc` – CoppeTeX build configuration (works locally and on Overleaf).

Run with: `latexmk example` (or `latexmk futuremanual2026`, your thesis...) `latexmk` reads this file automatically when it is in the build directory. On Overleaf, place this file at the project root; Overleaf runs `latexmk`.

Engine: pdfLaTeX by default (fast). For LuaLaTeX (recommended, full Unicode) set `$pdf_mode = 4` below, or use the `"% !TeX program = lualatex"` magic comment / the Overleaf "Menu > Compiler" setting.

Acknowledgments

Thanks to all COPPETEX users who have reported their experience with this class. We also thank to professor Fernando Lizarralde and Heiko Oberdiek for their

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Index

Numbers written in *italic* refer to the page where the corresponding entry is described; numbers underlined refer to the code line of the definition; numbers in roman refer to the code lines where the entry is used.

Symbols		
\%	 952	\@coppcoadvresumotru <i>1705</i>
\,	 918, 927	\@coppedoublespacefalse <i>1662</i>
\;	... 837, 839, 841, 851, 853, 855, 856	\@coppedoublespacetrue <i>1663</i>
\@M	 1386	\@coppeexamefalse <i>1590</i>
\@Roman	 3281	\@coppeexametrue .. <i>1635, 1644</i>
\@advisor	 2262, 2271, 2272, 2273, 2274, 2275, 2276, 2277, 2575, 2724, 2739, 2843, 2870, 3015, 3057, 3114	\@coppeloglossprefalse <i>1565</i>
\@afterheading	... 1276	\@coppeloglosspretrue <i>3178, 3205</i>
\@afterindent	 1200	\@coppelistastocfalse <i>1717</i>
\@afterindentfalse	1201	\@coppelistastoctrue <i>1718</i>
\@authname	... 2298, 2512, 2546, 2680, 2811, 2827, 2829, 3005, 3047, 3104, 3507, 3564	\@coppemorewritesfalse .. <i>1768, 1796, 1809</i>
\@author	 3086	\@coppemorewritetrue <i>1765, 1767</i>
\@authsurn	2299, 2512, 2546, 2680, 2811, 2826, 2829, 3005, 3047, 3104, 3564	\@coppemwforcedfalse <i>1766</i>
\@chapapp	 1253	\@coppemwforcedtrue <i>1767</i>
\@chapter	 1249	\@coppnumericfalse <i>1656</i>
\@coadvisor	 2226, 2228, 2229, 2230, 2231, 2232, 2233, 2236, 2243, 2591, 2598, 2725, 2752, 2973, 2979	\@coppnumerictrue <i>1657</i>
\@coppeannexfalse	1242	\@coppeorientexamfalse <i>1694</i>
\@coppeannextrue	. 3625	\@coppeorientexamtrue <i>1695</i>
\@coppeappendixfalse	 1216	\@coppepdfafalse . <i>1676</i>
\@coppeappendixtrue	 1237	\@coppepdfatrue .. <i>1677</i>
\@coppcoadvresumofalse	 1704	\@copperascunhofalse <i>1719</i>
		\@copperascunhotrue <i>1720</i>
		\@copperomanfalse <i>1660</i>
		\@copperomantrue . <i>1661</i>
		\@coppesansseriffalse <i>1777</i>
		\@coppesansseriftrue <i>1775, 1776</i>
		\@coppetwosidefalse <i>1721, 1723</i>
		\@coppetwosidettrue <i>1722</i>
		\@date 3087
		\@dedic 2963, 2968
		\@degree 1623, 1631, 1640, 1649, 2365, 2366, 3448, 3453
		\@degreename .. 1624, 1632, 1641, 1650, 2367, 2368, 2813, 2859, 3414, 3568
		\@dotsep 1392, 1396, 1398
		\@dottedtocline .. 1328
		\@examiner 2282, 2290, 2291, 2292, 2293, 2294, 2295, 2722, 2764
		\@firstoftwo 3801
		\@footnotetext 1489, 1490
		\@for 1984
		\@highpenalty 1408, 1410
		\@ifstar 3251
		\@labfile 3209, 3210, 3222, 3227, 3255, 3260
		\@losfile 3182, 3183, 3195, 3200
		\@lst .. 2050, 2077, 2086
		\@mainaux .. 3513, 3540
		\@makechapterhead 1273, 1275
		\@nameuse 1849, 1858, 1859, 1860, 2203, 2214, 3236, 3238, 3247, 3524
		\@pnumwidth 1394
		\@roman 3272
		\@secondoftwo 3801
		\@tempdima 1353, 1354, 1360, 1361, 1362, 1363, 1371, 1372, 1373
		\@tempsymb1 ... 3185, 3187, 3212, 3214
		\@tocrmarg 1385
		\@topnewpage 1273
		\@twosidettrue 1816, 1822

\bibnamedelimb ... 3747	\citepapud 247, 722, 4333, 4338	\coppe@babelopts .. 1832,
\bibnamedelimi ... 3747		1844, 1847, 1984
\bibparsep 4673	\citet 243, 654,	\coppe@bancalist .. 2708,
\bibstring ... 3592,	694, 696, 697,	2734, 2746,
3839, 3842,	698, 699, 701,	2759, 2770, 2774
3848, 3932,	702, 703, 705,	
3945, 3946,	706, 714, 948,	
3947, 3948,	3580, 4318, 4685	\coppe@bancasize .. 2672, 2722,
4088, 4121,	\citetapud 247, 722, 4331, 4336	2724, 2725,
4143, 4147, 4324		2727, 2729, 2731
\bottomrule ... 379,	\citetitle 153	\coppe@bibBegin 3288,
413, 431, 452, 707	\citeyear 4340	3304, 3319, 3602
brazilianabstract	\ClassInfo . 1795, 1989	\coppe@bibEnd . 3284,
(env.) 3095	\ClassWarningNoLine	3301, 3319, 3606
\braziliankeyword 2329	... 1103, 1143,	\coppe@campo .. 2888,
	1696, 1805, 2400	2892, 2933, 2935
	\clearfield . 3856, 4107	\coppe@capescelula . 2895,
C	\clearname 3708, 3717, 3905	2920, 2927,
\c@braziliankeywords		2932, 2934, 2936
..... 2330	\closeout 3574	\coppe@capesgap 2890,
\c@coppeagencias .. 2361, 2362	\coadvisor ... 49, 2227	2891, 2893, 2938
\c@foreignkeywords 2321	\color 2043,	\coppe@checatipo .. 2899, 2912
\c@keywords 2312	2044, 2045,	\coppe@checkwrites . 1513, 1541
\c@page . 1574, 2469,	2046, 2056,	
3277, 3287,	2058, 2064,	\coppe@coadvisorrows 2971,
3291, 3295,	2065, 2070,	3031, 3073, 3130
3299, 3303, 3306	2071, 2072,	\coppe@dataaprov .. 2349,
\c@secnumdepth ... 1250	2079, 2080, 2081	2350, 2700, 2701
\c@tocdepth 1383	\convertcolorsU .. 2031	\coppe@date . 1060, 1063
\caption .. 255, 266,	\coppe@@algolang .. 2101, 2102, 2105	\coppe@degword 3943, 3951
290, 300, 318,	\coppe@@brazilianstr .. 1826, 1865, 1986	\coppe@deptcode 2128,
338, 351, 369,	\coppe@@cur 1864, 1865,	2141, 3566, 3568
403, 422, 444,	1866, 1985,	\coppe@docmonth 1068,
462, 600, 605,	1866, 1987,	2302, 2304, 2307
607, 636, 645,	1886, 1989, 1991	\coppe@docyear 1069, 2303,
657, 685, 783,		2518, 2624,
794, 833, 847, 927	\coppe@@englishstr . 1609,	2830, 2860,
\captionsetup 1428, 2108	1827, 1866, 1987	3007, 3049,
\chaptermark 1269	\coppe@@ext . 1543, 1544	3106, 3439, 3572
\citacao ... 1994, 1997	\coppe@@lang 1984, 1985	\coppe@fichafila .. 2876,
\cite . 654, 666, 668,	\coppe@@toc . 1543, 1544	2879, 2949, 2959
669, 670, 671,	\coppe@abntdate ... 3767, 3776, 3777	\coppe@fichapage .. 2877, 2879, 2959
672, 674, 675,	\coppe@appendixheadings .. 1229, 1237, 3628	\coppe@folio .. 1566,
676, 678, 694,	\coppe@areaconc ... 2347, 2348,	1567, 2419, 2431
696, 697, 698,	2935, 3420, 3422	
699, 700, 702,	\coppe@babelextra@ 1842	
703, 704, 706, 714	\coppe@babelload .. 1847, 1849, 1850	
\citeauthor 4340		
\citep 240, 243,		
654, 666, 668,		
669, 670, 671,		
673, 674, 675,		
677, 678, 714,		
718, 948, 3580, 4684		

\coppe@foreignlang .	\coppe@orig@printbibliography	1351, 1354, 3603, 3605 1366, 1369, 1371
... 1611, 1613, 1615, 1617, 1619, 1621, 1832, 1860, 3041	\coppe@orig@starttoc	1536, 1545
\coppe@glosspre .. 1561	\coppe@printkeywords	2336, 3035, 3078, 3081, 3137, 3140
\coppe@hasLof	\coppe@projetovinc .	2355, 2356, 2883
.. 3146, 3296, 3310	\coppe@selectsubtitle	2214, 2220, 2684
\coppe@hypersetup .	\coppe@selecttitle .	2203, 2682, 2829, 3523, 3561
..... 2528, 2780	\coppe@sigla@first .	3242, 3246
\coppe@linhapesq ..	\coppe@sigla@full .	3237, 3243, 3250
..... 2351, 2352, 2567, 2568	\coppe@sigla@norm .	3244, 3251
\coppe@listtoc	\coppe@sigla@reg ..	3234, 3243
... 1453, 1484, 1559, 2023, 3153, 3166, 3353	\coppe@sigla@star .	3248, 3251
\coppe@lstarrow ...	\coppe@sigla@undef .	3239, 3245, 3249
... 2040, 2041, 2048, 2060, 2067, 2075, 2084	\coppe@starttoc .. 1736	
\coppe@mainBegin ..	\coppe@thisvolume .	2249, 2251, 2255
.. 2428, 3292, 3307	\coppe@tipoproducao	2353, 2354, 2881, 2900, 2901, 2902, 2903, 2904, 2906
\coppe@mainlang ...	\coppe@title@brazilian	3527
... 1609, 1610, 1613, 1615, 1617, 1619, 1621, 1832, 1848, 1849, 1859, 2203, 2214, 2220, 2683, 3390, 3560	\coppe@titleblock .	2218, 2514, 2548
\coppe@marcatipo ..	\coppe@tmp ... 2740,	2747, 2753, 2760, 2765, 2771, 3553, 3555
... 2880, 2922, 2924, 2925, 2926	\coppe@tocanx 1246, 1256	
\coppe@marcavinc ..	\coppe@tocapp 1243, 1258	
.. 2882, 2930, 2931	\coppe@tocline 1382, 1409, 1412, 1414, 1416, 1418	
\coppe@membro . 2673,	\coppe@tocnumberline	1336, 1352, 1537, 1544
2740, 2753, 2765	\coppe@tocnummax ..	1342,
\coppe@membrogap ..		
..... 2665, 2670, 2671, 2728, 2730, 2732		
\coppe@membroid ...		
.. 2654, 2660, 2676		
\coppe@nomeprojeto .		
.. 2357, 2358, 2933		
\coppe@nvolumes 2248, 2250, 2253, 2256		
\coppe@orig@appendix		
..... 1235, 1237		
\coppe@orig@footnotetext		
..... 1489, 1490		
	\coppe@tocnumsaved .	1360, 1368
	\coppe@tocnumsep ..	1343, 1345, 1361, 1372
	\coppe@tocnumwidth .	1341, 1350, 1355, 1362, 1363, 1373, 1389
	\coppe@ucfamily ...	3754, 3774
	\coppe@ucscan 3757, 3758	
	\coppe@ucscanp	3760, 3762
	\coppe@ucstop . 3757,	3758, 3760, 3762
	\coppe@ucstr	3756, 3757, 3760
	\coppe@ucstrip ... 3734	
	\coppe@uctmp 3755, 3756	
	\coppe@version	1060, 1062, 2484
	\coppe@volumeline .	2252, 2515, 2549
	\coppe@wrapstarttoc	1535, 1547
	\coppe@xmphook	3541, 3544, 3546
	\coppe@xmpkeys 3547,	3551, 3554, 3565
	\coppe@xmpout . 3540,	3543, 3557, 3561, 3563, 3565, 3567, 3570, 3572, 3574
	\coppefinalengine .	2448, 2481
	\coppefinalfont ...	2455, 2482
	\coppefinalmanual .	2461, 2486
	\coppeforeignstring	1855, 1860, 2244, 2246, 3078, 3081
	\coppemainstring ..	1855, 1859, 2098, 2101, 2237, 2239, 2255, 2256, 2568, 2701, 3035,

3383,	3384,	4480,	4481,	4607,	4608,
3385,	3393,	4482,	4483,	4609,	4610,
3395,	3398, 3400	4484,	4485,	4611,	4612,
\copperdefstring ..		4486,	4487,	4613,	4614,
..... 1852,		4488,	4489,	4615,	4616,
1856,	1879,	4490,	4491,	4617,	4618,
1880,	1881,	4492,	4493,	4619,	4620,
1882,	1883,	4494,	4495,	4621,	4622,
1884,	1885,	4496,	4497,	4623,	4624,
1886,	1887,	4498,	4499,	4625,	4626,
1888,	1889,	4500,	4501,	4627,	4628,
1890,	1891,	4502,	4503,	4629,	4630,
1892,	1893,	4504,	4505,	4631,	4632,
1894,	1895,	4506,	4507,	4633,	4634,
1896,	1897,	4508,	4509,	4635,	4636,
1898,	1899,	4510,	4511,	4637,	4638,
1900,	1901,	4512,	4513,	4639, 4640, 4641	
1902,	1903,	4514,	4515,	\coppestring	
1904,	1905,	4516,	4517,	... 1244, 1247,	
1906,	1907,	4518,	4519,	1443, 1444,	
1908,	1909,	4520,	4521,	1460, 1854,	
1910,	1911,	4522,	4523,	1858, 2018,	
1912,	1913,	4524,	4525,	2019, 3134,	
1914,	1915,	4526,	4537,	3137, 3140,	
1916,	1917,	4538,	4539,	3591, 3595,	
1918,	1919,	4540,	4541,	3596, 3623,	
1920,	1921,	4542,	4543,	3951, 3981,	
1922,	1923,	4544,	4545,	3983, 3985, 3987	
1924,	1925,	4546,	4547,	\CoppeTeX 191, 1040,	
1926,	1927,	4548,	4549,	1086, 1579, 2483	
1928,	1929,	4550,	4551,	\coppetexfinalpage .	
1930,	1931,	4552,	4553,	.. 1046, 1061, <u>2448</u>	
1932,	1933,	4554,	4555,	\coppeUC@w	
1934,	1935,	4556,	4557,	.. 3798, 3799, 3800	
1936,	1937,	4558,	4559,	\coppeUCifart	
1938,	1940,	4560,	4561,	.. 3789, 3795, 3796	
1941,	1942,	4562,	4563,	\coppeUCkeep .. 3785,	
1943,	1944,	4564,	4565,	3791, 3793, 3794	
1945,	1946,	4566,	4567,	\coppeUCstop	
1947,	1949,	4568,	4569,	... 3787, 3788,	
1950,	1951,	4570,	4571,	3790, 3791,	
1952,	1953,	4572,	4573,	3792, 3793, 3794	
1954,	1955,	4574,	4575,	\coppeUCtitle . 3782,	
1956,	1957,	4576,	4577,	3787, 3809, 3813	
1958,	1959,	4578,	4579,	\coppeUCtitleA	
1960,	1961,	4580,	4581,	... 3787, 3788	
1962,	1963,	4582,	4583,	\coppeUCtitleB	
1964,	1965,	4584,	4594,	... 3790, 3792	
1966,	1967,	4595,	4596,	\cpfonte 255, 1458, 1464	
1968,	1969,	4597,	4598,	\cpsource	
1970,	1971,	4599,	4600,	255, 1458, 1461,	
1972,	1974,	4601,	4602,	1464, 1465, 1466	
1975,	1977,	4603,	4604,	\cpsourcenam 1460, 1463	
1978,	4479,	4605,	4606,		

D		4037,	4042,	2731,	2795,
\Dados	835, 849	4043,	4044,	2848,	2952,
\dataaprovacao		4088,	4658, 4659	2984,	3020,
.	66, 2350, 2695	\DeclareFixedFont		3062,	3080,
\Date	3572	 2005, 2006		3119,	3139,
\date	63, 1064, 1066,	\DeclareLanguageMapping		3149,	3162,
2301, 2620, 3094		 3682,		3346,	3354,
\DeclareBibliographyDriver	3869,	3683,	4529,	3370,	3382,
.	3928, 3929,	4530,	4585, 4642	3459,	3490,
3956,	4012,	\DeclareMathOperator		3542,	3584, 3801
4029,	4050,	 12, 13, 900, 901		\emergencystretch	
4063,	4081,	\DeclareNameAlias		 1206, 1209	
4082,	4083,	 3686,		\endfirsthead	
4084,	4085,	3687, 3688, 3689		 464	
4097,	4127,	\DeclareRobustCommand		\endfoot	
4161,	4162,	 4336, 4338		 472	
4190,	4202,	\DeclareSourcemap		\endhead	
4203,	4204,	4216		 469	
4207,	4208,	\dedication		\endlastfoot	
4209,	4210,	94, 2962		 477	
4211,	4212, 4213	\defbibenvironment		\endttitlepage	
\DeclareBibliographyStrings		4663		 2632	
.	4348, 4377,	\defbibheading		\endwrite	
.	4403, 4429, 4455	 3594		 3268	
\DeclareDatamodelEntryfield		\DefineVerbatimEnvironment		\Enquanto	
.	3651,	 2090, 2096		 830, 844, 852	
3652,	3654,	 2699, 2922,		\enquote	
3655, 3656, 3657		 2924, 2925,		 229, 1994, 1997	
\DeclareDatamodelEntrytypes		 62, 2128, 3505		\enspace	
.	3638	\det		 2339,	
\DeclareDatamodelFields		 875		 2699, 2922,	
.	3640, 3641,	\detokenize		 2924, 2925,	
3642,	3643,	 3269,		 2926, 2930, 2931	
3644,	3645,	 3286, 3287,		environments:	
3646,	3647,	 3290, 3291,		abstract	
3648, 3649, 3650		 3294, 3295,		brazilianabstract	
\DeclareDelimFormat		 3298, 3299,		 3095	
.	3690,	 3303, 3306,		foreignabstract	
3691,	4321,	 3309, 3312, 3756		longquote	
4322, 4323, 4324		\diag		 3320	
\DeclareFieldFormat		 13, 901, 906, 907		\es@scroman	
.	3803,	\dimexpr		 3273	
3811,	3815,	 2675, 2896, 2918		\examiner	
3816,	3819,	\directlua		 55,	
3822,	3823,	 1514, 1791		56, 57, 58, 61, 2282	
3824,	3825,	\document		\ExecuteBibliographyOptions	
3826,	3827,	 1356		 3672, 4655	
3830,	3831,	\dots		F	
3832,	3835,	 966		\familydefault	
3836,	3842,	\draw		 28, 31, 1783, 1786	
3848,	3932,	 341, 342, 343,		\fancyhead	
3950,	4009,	 354, 355, 356, 634		 1567,	
4010,	4011,	E		1818, 2419, 2431	
		\else		\fancyhf	
		1134, 1219, 1257,		 1567,	
		1260, 1263,		2419, 2423, 2431	
		1266, 1274,		\fancypagestyle	
		1383, 1514,		 1567,	
		1515, 1574,		2419, 2423, 2431	
		1797, 1804,		\fi	
		1865, 1866,		 1105, 1152,	
		1986, 1987,		1219, 1259,	
		2104, 2238,		1262, 1265,	
		2245, 2422,		1268, 1277,	
		2444, 2458,		1354, 1364,	
		2580, 2603,		1370, 1375,	
		2644, 2729,		1376, 1395,	
				1527, 1528,	
				1529, 1544,	

<code>\ifcoppennumeric</code> ..	<code>\ifstrempy</code> . 3185, 3212	<code>\leavevmode</code> 1387
..... 1656, 3581	<code>\ifuseeditor</code> 3704, 3713	<code>\left</code> 890
<code>\ifcoppeorientexam</code>	<code>\ifXeTeX</code> 2451	<code>\leftarrow</code> ... 837,
.. 1694, 2723, 2737	<code>\ignorespaces</code> 3338	838, 839, 851, 853
<code>\ifcoppepdfa</code> 1676, 3466	<code>\ignorespacesafterend</code>	<code>\leftskip</code> 1378, 1379,
<code>\ifcopperascunho</code> .	 3341	1385, 1389, 1390
..... 1719, 2950	<code>\in@</code> 3800	<code>\leq</code> 852
<code>\ifcopperoman</code> 1660,	<code>\index</code> 944	<code>\linewidth</code> 599, 604,
2399, 2418, 2440	<code>\InheritBibliographyExtras</code>	2502, 2540, 2896
<code>\ifcoppesansserif</code> .	... 4345, 4374,	<code>\linhapesquisa</code> 79, 2347
.. 1775, 1785, 2456	4400, 4426, 4452	<code>\listabbreviationname</code>
<code>\ifcoppetwoside</code> ..	<code>\InputIfFileExists</code> .	 3206, 3383
..... 1721, 1821	 1867, 1988	<code>\listof</code> 257, 1468
<code>\if@english</code> . 1605, 1831	<code>\int</code> 918	<code>\listofalgorithms</code> . 142
<code>\if@files</code> 1367	<code>\interlinepenalty</code> 1386	<code>\listoffigures</code> 137, 3145
<code>\if@mainmatter</code> ... 1251	<code>\itemsep</code> 4672	<code>\listofmapas</code> 24, 140, 334
<code>\if@openright</code> 2442	<code>\itshape</code> 1295, 1304,	<code>\listofprogramas</code> ..
<code>\if@twoside</code> . 1573, 2468	1416, 1418, 3336	 141, 801,
<code>\ifbibstring</code>		2000, 2020, 2021
.. 3932, 3940, 3941	J	<code>\listofquadros</code>
<code>\ifblank</code> 3759,	<code>\javaexternal</code> 2088	... 139, 257, 1451
3763, 3790,	<code>\javainline</code>	<code>\listoftables</code> 138, 3159
3791, 3793, 4341	... 829, 830, 2087	<code>\listquadroname</code> 1444,
<code>\ifboolexpr</code>	<code>\javastyle</code> ... 2078,	1452, 1453, 1454
... 3704, 3713,	2085, 2087, 2088	<code>\listsymbolname</code> ...
3804, 3812, 4148		 3179, 3384
<code>\ifdefined</code> 1514, 1791,	K	<code>\local@advisorstring</code>
3488, 3491, 3539	<code>\keyword</code> 72, 73,	 2278,
<code>\ifdim</code> 1354,	74, 2310, 3497, 3505	2572, 2840,
1362, 1366, 1373	<code>\Keywords</code> 3565	3012, 3111, 3387
<code>\iffieldequalstr</code> ..	L	<code>\local@approvedname</code>
... 3880, 3945,	<code>\l@</code> 1475	.. 2773, 3393, 3395
3946, 3947, 3948	<code>\l@chapter</code> 1407	<code>\local@approvedonname</code>
<code>\iffieldundef</code> . 3768,	<code>\l@figure</code>	.. 2699, 3398, 3400
3769, 3770,	... 1450, 1475, 1481	<code>\local@cityname</code> ...
3854, 3858,	<code>\l@paragraph</code> 1417	... 2517, 2623,
3861, 3881,	<code>\l@quadro</code> .. 1449, 1450	2829, 3405, 3438
3944, 3982,	<code>\l@section</code> 1411	<code>\local@coadvisorstring</code>
4057, 4072,	<code>\l@subsection</code> 1413	 2235,
4107, 4141,	<code>\l@subsubsection</code> . 1415	2595, 3031, 3130
4145, 4148, 4172	<code>\labelnamepunct</code> ...	<code>\local@countryname</code> .
<code>\IfFileExists</code> 1798	... 3873, 3910,	 3407, 3438
<code>\ifin@</code> 3801	3959, 3991,	<code>\local@degname</code>
<code>\iflistundef</code> .. 3879,	4066, 4100,	... 1625, 1633,
3967, 3980,	4131, 4168, 4193	1642, 1651,
3984, 3986,	<code>\labelnumberwidth</code> 4668	2369, 2370,
4020, 4093, 4178	<code>\lambda</code> 906, 907	3419, 3432, 3448
<code>\ifLuaTeX</code> 2450	<code>\language</code> 1858, 3135	<code>\local@deptname</code> ...
<code>\ifnameundef</code> .. 3704,	<code>\Large</code> . 1218, 1231, 1281	... 2132, 2143,
3713, 3804,	<code>\large</code> . 1289, 2915, 2947	2147, 2151,
3812, 3901, 4119	<code>\LaTeXe</code> 1089	2155, 2159,
<code>\ifodd</code> 1574, 2469	<code>\ldots</code> 861	2163, 2167,
<code>\ifPDFTeX</code>	<code>\leaders</code> 1392	2171, 2175,
.. 1101, 1132, 2449		2179, 2183,

4100,	4102,	\node	344, 357,	3964,	3966,
4104,	4108,		632, 633, 773, 776	3973,	3983,
4110,	4112,	\nomeprojeto . .	82, <u>2347</u>	3998,	4000,
4131,	4133,	\normalfont		4017,	4021,
4135,	4138,		. . . 1218, 1231,	4022,	4025,
4151,	4168,		1281, 1289,	4046,	4047,
4170,	4172,		1292, 1295,	4049,	4057,
4174,	4183,		1304, 1412,	4072,	4076,
4185,	4193,		1414, 1416, 1418	4077,	4107,
4195,	4197, 4199	\not	3335	4109,	4111,
\newcoppefloat .	25,	\null	1390, 2474	4136,	4137,
	257, 334, 1467, 1478	\numberline		4143,	4147,
\newfloat			. . . 1261, 1378,	4172,	4173,
	. . 1445, 1469, 1479		1537, 1544, 1835	4179,	4180,
\newif	1216,			4181,	4184,
	1242, 1565,			4198,	4666, 4667
	1590, 1656,	O			
	1660, 1662,	\OR . . .	2901, 2902, 2903	\printindex	944
	1676, 1694,	\or 2304, 2305, 2306,		\printlist . . .	3879,
	1704, 1717,		2307, 2308, 2309	3919,	3920,
	1719, 1721,			3969,	3970,
	1765, 1766, 1775	P		3981,	3985,
\newlength . . .	1341,	\PackageWarning . .	3240	3987,	4019,
	1342, 1343,	\Para	830, 838	4020,	4092,
	2670, 2890, 3320	\paragraph . . .	1300,	4095,	4175, 4178
			1303, 1305, 1306	\printloabbreviations	
\newlist	1495	\parbox	2953		 143, <u>3204</u>
\newsavebox	2040	\parfillskip	1385	\printlosymbols . . .	
\newsigla		\parsep	3329, 4673		 144, <u>3173</u>
	997, 998, 1000, <u>3231</u>	\pdfLaTeX	1091	\printnames . . .	3705,
\newunit 3875, 3891,		\pdfsuppressptexinfo		3714,	3903, 4125
	3893, 3912,		. . 3475, 3488, 3489	\printtext . . .	3707,
	3916, 3918,	\pdfTeX	1092	3716,	3839,
	3922, 3925,	\pdfvariable . .	3477,	3879,	3904,
	3961, 3963,		3484, 3491, 3492	3981,	3983,
	3965, 3972,	\penalty	1410	3985,	3987,
	3974, 3993,	\phantom . . .	2881, 2883	4057,	4072,
	3995, 3997,	\printbibliography .		4121,	4143,
	3999, 4001,		 726, 1015,	4147,	4172, 4665
	4018, 4024,		3315, 3598,	\printurldate	3861
	4026, 4053,		3601, 3603, 3604	\projetoavinculado .	
	4056, 4060,	\printdate . . .	3883,		 81, <u>2347</u>
	4068, 4071,		3884, 3890,	\prologstyle 2069, 2076	
	4075, 4078,		3921, 3962,	\protected 3754, 3758,	
	4102, 4104,		3971, 4023,	3762,	3767,
	4108, 4110,		4096, 4149, 4182	3787,	3788,
	4112, 4133,	\printeventdate . .	4048	3792,	3794, 3796
	4135, 4138,	\printfield . . .	3856,	\protected@edef 2711,	
	4151, 4170,		3860, 3876,	2740,	2753,
	4172, 4174,		3882, 3885,	2765,	3553, 3755
	4183, 4185,		3886, 3887,	\providecommand . . .	
	4195, 4197, 4199		3888, 3889,	. . . 1443, 1444,	
\nobreak 1390, 1391, 1393			3892, 3915,	1447,	1482,
\nobreakspace			3917, 3923,	1856,	1858,
	. . 1245, 1248, 3768		3924, 3949,	1859,	1860,

1863,	1874,	363,	384,	592,	\singlespacing	
2365,	2367,	615,	625,	643,	. . . 1463, 1490,	
2369,	2371,	712,	728,	799,	2339,	2414,
2373,	2375,	863,	870,	878,	2500,	2506,
2448,	2455,	887,	896,	910,	2535,	2558,
\ProvidesFile		925,	929,	933,	2566,	2691,
. . . 3632, 3660,		937,	942,	946,	2774,	2913,
4311,	4344,	952,	962,	965,	3334,	3661, 3772
4373,	4399,	968,	971,	977, 1288	\smallskipamount . 1462	
4425,	4451,	\selectcolormodel 2031			\source 255,	
4478,	4536,	\selectlanguage . . .			257, 293, 334,	
4593,	4645,	 1854, 3336			645, 810, 1420,	
\Publisher 3570		\sep 3551			1456, 1465, 1468	
\pythonexternal . . 2052		\SetAlCapFnt 2118			\step . . . 4219, 4220,	
\pythoninline		\SetAlCapNameFnt . 2119			4221,	4222,
. . . 812, 829, 2051		\SetAlCapNameSty . 2121			4223,	4224,
\pythonstyle . . 2042,		\SetAlCapSty 2120			4225,	4226,
2049,	2051,	2052	\SetAlgoCaptionSeparator		4227,	4228,
			 2117		4229,	4230,
Q		\SetKwFor . . 2122, 2123			4231,	4232,
\quadroautorefname 1447		\setlist . . . 1496, 1497			4233,	4234,
\quadroname		\setmathfont			4235,	4236,
. . 1443, 1446, 1447		 11, 916, 1149			4237,	4238,
		\setstretch 2477			4239,	4242,
R		\settowidth 1353			4243,	4244,
\raggedright		\setunit 3706, 3715,			4245,	4246,
. . . 2571, 2594,		3857,	3861,		4247,	4248,
2839,	2975,	3873,	3879,		4249,	4250,
3011,	3053,	3882,	3883,		4251,	4252,
\recuolongquote . . .		3884,	3885,		4253,	4254,
. . 3320, 3321, 3325		3886,	3887,		4255,	4256,
\refstepcounter . . 1252		3888,	3889,		4257,	4259,
\renewbibmacro 3703,		3890,	3904,		4260,	4261,
3712,	3839,	3853	3906,	3910,	4262,	4263,
\RequireBibliographyStyle		3920,	3921,		4264,	4265,
. 3668, 4650		3924,	3959,		4266,	4267,
\RequireCitationStyle		3966,	3969,		4268,	4270,
. 4314, 4686		3970,	3971,		4271,	4272,
\resizebox . . . 418, 424		3980,	3982,		4273,	4274,
\restylefloat 1438, 1439		3984,	3986,		4275,	4276,
\Resultado . . . 836, 850		3991,	4020,		4277,	4278,
\Retorna . . 841, 855, 856		4021,	4022,		4279,	4280,
\right 893		4023,	4047,		4281,	4282,
\rightmargin 3326		4048,	4049,		4283,	4284,
\rightskip 1385		4058,	4066,		4285,	4286,
\RMSE 12, 900, 905		4073,	4077,		4287,	4288,
\romannumeral		4095,	4096,		4289,	4290,
. . 3277, 3295, 3299		4100,	4131,		4291,	4292,
		4137,	4144,		4293,	4294,
S		4168,	4178,		4295,	4296,
\sbox 2038, 2041		4179,	4180,		4297,	4298,
\Se 855		4181,	4182, 4193		4299,	4300, 4301
\section 183, 194, 202,		\sfdefault . . . 31, 1786			\stretch 2257,	
226,	260,	332,	\sigla 1000, 3231		2511,	2513, 2516

<code>\stripzeros</code> . 3768, 3770	<code>\title</code> 35, 1872, 4030, 4051,
<code>\Subject</code> 3567	2195, 3496, 3504 4052, 4059,
<code>\subsection</code> ... 395,	<code>\titleformat</code> .. 1217, 4061, 4062,
716, 720, 724,	1230, 1280, 4064, 4065,
732, 743, 752, 1291	1288, 1291, 4067, 4069,
<code>\subsubsection</code>	1294, 1303, 1307 4070, 4074,
..... 1294, 1306	<code>\titlein</code> ... 1870, 1874 4079, 4080,
<code>\subtitle</code> .. 2204, 2208	<code>\titlespacing</code> 4081, 4082,
<code>\subtitlein</code> . 2205, 2212	... 1221, 1234, 4083, 4084,
<code>\syml</code> 219, 220, 980,	1282, 1302, 1305 4085, 4098,
981, 982, 983,	<code>\today</code> 1065, 2482 4099, 4101,
984, 985, 986, 3184	<code>\topmargin</code> . 1166, 1173 4103, 4113,
	<code>\toprule</code> 372, 4114, 4128,
	407, 426, 447, 693 4129, 4130,
T	<code>\topsep</code> 3330 4132, 4134,
<code>\tabcolsep</code> . 2896, 2918	<code>\ttb</code> 2005, 2044, 2045, 4152, 4153,
<code>\tablename</code> 467	2056, 2058, 4161, 4162,
<code>\tableofcontents</code> ..	2064, 2071, 4166, 4167,
..... 145, 1734	2072, 2080, 2081 4169, 4186,
<code>\text</code> 891, 892	<code>\ttfamily</code> 2007, 2043, 4187, 4191,
<code>\textcite</code>	2054, 2063, 4192, 4194,
654, 714, 4318, 4337	2070, 2079, 2097 4196, 4200,
<code>\textcolor</code> . 2029, 2041	<code>\ttm</code> 2006, 2042, 2053, 4201, 4202,
<code>\textemdash</code> 3969	2062, 2069, 2078 4203, 4204,
<code>\textendash</code>	4207, 4208,
... 1219, 1232,	U
1245, 1248, 2117	<code>\uppercase</code> 3274 4209, 4210,
<code>\textsc</code> 2478	4211, 4212, 4213
<code>\textsuperscript</code> .. 329	<code>\url</code> 157, 939, 2485, 3842
<code>\thechapter</code> ... 1219,	<code>\usebibmacro</code> .. 3707,
1232, 1253,	3716, 3870,
1256, 1258, 1261	3871, 3872,
<code>\theFancyVerbLine</code> 2097	3874, 3894,
<code>\thefield</code> 3768,	3895, 3904,
3769, 3770, 3941	3908, 3909,
<code>\theparagraph</code> 1304	3911, 3913,
<code>\thesection</code> 1289	3914, 3926,
<code>\thesubsection</code> ... 1292	3927, 3928,
<code>\thesubsubsection</code> .	3929, 3957,
..... 1295, 1307	3958, 3960,
<code>\thetable</code> 467	3975, 3976,
<code>\times</code> 2881, 2883	3989, 3990,
<code>\tipoproducao</code>	3992, 3994,
... 80, 2347, 2905	3996, 4002,
<code>\Title</code> 3561	4003, 4013,
	4027, 4028,
	V
	<code>\varnothing</code> 8, 861, 1146
	<code>\volume</code> 2251
	<code>\volumes</code> 2250
	X
	<code>\XeLaTeX</code> 1094
	<code>\xmlstyle</code> .. 2053, 2061
	<code>\xspace</code> 963

Change History

v0.0		v3.1	
General: Creation Date.	1	<code>\department</code> : Included a sort key in <code>symb</code>	59
v0.1		v3.2	
General: Documentation:		General: Fixed version problem between <code>cls</code> and <code>dtx</code>	1
bibliography fixed, title translation.	1	v3.3	
Sourceforge submission.	1	General: Extend <code>abrev</code> to work like <code>symb</code> and accept a sorting key .	1
v0.2		v3.4	
General: Unification of the code for the list of symbols and abbreviations.	1	General: Some examples for figures, tables, longtables, etc. .	1
v0.3		v3.4.1	
General: Added 'draft' option. . . .	1	General: Fix english option to consider brazilian a second language, describe it in document	58
<code>\maketitle</code> : Added number of examiners test.	69	v3.5	
Generalization.	69	General: New command <code>annex</code> . .	110
v0.4		v3.5.1	
General: Beta documentation. . . .	1	General: New command <code>annex</code> now supports brazilian or english	110
v0.5		v4.0	
<code>abstract</code> : Changed from macro to environment.	77	General: Catalog-label writers (<code>coppe@mainBegin</code> , <code>coppe@bibBegin</code> , <code>coppe@bibEnd</code> , <code>coppe@hasLof</code>) made language-agnostic: <code>detokenize</code> wrap on label names protects ':' and '.' from babel shorthand activation, and the raw <code>etex</code> primitives <code>romannumeral</code> / <code>number</code> replace LaTeX's <code>roman</code> / <code>arabic</code> to bypass <code>spanish-babel</code> 's <code>small-caps @roman</code> redefinition that would otherwise crash <code>pdftex</code> with "Unbalanced write command" on every <code>es-main</code> document.	1
<code>\backmatter</code> : Added <code>mainmatter</code> pages counter.	65	Class-wide string dispatcher (<code>copperdefstring</code> , <code>coppestring</code> , <code>coppemainstring</code> , <code>coppeforeignstring</code>) replaces scattered <code>iflanguage</code> and <code>if</code> english branches; <code>pt/en</code> outputs remain byte-identical. .	1
<code>foreignabstract</code> : Changed from macro to environment.	78	Multilingual model: three fixed slots (<code>main</code> , <code>foreign</code> , <code>optional</code> <code>third</code>); five built-in language options (<code>brazilian</code> , <code>english</code> ,	
v1.0			
General: First COPPETeX release. . .	1		
v2.0			
General: COPPETeX release 2.0. . . .	1		
v2.1			
General: COPPETeX release 2.1:			
Matching the new rules.	1		
<code>\advisor</code> : Advisors, co-advisors, co-co-advisors, etc., all of them are simply considered advisors. .	61		
v2.2			
General: Matching new guidelines, including new logo.	1		
v2.2.2			
General: Fixed some text constants in <code>.bib</code> and documented it here. .	1		
v3.0			
General: Added support for monographs in English.	58		
New approval page layout.	72		
<code>\department</code> : Added new course on Nanotechnology.	59		
<code>\examiner</code> : Examiners expansion without degree.	62		

spanish, french, italian); plug-in mechanism for any other Babel language via coppe-lang-lang.def plus lang-coppe.lbx.	1	sumario indicative.	1
New user API: titlein, usecoppelanguage, brazilianabstract environment. .	1	New options: pdfa (PDF/A-2b via pdfx, XMP built from the document), assinaturas, coorientador, rascunhoficha, listasnosumario, morewrites/semmorewrites. New command coadvisor. Under pdfTeX the class loads morewrites by itself when installed, because a work that asks for every list needs seventeen of TeX's sixteen output streams.	1
<code>\coppetexfinalpage</code> : Initial coppetexfinalpage colophon page.	66	newcoppfloat fixed twice: newfloat was wrapped in a group, and since newfloat defines the environment locally the new float vanished when the group closed; and its list came out unnumbered and without leaders because l@figure was not inherited. Author-declared floats are now numbered within the chapter like every other illustration. . . .	1
v4.1 General: Breaking change in how the board is declared: advisor, coadvisor and examiner now take the TREATMENT as the optional argument, empty by default, and the INSTITUTION as the last mandatory one, which may be left empty. 3.1.2.1.3(e) asks for name, titulacao and institution and asks for no treatment, so that is what the folha de aprovacao prints.	1	No signature rules on the folha de aprovacao: the deposit has been digital only since Resolucao CEPG n. 246/2023, so there is nothing to sign by hand. The assinaturas option is still accepted and only warns. .	1
Conformance with the revised 9th edition (2026) of the UFRJ/SiBI manual: continuous pagination from the folha de rosto, folio in 10pt at 2cm from the top and right edges, one-sided layout with a 3cm left margin, additional sheet with the catalog card and the CAPES collection fields, approval sheet of 3.1.2.1.3, keywords closing the three abstracts, pre-textual lists out of the sumario, Apendice and Anexo headings centred, fifth heading level, and Latin Modern instead of the bitmap fonts.	1	Single source: pdfatex coppe.ins generates everything distributed – class, biblatex styles, language packs, .bib bases, .ist, the five per-language examples, example_pdfa, the cover montage and latexmkrc. The regression suite, the adversarial documents and the harness prove the class and are NOT distributed.	1
Fixes found by the adversarial review: cover and folha de rosto compose in single spacing so they do not overflow under doublespacing, no folio is printed on a pre-textual sheet when something does overflow, the approval sheet holds a board of eight, and es-nosectiondot removes the period babel-spanish put in the		The class sets the document in the sans-serif family by DEFAULT; comserifa restores the serif one. Neither the manual nor the COPPE norm prescribes a family, and the SiBI manual that carries the rule is itself set in Arial. semserifa is still accepted and	

now does nothing.	1	this there was no choice, and the only way out was not to declare the advisor at all, which also removed them from the capa and the folha de rosto. 1
The folha adicional draws the Coleta CAPES fields as the closed frame of Annex H – one cell per field, a rule between cells, and no rules to write on – instead of loose paragraphs with dotted fill rules.	1	The research-line label on the folha de rosto follows the main language of the work instead of being hard-coded in Portuguese.
The folha de aprovacao lists only the EXAMINERS. The sheet records who examined the work, and the advisor conducted it; the new orientadorexamina option puts advisors and coadvisors back, ahead of the examiners, for the Programas that seat the advisor on the board. Before		With no dataaprovacao given, the folha de aprovacao writes the date as "a ser determinada" in the main language instead of drawing a rule to fill in by hand: the deposit is digital and nobody writes on a PDF.